

AOS-CX 10.06 Security Guide

6100 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: <code><example-text></code> <i>example-text</i> <code>example-text</code> <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Table Continued

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

This AOS-CX Switch provides the following security features:

- Local user and group management.
- Authentication, Authorization, and Accounting (AAA), either local (password or SSH public key-based), or remote password-based TACACS+ or RADIUS.
- SSH server. SSH is a cryptographic protocol that encrypts all communication between devices.
- Ability to use enhanced security as described in [Configuring enhanced security](#).
- Making sensitive switch configuration information available for secure export/import between switches. For information, see [service export-password](#).

About Authentication, Authorization, and Accounting (AAA)

- **Authentication:** identifies users, validates their credentials, and grants switch access.
- **Authorization:** controls authenticated users command execution and switch interaction privileges.
- **Accounting:** collects and manages user session activity logs for auditing and reporting purposes.

Local AAA on your Aruba switch provides:

- Authentication using local password or SSH public key.
- Authorization using role-based access control (RBAC), and optionally, using user-defined local user groups with command authorization rules defined per group.
- Accounting of user activity on the switch using accounting logs.

Remote AAA provides the following for your Aruba switch:

- Authentication using remote AAA servers with either TACACS+ or RADIUS.
- Authorization using remote AAA servers with TACACS+ fine-grained command authorization. Local RBAC or local rule-based authorization is also possible.
- Transmission of locally collected accounting information to remote TACACS+ and RADIUS servers.



NOTE: TACACS+ (Terminal Access Controller Access-Control System Plus) and RADIUS (Remote Authentication Dial-In User Service) server software is readily available as either open source or from various vendors.



NOTE: For switches that support multiple management modules such as the Aruba 8400, all AAA functionality discussed only applies to the active management module. See also *AAA on switches with multiple management modules* in the *High Availability Guide*.

Default user `admin`

A factory-default switch comes with a single user named `admin`.

The `admin` user:

- Has an empty password. Press **Enter** in response to the `admin` password prompt. At initial boot, you are prompted to define a password for the `admin` user. Although empty (blank) passwords are allowed, it is recommended that you use strong passwords for all production switches.
- Is a member of the `administrators` group.
- Cannot be removed from the switch.



NOTE: The switch `admin` user is distinct from the Service OS `admin` user. The Service OS acts as the `bootloader` and recovery operating system. The Service OS has its own CLI.

Example of first login with password setting

```
switch login: admin
Password:

Please configure the 'admin' user account password.
Enter new password: *****
Confirm new password: *****
switch#
```

Built-in user groups and their privileges

The switch provides the following built-in user groups with corresponding roles. Each of these roles comes with a set of privileges.

Group/Role	Privileges
administrators	Administrators have full privileges, including: • Full CLI access. • Performing firmware upgrades. • Viewing switch configuration information, including sensitive information such as passwords which are displayed as ciphertext. • Performing switch configuration. • Adding/removing user accounts. • Configuring users accounts, including passwords. Once set, a password cannot be deleted or set to empty. • REST API: All methods (GET, PUT, POST, DELETE) and switch resources are available. The privilege level for <code>administrators</code> is 15.
operators	Operators have no switch configuration privileges. Operators are restricted to: • Basic display-only CLI access. • Viewing of nonsensitive switch configuration information. • REST API: Other than the <code>\login</code> and <code>\logout</code> resources, only the GET method is available. The privilege level for <code>operators</code> is 1.
auditors	Auditors are restricted to functions related to auditing only: • CLI: Access to commands in the auditor context (<code>auditor></code>) only. • Web UI: Access to the System > Log page only. • REST API: POST method available for the <code>\login</code> and <code>\logout</code> resources. GET method available for the following resources only: ◦ Audit log: <code>/logs/audit</code> ◦ Event log: <code>/logs/event</code> The privilege level for <code>auditors</code> is 19.

User-defined user groups

The switch enables you to create up to 29 user-defined local user groups, for the purpose of configuring local authorization. Each of the 29 user-defined groups support up to 1024 CLI command authorization rules that define what CLI commands can be executed by members of the group. See [user-group](#).

The local user groups with their command execution rules are useful for the following:

- Providing authorization for use with RADIUS servers.
- Providing fallback authorization for use with TACACS+ servers.
- Providing authorization when neither RADIUS or TACACS+ servers are used.

User name requirements

<USERNAME>

Specifies the user name. Requirements:

- Must start with a lowercase letter.
- Can contain numbers and lowercase letters.
- Can include only these three special characters: hyphens (-), dots (.), and underscores (_).
- Can have a maximum of 32 characters.
- Cannot be empty.
- Cannot contain uppercase letters.
- Cannot be: `admin`, `root`, or `remote_user`.
- Cannot be Linux reserved names such as:

`daemon`, `bin`, `sys`, `sync`, `proxy`, `www-data`, `backup`, `list`, `irc`, `gnats`, `nobody`, `systemd-bus-proxy`, `sshd`, `messagebus`, `rpc`, `systemd-journal-gateway`, `systemd-journal-remote`, `systemd-journal-upload`, `systemd-timesync`, `systemd-coredump`, `systemd-resolve`, `rpcuser`, `vagrant`, `opds`, `rdanet`, `_lldpd`, `rdaadmin`, `rdaweb`, `docker_container`, `tss`.

Password requirements

Passwords must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
- Contain at most 32 characters.
- Contain at least the number of characters configured (optionally) for `minimum-password-length`.



CAUTION: Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



NOTE: Only an administrator can change the password of a user assigned to the `operators` role.

User and user group management tasks

User and user group management common tasks are as follows. Simple tasks link to the relevant command description. Larger tasks link to the relevant multistep procedure.

Task	Command or procedure	Example
Creating a user	<u>user</u>	user jamie group administrators password
Changing a user password	<u>user password</u>	user jamie password
Removing a user	<u>user</u>	no user jamie
Setting a user account password	<u>user password</u>	user admin password
Resetting the admin password using the Service OS	(<u>procedure</u>)	
Resetting the admin password by reverting the switch to factory defaults	(<u>procedure</u>)	erase startup-config boot system
Showing a list of all users	<u>show user-list</u>	show user-list
Showing information for the logged-in user	<u>show user information</u>	show user information
Creating a user group	<u>user-group</u>	user-group admuser2
Adding command authorization rules to a user group	or (within <u>user-group</u>)	10 deny cli command "show aaa .*" 20 permit cli command "show .*"
Adding comments to rules in a user group	(within <u>user-group</u>)	10 comment Deny all show aaa commands. 20 comment Permit all other show commands.
Resequencing rules in a user group	(within <u>user-group</u>)	resequence 100 20
Showing a list of all user groups	<u>show user-group</u>	show user-group

Resetting the switch admin password using the Service OS console

Perform this task only when the switch (Product OS) admin user password has been forgotten.

Prerequisites

- You are connected to the switch through the console port.
- You know the Service OS password (if configured).



NOTE: If you forget the Service OS password, the only recourse is to zeroize the switch, reverting it to factory defaults. For information see *Zeroization* in the *Diagnostics and Supportability Guide*.

Procedure

1. Reboot the switch.
2. At the boot prompt, select 0. Service OS Console.

0. Service OS Console

1. Primary Software Image [XL.01.01.0001]
2. Secondary Software Image [XL.01.01.0002]

3. At the Switch Login prompt, enter `admin` and press `Enter`. If prompted for a Service OS password, enter it and press `Enter`.

```
Switch login: admin
Password: *****
Hewlett Packard Enterprise
SVOS>
```

4. At the `SVOS>` prompt, enter `password` and press `Enter`.
5. Enter the new switch (Product OS) password at both password prompts.

```
SVOS> password
Enter password: *****
Confirm password: *****
SVOS>
```

6. Enter `boot` and press `Enter`.

```
SVOS> boot

ServiceOS Information:
  Version:          PL.10.06.0001
  Build Date:       2020-12-18 05:25:54 PDT
  Build ID:         ServiceOS:PL.01.07.0003-internal:ab08328b9622:202009090525
  SHA:              ab08328b96224e8d56b3e629da9de40bd0f02677

Boot Profiles:

0. Service OS Console
1. Primary Software Image [PL.10.06.0001]
2. Secondary Software Image [PL.10.06.0001]

Select profile(primary):
```

7. To boot with the primary switch image press 1 and then `Enter`. To boot with the secondary switch image, press 2 and then `Enter`. If you make no selection for approximately 10 seconds, the switch boots the default image. The default is shown in parentheses to the right of `Select profile`, for example:
`Select profile(primary):.`
8. Once in AOS-CX, save the configuration to make the `admin` login user account password setting persistent.

Resetting the admin password by reverting the switch to factory defaults



CAUTION: This task erases all switch configuration, reverting the switch to its factory default state. Consider using other less-impacting techniques for admin password reset. For example, another administrator user can reset the admin user password to a known value. See also [Resetting the switch admin password using the Service OS console](#).

Prerequisites

If wanted, you have saved a copy of the switch configuration.

Procedure

1. At the manager command prompt, enter `erase startup-config`.

```
switch# erase startup-config
```

2. Enter `boot system`, responding `n` to the Do you want to save the current configuration prompt and then responding `y` to the Continue prompt.

```
switch# boot system
Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

3. At the login prompt, enter `admin` and press Enter. The admin password remains empty until it is set.

User and group commands

user

Syntax

```
user <USERNAME> group {administrators | operators | auditors | <USER-GROUP>}
    password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

```
no user <USERNAME>
```

Description

Creates a user and adds the user to one of the user groups. Users are given the privileges of their group. For the three built-in user groups (`administrators`, `operators`, `auditors`), the privileges are fixed. For user-defined local user groups, the privileges are defined by the CLI command authorization rules of the group.

When entered without either optional `ciphertext` or `plaintext` parameters, the cleartext password is prompted for twice, with the characters entered masked with "*" symbols.

The `no` form of this command removes a user account from the switch. The administrator cannot delete the user account from which they are logged in. The `admin` user cannot be deleted.

Command context

config

Parameters

<USERNAME>

Specifies the user name. Requirements:

- Must start with a lowercase letter.
- Can contain numbers and lowercase letters.
- Can include only these three special characters: hyphens (-), dots (.), and underscores (_).
- Can have a maximum of 32 characters.
- Cannot be empty.
- Cannot contain uppercase letters.
- Cannot be: `admin`, `root`, or `remote_user`.
- Cannot be Linux reserved names such as:

`daemon`, `bin`, `sys`, `sync`, `proxy`, `www-data`, `backup`, `list`, `irc`, `gnats`, `nobody`, `systemd-bus-proxy`, `sshd`, `messagebus`, `rpc`, `systemd-journal-gateway`, `systemd-journal-remote`, `systemd-journal-upload`, `systemd-timesync`, `systemd-coredump`, `systemd-resolve`, `rpcuser`, `vagrant`, `opds`, `rdanet`, `_lldpd`, `rdaadmin`, `rdaweb`, `docker_container`, `tss`.

group

Selects the local user group to which the new user will be assigned.

administrators | operators | auditors

Selects one of three built-in local user groups.

<USER-GROUP>

Specifies an existing user-defined local user group.

ciphertext <CIPHERTEXT-PASSWORD>

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable **<CIPHERTEXT-PASSWORD>** is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



NOTE: The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Up to 63 local users can be added, for a total of 64 users including the default user `admin`. A user can belong to only one group.
- The switch ships with the `admin` user account and three built-in local user groups: `administrators`, `operators`, and `auditors`. The `admin` account belongs to the `administrators` group. The Service OS also includes the administrator user `admin`. The two `admin` users are entirely distinct.
- When a local user account is removed, the user loses all active login/SSH sessions. Any calls on the existing REST session with that local user account fail with a permissions issue as soon as the user is deleted. Soon afterwards, the existing REST sessions with the deleted local user account become invalidated. If a user is viewing the GUI while their account is deleted, the user is redirected to the login page within 60 seconds. The home directory associated with the user is also removed from the switch.
- Cleartext passwords (whether entered with prompting or entered directly) must:
 - Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
 - Contain at most 32 characters.
 - Contain at least the number of characters configured (optionally) for `minimum-password-length`.



CAUTION: Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



NOTE: Only an administrator can change the password of a user assigned to the `operators` role.

Examples

Creating local user `jamie` in the `administrators` group with a prompted password:

```
switch(config)# user jamie group administrators password
Adding user jamie
Enter password:*****
Confirm password:*****
```

Creating user `chris` in the existing user-defined local user group `admuser2` with a cleartext password, using direct entry without prompting:

```
switch(config)# user chris group admuser2 password plaintext passWORDxJ|989
```

Creating user `alex` in the `operators` group with a ciphertext password (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex group operators password ciphertext NDcDI2...8igJfA=
```

Removing user `jamie`:

```
switch(config)# no user jamie
User jamie's home directory and active sessions will be deleted.
Do you want to continue [y/n]?y
```

user-group

Syntax

```
user-group <GROUP-NAME>
```

```
no user-group <GROUP-NAME>
```

Description

If `<GROUP-NAME>` does not exist, this command creates a local user group and then enters its context. If `<GROUP-NAME>` exists, this command enters the context for the specified `<GROUP-NAME>`. Within the `<GROUP-NAME>` context, several subcommands are available for working with rules that specify what CLI commands are permitted or denied for all members of the local group.

In addition to the three built-in user groups `administrators`, `operators`, and `auditors`, up to 29 user-defined local user groups can be defined. All users can be members of only one of the up to 32 groups.

The `no` form of this command deletes the specified user group. All members of the deleted group lose all command authorization privilege.



CAUTION: Do not causally delete user-defined local user groups without understanding the implications. Although user-defined local user groups can be deleted with the respective members losing all privileges, the three built-in groups `administrators`, `operators`, and `auditors` are always available and their privileges are unchangeable.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Subcommands

These subcommands are available within the user-defined local user group context (shown in the switch prompt as `config-usr-grp-<GROUP-NAME>`).

```
[<SEQ-NUM>] {permit | deny} cli command "<REGEX>"
```

```
no <SEQ-NUM>
```

Defines a CLI command privilege `permit` or `deny` rule. There is an implicit `"deny ."` rule at the end of every user-defined group rule list. Members of a user-defined group without any `permit` rules have no CLI command privileges.

The `no` form of this subcommand deletes the specified (by sequence number) rule from the group.



NOTE: Rule evaluation proceeds from lowest to highest sequence number until the first successful match, resulting in either CLI command permission or denial. Rule evaluation ceases upon first match. Therefore, rules for related CLI commands must be defined in most restrictive to least restrictive order.

<SEQ-NUM>

Specifies the CLI command rule sequence number. When omitted, a sequence number that is 10 greater the highest existing sequence number is auto-assigned. When no rules exist, the first auto-assigned sequence number is 10.

{permit | deny}

Sets the rule type as either `permit` or `deny`. Rule order is important. For example, these two related rules together authorize all `show` commands except for the `show aaa` commands.

```
switch(config-usr-grp-admuser2)#10 deny cli command "show aaa .*"
switch(config-usr-grp-admuser2)#20 permit cli command "show ."
```

To achieve the wanted effect in this example, the `deny` rule must precede the `permit` rule. These two rules together achieve the following:

- All `show aaa` commands match on rule 10, triggering command denial, and the immediate cessation of further rule evaluation. Matching on rule 20 is never attempted.
- All other `show` commands (excluding `show aaa` commands) match on rule 20 and are therefore permitted.

<REGEX>

Specifies the CLI command matching criteria of the rule. The criteria can be expressed as `".*"` which matches all commands. Otherwise, the criteria is expressed as a POSIX-compliant regular expression (regex) string starting with an exact match command token (for example `show`) followed by a regex representing command arguments. The first word must be a string that contains only alphanumeric or hyphen characters.

For example, to allow all commands starting with the word `interface`, the regex must be `"interface .*"` or just `"interface"`. Using `"interface.*"` (without the space) is not supported. For example, `"show .*"` matches every `show` command. Consult the Extended regular expression information available at: https://pubs.opengroup.org/onlinepubs/9699919799/basedefs/V1_chap09.html#tag_09_04.

Sample matching criteria	Sample matched CLI command or specifier	Matches
<code>show .*</code>	<code>show accounting log</code>	All <code>show</code> commands
<code>bgp .*</code>	<code>bgp router-id 1.1.1.1</code>	All <code>bgp</code> commands
<code>interface .*</code>	<code>interface 1/1/1</code>	All interface specifiers
<code>vlan (3 4)</code>	<code>vlan 3</code>	VLAN 3 or 4
<code>vlan [1-9]</code>	<code>vlan 5</code>	A single VLAN in the range 1 to 9
<code>vlan ([1-9] 1[0-9])</code>	<code>vlan 19</code>	A single VLAN in the range 1 to 19

[<SEQ-NUM>] comment <TEXT-STRING>

no <SEQ-NUM> comment

Adds a comment to an existing rule. The `no` form of this subcommand removes an existing comment.

```
switch(config-usr-grp-admuser2)# 10 comment Deny all show aaa commands.
switch(config-usr-grp-admuser2)# 20 comment Permit all other show commands.
switch(config-usr-grp-admuser2)#
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show ."
```


include <GROUP-NAME>

[no] include <GROUP-NAME>

Include all rules from the specified user-defined <GROUP-NAME>. Only one group can be included in the definition of another group. The content of the included group is effectively placed at the top of the rules list in the current group. If the specified <GROUP-NAME> does not exist, it is created.

The no form of this subcommand removes the specified included group from the current group. The specified included group must exist and must be included in the current group or else an error message is shown.

The name of the included group is shown at the top of the `show user-group` command for the group with the include.

In this example, group `admuser1` is included in group `admuser2`. So the `admuser1` rules are evaluated first and then the rules in the `admuser2` group are only evaluated if no CLI command match occurs for the rules in group `admuser1`.

```
switch(config-usr-grp-admuser2)# include admuser1
switch(config-usr-grp-admuser2)# show user-group admuser2
User Group Summary
=====
Name           : admuser2
Type           : configuration
Included Group  : admuser1
Number of Rules : 2
User Group Rules
=====

```

SEQUENCE NUM	ACTION	COMMAND	COMMENT
10	deny	show aaa .*	Deny all show aaa commands.
20	permit	show .*	Permit all other show commands.

resequence [**<STARTING-SEQ-NUM>** **<INCREMENT>**]

Resequences the CLI command authorization rules. When entered without the optional parameters the rules are resequenced with a <STARTING-SEQ-NUM> of 10 and an <INCREMENT> of 10.

<STARTING-SEQ-NUM>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Resequencing the rules to start at 100 with an increment of 20:

```
switch(config-usr-grp-admuser2)# resequence 100 20
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  100 comment Deny all show aaa commands.
  100 deny cli command "show aaa .*"
  120 comment Permit all other show commands.
  120 permit cli command "show .*"

```

Resequencing the rules to the default of starting at 10 with an increment of 10:

```
switch(config-usr-grp-admuser2)# resequence
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show .*"

```

show running-config current-context

Shows all the commands used to configure the rules in the current group context.

```
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show .*"

```

list

List the subcommands available within the user-defined group context.

exit

Exits the user-defined group context.

end

Exits the user-defined group context and then the config context.

user password

Syntax

```
user <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

Description

Changes a password for an account or enables the password for the admin account. When entered without either optional `ciphertext` or `plaintext` parameters, the cleartext password is prompted for twice, with the characters entered masked with "*" symbols.

Command context

config

Parameters

<USERNAME>

Specifies the corresponding user name for the password you want to change.

ciphertext <CIPHERTEXT-PASSWORD>

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable `<CIPHERTEXT-PASSWORD>` is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



NOTE: The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The admin account is available on the switch without a password by default.

Cleartext passwords (whether entered with prompting or entered directly) must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [\x21-\x7E] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [\x3F] (decimal 63)) is not permitted.
- Contain at most 32 characters.
- Contain at least the number of characters configured (optionally) for `minimum-password-length`.



CAUTION: Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



NOTE: Only an administrator can change the password of a user assigned to the `operators` role.

Examples

Enabling (or changing) a cleartext password for `admin`:

```
switch(config)# user admin password
Changing password for user admin
Enter password:*****
Confirm password:*****
```

Changing the cleartext password for user `chris`, using direct entry without prompting:

```
switch(config)# user chris password plaintext PASSwordZQ#@67
```

Changing the ciphertext password for user `alex` (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex password ciphertext XqYJ36...W83D4Y=
```

service export-password

Syntax

```
service export-password
```

```
no service export-password
```

Description

Configures a nondefault export password. The export password is used to transform critical security parameters (such as password hashes) into ciphertext suitable for exporting and showing by commands such as `show running-config`. This transformation enables safe switch configuration import and export.

The `no` form of this command reverts the export password to its factory default.



NOTE: All factory-default switches have identical default export passwords. For security, it is recommended that you set the same nondefault export password on every switch in a group that will exchange configuration information. Only switches with identical export passwords can exchange configuration information.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Usage

Prompts you twice for the new export password.

The export password must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed.
- Contain at most 32 characters.
- Not be blank.

Examples

Configuring a new export password:

```
switch(config)# service export-password
Enter password:*****
Confirm password:*****
```

Reverting the export password to its factory default:

```
switch(config)# no service export-password
```

show user-group

Syntax

```
show user-group [<GROUP-NAME>]
```

Description

Shows user group information for the built-in groups plus any user-defined local user groups. When entered without `<GROUP-NAME>`, summary information is shown for all groups.

Command context

Manager (#)

Parameters

<GROUP-NAME>

Narrows the `show` command output to that of the specified group, and for local user groups, adds the User Group Rules list.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Show the list of all user groups, including built-in groups and local user groups.

```
switch# show user-group
GROUP NAME      GROUP TYPE      INCLUDED GROUP  NUMBER OF RULES
-----
administrators  built-in        n/a             n/a
admuser1        configuration    --             5
admuser2        configuration    admuser1        2
auditors        built-in        n/a             n/a
operators       built-in        n/a             n/a
```

Show detailed information for local user group `admuser2`.

```
switch(config-usr-grp-admuser2)# show user-group admuser2
User Group Summary
=====
Name           : admuser2
Type           : configuration
Included Group : admuser1
Number of Rules : 2
User Group Rules
=====
SEQUENCE NUM  ACTION      COMMAND          COMMENT
-----
10            deny        show aaa .*      Deny all show aaa commands.
20            permit      show .*          Permit all other show commands.
```

show user information

Syntax

```
show user information
```

Description

Shows the following information for the logged-in user:

- User name.
- User authentication type: `local`, `RADIUS`, or `TACACS+`.
- User group: `administrators`, `operators`, or `<GROUP-NAME>`.
- User privilege level: For the built-in user groups and `RADIUS` or `TACACS+`, the role privilege level value is shown. For user-defined user groups, `N/A` is shown.

Command context

Operator (`>`) or Manager (`#`)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (`>`) only.

Examples

Showing information for the `admin` user:

```
switch# show user information
Username           : admin
Authentication type : Local
User group         : administrators
User privilege level : 15
```

Showing information for a member of the user-defined local user group `admuser2`:

```
switch# show user information
Username       : admin2-b
Authentication type : Local
User group     : admuser2
User privilege level : N/A
```

Showing information for a member of operators:

```
switch# show user information
Username       : operator
Authentication type : Local
User group     : operators
User privilege level : 1
```

Showing information for remote RADIUS user `rad_user1` mapped to local user group `administrators`:

```
switch# show user information
Username       : rad_user1
Authentication type : RADIUS
User group     : administrators
User privilege level : 15
```

Showing information for remote RADIUS user `rad_user2` mapped to local user group `operators`:

```
switch# show user information
Username       : rad_user2
Authentication type : RADIUS
User group     : operators
User privilege level : 1
```

Showing information for remote TACACS+ `tac_user1` logged in with `priv-lvl 15` (mapped to user group `administrators`):

```
switch# show user information
Username       : tac_user1
Authentication type : TACACS+
User group     : administrators
User privilege level : 15
```

show user-list

Syntax

```
show user-list
```

Description

Shows all configured users and their corresponding group names.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Show the user list from a switch with only the admin user defined.

```
switch# show user-list
```

USER	GROUP
admin	administrators

Show the user list after adding a user to the operators built-in group.

```
switch# show user-list
```

USER	GROUP
admin	administrators
oper1	operators

Show the user list after adding a user to the auditors built-in group.

```
switch# show user-list
```

USER	GROUP
admin	administrators
oper1	operators
audit1	auditors

Show the user list after adding a total of three users to two user-defined user groups.

```
switch# show user-list
```

USER	GROUP
admin	administrators
oper1	operators
audit1	auditors
adm1a	admuser1
admin2-a	admuser2
admin2-b	admuser2

About the SSH server



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

SSH (Secure Shell) is a cryptographic protocol that encrypts all communication between devices.

Each switch VRF includes an SSH server. The SSH server on the `mgmt` VRF is enabled by default in software version 10.02 and higher, and disabled in version 10.01 and lower. Only the SSH servers included in the switch are supported.

The SSH server provides SSH client to switch communications, enabling SSH clients (at least SSH v2.0) to connect to the switch for the purpose of managing it. The SSH server interfaces with the authentication service that provides local and/or remote AAA.



NOTE: The SSH server will perform a rekey operation for all open SSH sessions at every hour or after 1 GB of data transferred, whichever occurs first. The rekey is performed to address a common security concern that encryption/decryption keys not be used for long periods of time. This limits the amount of data exposed in the unfortunate case where a key is exposed or refactored.



NOTE: SSH public key authentication is separate from SSH server. Look for information on *SSH public key* under **Local authentication**.

SSH defaults

Setting	Default value
Maximum SSH password retries	3 password retries.
Password-based (with SSH client) authentication	Enabled.
SSH password-based login grace period timeout	120 seconds.
SSH public key authentication	Enabled.
SSH idle session timeout	60 seconds.

SSH server tasks

SSH server tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Enabling the SSH server	<u>ssh server vrf</u>	ssh server vrf default
Disabling the SSH server	<u>no ssh server vrf</u>	no ssh server vrf default
Generating an SSH host-key pair	<u>ssh host-key</u>	ssh host-key rsa bits 2048
Clearing the list of trusted SSH servers for your user account	<u>ssh known-host remove</u>	ssh known-host remove 1.1.1.1
Configuring SSH to use a set of ciphers	<u>ssh ciphers</u>	ssh ciphers chacha20-poly1305@openssh.com aes256-ctr aes256-cbc
Configuring SSH to use a set of host key algorithms	<u>ssh host-key-algorithms</u>	ssh host-key-algorithms ssh-rsa ssh-ed25519 ecdsa-sha2-nistp521
Configuring SSH to use a set of MACs	<u>ssh macs</u>	ssh macs hmac-sha2-256 hmac-sha2-512
Configuring SSH to use a set of key exchange algorithms	<u>ssh key-exchange-algorithms</u>	ssh key-exchange-algorithms ecdh-sha2-nistp256
Configuring SSH to use a set of public key algorithms	<u>ssh public-key-algorithms</u>	ssh public-key-algorithms x509v3-ssh-rsa ssh-rsa rsa-sha2-256
Configuring SSH idle session timeout	<u>cli-session</u>	switch(config)# cli-session switch(config-cli-session)# timeout 20
Showing the SSH server configuration	<u>show ssh server</u>	show ssh server all-vrfs
Showing the active SSH sessions	<u>show ssh server sessions</u>	show ssh server sessions all-vrfs
Showing the SSH server host keys	<u>show ssh host-key</u>	show ssh host-key ecdsa

SSH server commands



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

show ssh host-key

Syntax

```
show ssh host-key [ecdsa | ed25519 | rsa]
```

Description

Shows the public host keys for the SSH server. If the key type is not provided, all available host-keys are shown.

Command context

Manager (#)

Parameters

ecdsa

Selects the ECDSA host-key pair.

ed25519

Selects the ED25519 host-key pair.

rsa

Selects the RSA host-key pair.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the ECDSA public host-key:

```
switch# show ssh host-key ecdsa

Key Type : ECDSA      Curve : ecdsa-sha2-nistp256

ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
04mjVFGMVkZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=
```

Showing all public host keys:

```
switch# show ssh host-key

Key Type : ECDSA      Curve : ecdsa-sha2-nistp256
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
04mjVFGMVkZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=

Key Type : ED25519
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGb6910Jwoe8Hkl9K5YhqijrWI3yovNbiJVq6tw4WjJr4

Key Type : RSA        Key Size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDdVCXlw43h4n1bwg9jI6DSBMngymCdPD0JUG42Sn9IS
...
nGSXtrNy6OmlFDJTAY+zz5Kd8d21ZLuhf07IHNgF3pff65Xc8qNJBv
```

show ssh server

Syntax

```
show ssh server [vrf default]
```

Description

Shows the SSH server configuration for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

vrf default

Specifies the default VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing the SSH server configuration on the default VRF:

```
switch# show ssh server
```

SSH server configuration on VRF default :

```
IP Version      : IPv4 and IPv6      SSH Version      : 2.0
TCP Port        : 22                  Grace Timeout (sec) : 120
Max Auth Attempts : 6
```

Ciphers:

```
chacha20-poly1305@openssh.com, aes128-ctr, aes192-cbc,
aes128-cbc, aes192-ctr, aes256-gcm@openssh.com,
aes128-gcm@openssh.com, aes256-ctr, aes256-cbc
```

Host Key Algorithms:

```
ecdsa-sha2-nistp256, ecdsa-sha2-nistp384, ecdsa-sha2-nistp521,
ssh-ed25519, rsa-sha2-256, rsa-sha2-512, ssh-rsa
```

Key Exchange Algorithms:

```
curve25519-sha256, curve25519-sha256@libssh.org,
ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521,
diffie-hellman-group-exchange-sha256, diffie-hellman-group16-sha512,
diffie-hellman-group18-sha512, diffie-hellman-group14-sha256,
diffie-hellman-group14-sha1
```

MACs:

```
hmac-sha1-etm@openssh.com, umac-64@openssh.com,
umac-128@openssh.com, hmac-sha2-256, hmac-sha2-512, hmac-sha1
```

Public Key Algorithms:

```
rsa-sha2-256, rsa-sha2-512ssh-rsa, ecdsa-sha2-nistp256,
ecdsa-sha2-nistp384, ecdsa-sha2-nistp521, ssh-ed25519,
x509v3-rsa2048-sha256, x509v3-ssh-rsa, x509v3-sign-rsa,
x509v3-ecdsa-sha2-nistp256, x509v3-ecdsa-sha2-nistp384,
x509v3-ecdsa-sha2-nistp521
```

show ssh server sessions

Syntax

```
show ssh server sessions [vrf default]
```

Description

Shows the active SSH sessions on the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

vrf default

Specifies the default VRF.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If you provide the command with a VRF name, the command shows the active SSH session for the specified VRF. Any user can show sessions of all VRFs by using the `all-vrfs` parameter. The maximum number of sessions per VRF is five. The maximum SSH idle session timeout is 60 seconds.

Examples

Showing the active SSH sessions on the default VRF:

```
switch# show ssh server sessions

SSH sessions on VRF default
  IPv4 SSH Sessions
    Server IP      : 10.1.1.1
    Client IP      : 10.1.1.2
    Client Port    : 58835

    IPv6 SSH Sessions
    Server IP      : FF01:0:0:0:0:0:0:FB
    Client IP      : FF01:0:0:0:0:0:0:FC
    Client Port    : 58836
```

ssh ciphers

Syntax

```
ssh ciphers <CIPHERS-LIST>
```

```
no ssh ciphers
```

Description

Configures SSH to use a set of ciphers in the specified priority order. Ciphers in SSH are used for privacy of data being transported over the connection. The first cipher type entered in the CLI is considered a first priority. Each option is an algorithm that is used to encrypt the link and each name indicates the algorithm and cryptographic parameters that are used. Only ciphers that are entered by the user are configured.

The `no` form of this command removes the configuration of ciphers and reverts SSH to use the default set of ciphers.

Command context

config

Parameters

<CIPHERS-LIST>

Valid cipher types are:

- aes128-cbc
- aes192-cbc
- aes256-cbc
- aes128-ctr
- aes192-ctr
- aes256-ctr
- aes128-gcm@openssh.com
- aes256-gcm@openssh.com
- chacha20-poly1305@openssh.com

Default set of ciphers in priority order:

1. chacha20-1305@openssh.com
2. aes128-ctr
3. aes192-ctr
4. aes256-ctr
5. aes128-gcm@openssh.com
6. aes256-gcm@openssh.com

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use only specified ciphers in the priority order:

```
switch(config)# ssh ciphers chacha20-poly1305@openssh.com aes256-ctr aes256-cbc
```

Reverting SSH to use the default set of ciphers:

```
switch(config)# no ssh ciphers
```

ssh host-key

Syntax

```
ssh host-key {ecdsa [ecdsa-sha2-nistp256 | ecdsa-sha2-nistp384 | ecdsa-sha2-nistp521] |  
              ed25519 | rsa [bits {2048 | 4096}] }
```

Description

Generates an SSH host-key pair.

Command context

config

Parameters

ecdsa

Selects the ECDSA host-key pair type as `ecdsa-sha2-nistp256` (the default), `ecdsa-sha2-nistp384`, or `ecdsa-sha2-nistp521`.

ed25519

Selects the ED25519 host-key pair.

rsa

Selects the RSA host-key pair. Optionally, the key bit length is selected with either `bits 2048` (the default) or `bits 4096`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When an SSH server is enabled on a VRF for the first time, host-keys are generated.

If the host-key of the given type exists, a warning message is displayed with a request to overwrite the previous host-key with the new key.

Examples

Overwriting an old ECDSA host-key with a new `ecdsa-sha2-nistp384` host-key:

```
switch(config)# ssh host-key ecdsa ecdsa-sha2-nistp384  
ecdsa host-key will be overwritten.  
Do you want to continue (y/n)?
```

Overwriting an old RSA host-key with a new RSA host-key with 2048 bits:

```
switch(config)# ssh host-key rsa bits 2048  
rsa host-key will be overwritten.  
Do you want to continue (y/n)?
```

Overwriting an ECDSA host-key with an ED25519 host-key pair:

```
switch(config)# ssh host-key ed25519  
ed25519 host-key will be overwritten.  
Do you want to continue (y/n)?
```

ssh host-key-algorithms

Syntax

```
ssh host-key-algorithms <HOST-KEY-ALGORITHMS-LIST>
```

```
no ssh host-key-algorithms
```

Description

Configures SSH to use a set of host key algorithms in the specified priority order. Host key algorithms specify which host key types are allowed to be used for the SSH connection. The first host key entered in the CLI is considered a first priority. Each option represents a type of key that can be used. Host keys are used to verify the host that you are connecting to. This configuration allows you to control which host key types are presented to incoming clients, or which host key types to receive first from hosts. Only the host key algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of host key algorithms and reverts SSH to use the default set of algorithms.

Command context

`config`

Parameters

<HOST-KEY-ALGORITHMS-LIST>

Valid host key algorithms are:

- `ecdsa-sha2-nistp256`
- `ecdsa-sha2-nistp384`
- `ecdsa-sha2-nistp521`
- `rsa-sha2-256`
- `rsa-sha2-512`
- `ssh-ed25519`
- `ssh-rsa`

Default set of host key algorithms in priority order:

1. `ecdsa-sha2-nistp256`
2. `ecdsa-sha2-nistp384`
3. `ecdsa-sha2-nistp521`
4. `ssh-ed25519`
5. `rsa-sha2-256`
6. `rsa-sha2-512`
7. `ssh-rsa`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use only specified host key algorithms:

```
switch(config)# ssh host-key-algorithms ssh-rsa ssh-ed25519 ecdsa-sha2-nistp521
```

Reverting SSH to use the default set of host key algorithms:

```
switch(config)# no host-key-algorithms
```

ssh key-exchange-algorithms

Syntax

```
ssh key-exchange-algorithms <KEY-EXCHANGE-ALGORITHMS-LIST>
```

```
no ssh key-exchange-algorithms
```

Description

Configures SSH to use a set of key exchange algorithm types in the specified priority order. The first key exchange type entered in the CLI is considered a first priority. Key exchange algorithms are used to exchange a shared session key with a peer securely. Each option represents an algorithm that is used to distribute a shared key in a way that prevents outside interference, manipulation, or recovery. Only the key exchange algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of key exchange algorithms and reverts SSH to use the default set of algorithms.

Command context

```
config
```

Parameters

<KEY-EXCHANGE-ALGORITHMS-LIST>

Valid key exchange algorithms are:

- `curve25519-sha256`
- `curve25519-sha256@libssh.org`
- `diffie-hellman-group-exchange-sha1`
- `diffie-hellman-group-exchange-sha256`
- `diffie-hellman-group14-sha1`
- `diffie-hellman-group14-sha256`
- `diffie-hellman-group16-sha512`
- `diffie-hellman-group18-sha512`
- `ecdh-sha2-nistp256`
- `ecdh-sha2-nistp384`
- `ecdh-sha2-nistp521`

Default set of key exchange algorithms in priority order:

1. `curve25519-sha256`
2. `curve25519-sha256@libssh.org`
3. `ecdh-sha2-nistp256`
4. `ecdh-sha2-nistp384`
5. `ecdh-sha2-nistp521`

6. `diffie-hellman-group-exchange-sha256`
7. `diffie-hellman-group16-sha512`
8. `diffie-hellman-group18-sha512`
9. `diffie-hellman-group14-sha256`
10. `diffie-hellman-group-exchange-sha1`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified key exchange algorithms:

```
switch(config)# ssh key-exchange-algorithms ecdh-sha2-nistp256 curve25519-sha256  
diffie-hellman-group-exchange-sha256
```

Reverting SSH to use the default set of key-exchange-algorithms:

```
switch(config)# no key-exchange-algorithms
```

ssh known-host remove

Syntax

```
ssh known-host remove {all | {<IPv4-ADDRESS> | <HOSTNAME> | <IPv6-ADDRESS>} }
```

Description

Clears the list of trusted SSH servers for your user account. When you download or upload a file to or from a server using SFTP, you establish a trusted SSH relationship with that server. Each user account maintains its own set of SSH server host-keys for every server to which the user previously connected.

Command context

config

Parameters

all

Clears the trusted servers list.

<IPv4-ADDRESS>

Specifies the IPv4 address of the remote device.

<HOSTNAME>

Specifies the host name of the remote device. The length of the host name can be up to 255 characters.

<IPv6-ADDRESS>

Specifies the IPv6 address of the remote device.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing the trusted server list:

```
switch(config)# ssh known-host remove all
```

Removing a specified server from the trusted server list:

```
switch(config)# ssh known-host remove 1.1.1.1
```

ssh macs

Syntax

```
ssh macs <MACS-LIST>
```

```
no ssh macs
```

Description

Configures SSH to use a set of message authentication codes (MACs) in the specified priority order. The first MAC entered in the CLI is considered a first priority. MACs maintain the integrity of each message sent across an SSH connection. Each option represents an algorithm that can be used to provide integrity between peers. Only the MAC types that are specified by the user are configured.

The `no` form of this command removes the configuration of MACs and reverts SSH to use the default set of MACs.

Command context

```
config
```

Parameters

<MACS-LIST>

Valid MAC types are:

- hmac-sha1
- hmac-sha1-96
- hmac-sha1-etm@openssh.com
- hmac-sha2-256
- hmac-sha2-512
- hmac-sha2-256-etm@openssh.com
- hmac-sha2-512-etm@openssh.com

Default set of MACs in priority order:

1. hmac-sha2-256-etm@openssh.com
2. hmac-sha2-512-etm@openssh.com
3. hmac-sha1-etm@openssh.com
4. hmac-sha2-256

5. `hmac-sha2-512`

6. `hmac-sha1`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified MACs:

```
switch(config)# ssh macs hmac-sha2-256 hmac-sha2-512
```

Reverting SSH to use the default set of MACs:

```
switch(config)# no ssh macs
```

ssh maximum-auth-attempts

Syntax

```
ssh maximum-auth-attempts <ATTEMPTS>
```

```
no maximum-auth-attempts
```

Description

Sets the SSH maximum number of authentication attempts.

The `no` form of the command resets the maximum to its default of 6.

Command context

`config`

Parameters

<ATTEMPTS>

Specifies the maximum number of SSH authentication attempts. Range: 1 to 10. Default: 6.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the maximum number of authentication attempts:

```
switch(config)# ssh maximum-auth-attempts 3
```

Resetting the maximum number of authentication attempts to its default of 6:

```
switch(config)# no maximum-auth-attempts
```

ssh public-key-algorithms

Syntax

```
ssh public-key-algorithms <PUBLIC-KEY-ALGORITHMS-LIST>
```

```
no ssh public-key-algorithms
```

Description

Configures SSH to use a set of public key algorithms in the specified priority order. The first public key type entered in the CLI is considered a first priority. Public key algorithms specify which public key types can be used for public key authentication in SSH. Each option represents a public key type that the SSH server can accept or that the SSH client can present to a server. Only the public key algorithms that are chosen by the user are configured.

The `no` form of this command removes the configuration of public key algorithms and reverts SSH to use the default set.

Command context

```
config
```

Parameters

<PUBLIC-KEY-ALGORITHMS-LIST>

Valid public key algorithm types are:

- `ecdsa-sha2-nistp256`
- `ecdsa-sha2-nistp384`
- `ecdsa-sha2-nistp521`
- `ssh-ed25519`
- `ssh-rsa`
- `rsa-sha2-256`
- `rsa-sha2-512`
- `x509v3-ecdsa-sha2-nistp256`
- `x509v3-ecdsa-sha2-nistp384`
- `x509v3-ecdsa-sha2-nistp521`
- `x509v3-rsa2048-sha256`
- `x509v3-sign-rsa`
- `x509v3-ssh-rsa`

Default set of public key algorithms in priority order:

1. `rsa-sha2-256`
2. `rsa-sha2-512`
3. `ssh-rsa`

4. ecdsa-sha2-nistp256
5. ecdsa-sha2-nistp384
6. ecdsa-sha2-nistp521
7. ssh-ed25519
8. x509v3-rsa2048-sha256
9. x509v3-ssh-rsa
10. x509v3-sign-rsa
11. x509v3-ecdsa-sha2-nistp256
12. x509v3-ecdsa-sha2-nistp384
13. x509v3-ecdsa-sha2-nistp521

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified public key algorithms:

```
switch(config)# ssh public-key-algorithms x509v3-ssh-rsa ssh-rsa rsa-sha2-256
```

Reverting SSH to use the default set of public key algorithms:

```
switch(config)# no ssh public-key-algorithms
```

ssh server vrf

Syntax

```
ssh server vrf <VRF-NAME>  
no ssh server vrf <VRF-NAME>
```

Description

Enables the SSH server on the specified VRF.

The `no` form of the command disables the SSH server on the specified VRF.

Command context

```
config
```

Parameters

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the SSH server on the management VRF:

```
switch(config)# ssh server vrf mgmt
```

Disabling the SSH server on the management VRF:

```
switch(config)# no ssh server vrf mgmt
```

About the SSH client

The switch provides an SSH client that enables the switch to log in to an SSH server such as another switch, typically for command execution purposes. The SSH client provides secure encrypted communications between the switch and the SSH server over any network.

SSH client commands



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

ssh (client login)

Syntax

```
ssh [<USERNAME>@]{<IPV4> | <HOSTNAME>} [vrf <VRF-NAME>] [port <PORT-NUMBER>]
```

Description

Establishes a client session with an SSH server which is typically another switch.

Command context

Manager (#)

Parameters

<USERNAME>

Specifies the username that the client uses to log in to an SSH server. When omitted, the username of the current session is used.

{<IPV4> | <HOSTNAME>}

Specifies the SSH server to which the SSH client will connect.

- `<IPV4>`: The IPv4 address.
- `<HOSTNAME>`: The host name.

vrf <VRF-NAME>

Specifies the VRF to be used for the SSH client session. When omitted, the default VRF named `default` is used.

port <PORT-NUMBER>

Specifies the SSH server TCP port number. When omitted, the default TCP port 22 is used.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Establishing an SSH client session (using the management VRF) with an SSH server:

```
switch# ssh admin@10.0.11.180 vrf mgmt
```

Establishing an SSH client session (using the default VRF and a specific port) with an SSH server:

```
switch# ssh admin@10.0.11.175 port 223
```

Configuring a test user on switch 1 and then connecting to switch 1 from switch 2 using the SSH client on the mgmt VRF:

**** Configuring a test user on switch 1 ****

```
switch(config)# user-group test
```

```
switch(config-usr-grp-test)# permit cli command ".*"
```

```
switch(config)# exit
```

```
switch(config)# user test-user group test password plaintext tst#9J%
```

**** On switch 2, connecting to switch 1 using the SSH client ****

```
switch# ssh test-user@10.0.11.177 vrf mgmt
```


About local AAA

Local AAA on your Aruba switch provides:

- Authentication using local password or SSH public key.
- Authorization using local role-based access control (RBAC). Optional per-command authorization is possible through configuration of user-defined local user groups, with command authorization rules applied to respective group members.
- Accounting of user activity on the switch using accounting logs.



NOTE: For switches that support multiple management modules such as the Aruba 8400, all AAA functionality discussed only applies to the active management module. See also *AAA on switches with multiple management modules* in the *High Availability Guide*.

Local AAA defaults and limits

Setting	Default value / limit
Local authentication	Enabled by default for all connection types: console, SSH, and REST.
Local role-based access control (RBAC) authorization	Enabled by default for all connection types: console, SSH, and REST.
Local accounting	Enabled.
Maximum number of local users	64 users, including the default <code>admin</code> user.
Maximum number of user-defined local user groups	32 groups, including the three built-in groups <code>administrators</code> , <code>operators</code> , <code>auditors</code> .
Password for default <code>admin</code> account	The password is empty by default.
SSH public key authentication	Enabled.

Local authentication

Local authentication overview

Authentication identifies users, validates their credentials, and grants switch access. Local authentication is either password-based or SSH public key-based.

Password-based local authentication

- Validates users with local user name and password credentials
- Is supported on all interfaces/channels (SSH, WebUI, Console, REST)
- Is enabled by default but can be superseded by remote authentication or with SSH client using SSH public key authentication

SSH public key-based local authentication

- Validates users identified with SSH public keys stored in the local user database
- Is supported on the SSH interface/channel with SSH client
- Takes precedence over password-based authentication whether local or remote
- Is enabled by default (also requires key configuration to work)

Local authentication tasks

The local authentication (local password and SSH public key) tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Enable authentication as local for the specified connection types	<u>aaa authentication login</u>	Enable local authentication for the default and console connection types: aaa authentication login default local aaa authentication login console local
Show authentication configuration	<u>show aaa authentication</u>	show aaa authentication
Enable password-based authentication minimum password length checking	<u>aaa authentication minimum-password-length</u>	aaa authentication minimum-password-length 12
Disable password-based authentication minimum password length checking	<u>aaa authentication minimum-password-length</u>	no aaa authentication minimum-password-length
Enable local password-based authentication login attempt limiting	<u>aaa authentication limit-login-attempts</u>	aaa authentication limit-login-attempts 4 lockout-time 20
Disable local password-based authentication login attempt limiting	<u>aaa authentication limit-login-attempts</u>	no aaa authentication limit-login-attempts

Table Continued

Task	Command or procedure	Example
Enable local password-based authentication for use with SSH clients (enabled by default)	<u>ssh password-authentication</u>	ssh password-authentication
Disable local password-based authentication for use with SSH clients	<u>ssh password-authentication</u>	no ssh password-authentication
Enable SSH public key authentication (enabled by default)	<u>ssh public-key-authentication</u>	ssh public-key-authentication
Disable SSH public key authentication	<u>ssh public-key-authentication</u>	no ssh public-key-authentication
Show state of local password-based (for SSH) and SSH public key authentication	<u>show ssh authentication-method</u>	show ssh authentication-method
Copying the client SSH public key into the key list	<u>user authorized-key</u>	user admin authorized-key ecdsa-sha2-nistp256 E2VjZH...QUiCAk= root@switch
Removing SSH public keys from the key list	<u>user authorized-key</u>	no user admin authorized-key 2
Showing the SSH client public key list	<u>show user</u>	show user admin authorized-key

Local authorization

Local authorization overview

Authorization controls authenticated users command execution and switch interaction privileges. Local authorization uses role-based access control (RBAC) to provide role-based privilege levels plus optional user-defined local user groups with command execution rules. Authorization occurs only after successful authentication.

- **Administrators** have full command execution and switch interaction privilege.
- **Operators** are limited to the use of several nonsensitive `show` commands.
- **Auditors** are limited to a few auditing-related commands.

Optional per-command authorization is available through configuration of user-defined local user groups with command authorization rules applied to respective group members. see [User-defined user groups](#).

Local authorization tasks

The local authorization tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Enable authorization as local RBAC for the specified connection types	<u>aaa authorization commands</u>	Enable local authorization for the default and console connection types: aaa authorization commands default local aaa authorization commands console local
Show authorization configuration	<u>show aaa authorization</u>	show aaa authorization

Local accounting

Local accounting overview

Local accounting is always active. It cannot be turned off.

This accounting information is captured and made available locally (using `show accounting log`) and, if desired, for sending to remote AAA servers:

- Exec Accounting: user login/logout events.
- Command accounting: commands executed by users.
- System accounting: remote accounting On/Off events.
- CLI show commands.
- Interactions on the non-CLI interfaces: REST and WebUI.

The following is not captured or made available as accounting information:

- CLI commands that reboot the switch.
- Interactions in the bash shell.



NOTE: See also the `show accounting log` command.

Local accounting tasks

The local accounting tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Enable accounting as local for the specified connection types	<u>aaa accounting all-mgmt</u>	Enable local accounting for the default and console connection types: aaa accounting all-mgmt default start-stop local aaa accounting all-mgmt console start-stop local
Show accounting configuration	<u>show aaa accounting</u>	show aaa accounting
Show local accounting log contents	<u>show accounting log</u>	show accounting log last 10

aaa accounting all-mgmt

Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}
```

```
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



NOTE: The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

`config`

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console...` to define the console accounting list, then this default configuration is used for console.

console

Defines a list of accounting server groups to be used for the `console` connection type.

https-server

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

ssh

Defines a list of accounting server groups to be used for the `ssh` connection type.

start-stop

Selects accounting information capture at both the beginning and end of a process.

local

Selects local-only accounting when used without the `group` parameter.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Local accounting is always active. It cannot be turned off.

Examples

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

Setting local accounting for the console connection type:

```
switch(config)# aaa accounting all-mgmt console start-stop local
```

aaa authentication limit-login-attempts

Syntax

```
aaa authentication limit-login-attempts <ATTEMPTS> lockout-time <LOCKOUT-TIME>
```

```
no aaa authentication limit-login-attempts
```

Description

Enables local login attempt limiting. If the number of failed local login attempts equals the configured threshold, the user is locked out for the configured duration.

The `no` form of this command disables local login attempt limits.



NOTE: This local login attempt limiting feature is only available when not using remote authentication through AAA servers (TACACS+ or RADIUS).

Command context

config

Parameters

<ATTEMPTS>

Specifies the threshold of failed local login attempts that triggers user logout. Range: 1 to 10. For example, if <ATTEMPTS> is set to 1, a single failed login attempt triggers immediate user logout.

<LOCKOUT-TIME>

Specifies the amount of time a user is locked out. Range: 1 to 3600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling local login attempt failure limiting with a 20 second logout being triggered upon the fourth consecutive login attempt failure.

```
switch(config)# aaa authentication limit-login-attempts 4 lockout-time 20
```

Disabling login attempt failure limiting:

```
switch(config)# no aaa authentication limit-login-attempts
```

aaa authentication login

Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}
```

```
no aaa authentication login <CONNECTION-TYPE>
```

Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

```
config
```

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for console.

console

Defines a list of authentication server groups to be used for the `console` connection type.

https-server

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

ssh

Defines a list of authentication server groups to be used for the `ssh` connection type.

local

Selects local-only authentication when used without the `group` parameter.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

Setting local authentication for the console connection type:

```
switch(config)# aaa authentication login console local
```

aaa authentication minimum-password-length

Syntax

```
aaa authentication minimum-password-length <LENGTH>
```

```
no aaa authentication minimum-password-length
```

Description

Enables minimum password length checking. Existing passwords shorter than the minimum length are unaffected. Length checking does not apply to ciphertext passwords. Length checking applies both to local and remote authentication.

The `no` form of this command disables minimum password length checking.

Command context

```
config
```

Parameters

<LENGTH>

Specifies the minimum password length. Range: 1 to 32.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling password length checking, with a minimum length of 12.

```
switch(config)# aaa authentication minimum-password-length 12
```

Disabling minimum password length checking:

```
switch(config)# no aaa authentication minimum-password-length
```

aaa authorization commands

Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}  
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>
```

```
no aaa authorization commands <CONNECTION-TYPE>
```

Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



NOTE: Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

console

Selects the `console` connection type for configuration.

ssh

Selects the `ssh` connection type for configuration.

local (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



NOTE: If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

none

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (administrators, operators, auditors).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



NOTE: With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```

show aaa accounting

Syntax

```
show aaa accounting
```

Description

Shows the accounting configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing local accounting for the default and console connection types:

```
switch(config)# aaa accounting all default start-stop local
switch(config)# aaa accounting all console start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop
```

Accounting for default channel:

GROUP NAME	GROUP PRIORITY
local	0

Accounting for console channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authentication

Syntax

```
show aaa authentication
```

Description

Shows the authentication configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing local authentication for the default and console connection types (channels):

```
switch(config)# aaa authentication login default local
switch(config)# aaa authentication login console local
switch(config)# exit
switch# show aaa authentication
```

```
AAA Authentication:
  Fail-through           : Disabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set
```

Authentication for default channel:

GROUP NAME	GROUP PRIORITY
local	0

Authentication for console channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authorization

Syntax

```
show aaa authorization
```

Description

Shows the authorization configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing full-fledged local RBAC authorization for the default and console connection types (channels):

```
switch(config)# aaa authorization commands default none
switch(config)#
switch(config)# aaa authorization commands console none
switch(config)# exit
```

```
switch#  
switch# show aaa authorization
```

Authorization for default channel:

GROUP NAME	GROUP PRIORITY
none	0

Authorization for console channel:

GROUP NAME	GROUP PRIORITY
none	0

show ssh authentication-method

Syntax

```
show ssh authentication-method
```

Description

Shows the status of the SSH public key method and the local password-based (through SSH client) authentication method.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the authentication methods.

```
switch# show ssh authentication-method  
SSH publickey authentication : Enabled  
SSH password authentication : Enabled
```

show user

Syntax

```
show user <USERNAME> authorized-key
```

Description

Shows the SSH client public key list for a specified user.

Command context

Operator (>) or Manager (#)

Parameters

<USERNAME>

Specifies the username for which you want to show the SSH client public key list.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

Any user can show their own public key list; however, administrators can also show a public key list of other users.

Examples

Showing a client public key:

```
switch# show user admin authorized-key

1. Key Type : RSA      Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDMtyMBmmAaF6r1zxf3DZNHSYVHBjhlbBlyAIqQ8DSHK
...
U+aE14UW/ifiukmK67sIHwK+FhhRYwPztQc5pjyOPk128a4pgKQaHCcOF169Z admin@switch
```

Showing two client public keys:

```
switch# show user admin authorized-key

1. Key Type : ECDSA    Curve : nistp256
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBEqEFevZ0
...
l76V+D0svdCJ9Wo32zqI9OeAdTJw/eZYp5qknhNgS81HjAI6J/4/kAqdZAJbqQUiCAk= admin@switch

2. Key Type : RSA      Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDXQHrQV7+/GcMdOhr//IRjJkX7TQKupW89j80bL7xq8
...
j8qKuHWSN0/h/HxjzQJuYDVmZN5vG3DhpXbBZU1ZNnchVod13QLCesqA3VLKN admin@switch
```

ssh password-authentication

Syntax

```
ssh password-authentication
```

```
no ssh password-authentication
```

Description

Enables the password-based authentication method for use with SSH clients.

The `no` form of this command disables the password-based authentication method for use with SSH clients.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch ships with password-based authentication (for SSH clients) enabled. The maximum number of password retries is three.

Examples

Enabling password authentication for use with SSH clients:

```
switch(config)# ssh password-authentication
```

Disabling password authentication for use with SSH clients:

```
switch(config)# no ssh password-authentication
```

ssh public-key-authentication

Syntax

```
ssh public-key-authentication
```

```
no ssh public-key-authentication
```

Description

Enables the SSH public key authentication method. The switch ships with SSH public key authentication enabled.

The `no` form of this command disables the SSH public key authentication method.



NOTE: Although SSH public key authentication is enabled by default, it cannot be used until SSH public keys are added with the `user authorized-key` command.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling SSH public key authentication:

```
switch(config)# ssh public-key-authentication
```

Disabling SSH public key authentication:

```
switch(config)# no ssh public-key-authentication
```

user authorized-key

Syntax

```
user <USERNAME> authorized-key <PUBKEY>
```

```
no user <USERNAME> authorized-key [<KEYNUM>]
```


Description

Copies an SSH client public key into the key list. If the key list and the public key do not exist, it creates a list with the public key. If the SSH client public key exists, the command appends the new key to the existing list. The client public key list holds a maximum of 32 client keys.

The `no` form of the command removes either one or all SSH public keys from the key list.

Command context

`config`

Parameters

<USERNAME>

Specifies the name of the user.

<PUBKEY>

Specifies the SSH client public key to be copied into the key list.

<KEYNUM>

Specifies the key number. The range is 1 to 32. Use the `show user <USERNAME> authorized-key` command to find the key number associated with the key.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Each key on the key list has a key identifier. The `show user <USERNAME> authorized-key` command displays the key identifier associated with the key.

Administrators can add and remove the public keys of themselves and other users. Operators can add and remove only their own public keys. If the public key authentication method is enabled, the client public key present is used by the SSH server to authenticate the client. The authentication method reverts to the password authentication method and prompts for a client password when one of the following occurs:

- The client public keys are not present.
- The server does not have the keys enabled.
- The public key method is disabled.

You can either remove all keys or a specific key. Each key on the key list has a key identifier. If you provide the key identifier in this command, the command removes the corresponding key from the list. If you provide no key identifier, the command removes all keys from the key list.

Examples

Adding a public key:

```
switch(config)#user admin authorized-key ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBEqEFevZ0176V+D0svdCJ9Wo32zqI9OeAIdTJwT/eZYp50qkA  
nhZNgS81HBjAI6QJ/4/kAyqdZ9oAjbiqQUiCAk= root@switch
```

Removing all SSH public keys from the list:

```
switch(config)# no user admin authorized-key
```

Removing the specified SSH public key from the list:

```
switch(config)# no user admin authorized-key 2
```

About remote AAA with TACACS+



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

Remote AAA provides the following for your Aruba switch:

- Authentication using remote TACACS+ AAA servers.
- Authorization using remote TACACS+ AAA servers, providing fine-grained command authorization. Optional user-defined local user groups with configured command authorization rules can be used to provide authorization fallback protection for when TACACS+ servers become temporarily unavailable.
- Transmission of locally collected accounting information to remote TACACS+ servers.



NOTE: For switches that support multiple management modules such as the Aruba 8400, all AAA functionality discussed only applies to the active management module. See also *AAA on switches with multiple management modules* in the *High Availability Guide*.

Default server groups

The switch always has these four default groups:

- `tacacs`: for remote AAA, always contains every configured TACACS+ server.
- `radius`: for remote AAA, always contains every configured RADIUS server.
- `local`: for local authentication.
- `none`: for local (RBAC) authorization.

User-defined AAA servers are always added to the matching default group, either `tacacs` or `radius`. Optionally, each server can also be added to exactly one additional user-defined (custom) group. A maximum of 28 user-defined groups can be created.

The order in which servers are added to a group is important. The server added first is accessed first, and if necessary, the second server is accessed second, and so on.

Remote AAA (TACACS+) defaults and limits

Setting	Default value / limit
Authentication of REST sessions with TACACS+	Disabled
Maximum number of TACACS+ servers in an AAA group	16

Table Continued

Setting	Default value / limit
Maximum number of TACACS+ servers that can be configured	16
Maximum number of user-defined AAA server groups that can be configured	28
TACACS+ authentication	Disabled
TACACS+ authentication global timeout	5 seconds
TACACS+ authentication passkey (shared secret)	None
TACACS+ authentication <code>tcp-port</code>	49
TACACS+ global authentication protocol	PAP
TACACS+ server tracking default interval	300 seconds
TACACS+ server access through the default VRF	default ¹

¹ The default value is `default`, unless another VRF is specified during the server configuration.

About global versus per-TACACS+ server passkeys (shared secrets)

To communicate with a TACACS+ AAA server, the switch must have a passkey (shared secret) configured that matches what is configured on the server. Use one of these commands to achieve the desired configuration:

- For a global passkey common to every TACACS+ server, use `tacacs-server key`.
- For a per-TACACS+ server passkey, use `tacacs-server host` with the `key` parameter.



NOTE: If both passkeys are configured on the switch, the per-TACACS+ server passkey is used.

Remote AAA TACACS+ server configuration requirements

The user-supplied TACACS+ server must:

- Have an IPv4/IPv6 address or fully qualified domain name (FQDN) that is visible to the switch.
- Have a passkey (shared secret) that matches what is configured on the switch.
- Provide username and password definitions for every switch user. Remote users do not require definition on the switch.

- Configure user role assignment using TACACS+ attributes.
- Have any needed command authorization configured to control what commands (per user or user role) will be executable on the switch.



NOTE: Consult your TACACS+ server documentation for installation and general configuration details.



NOTE: If SSH public key authentication is used, the key information is stored locally on the switch, making username and password definition on the TACACS+ server unnecessary.

User role assignment using TACACS+ attributes

User role assignment is configured on the TACACS+ server using VSAs (vendor-specific attributes) and TACACS+ specified attributes.

TACACS+ servers can return multiple attribute value pairs (AVPs) in response to an authentication request. The attributes are processed in this order of precedence to determine the user role assigned:

- If the `Aruba-Admin-Role` VSA is present, map the user to the matching corresponding local user-group name.
 - Else if the `priv-lvl` TACACS+ specified attribute is present, extract the privilege level (1, 15, or 19) and map the user to the local user-group corresponding to this privilege level (1=operators, 15=administrators, 19=auditors). Privilege levels 2 to 14 may also be used with matching local user groups named 2 to 14.
 - Otherwise, the user role cannot be determined, and authentication fails.

This information is summarized as follows:

Aruba-Admin-Role	priv-lvl	User role assigned
<GROUP-NAME>	Do not care	Matching local user <GROUP-NAME>
Not present	1	Operators
Not present	15	Administrators
Not present	19	Auditors
Not present	2 to 14	Matching local user groups named 2 to 14
Not present	Not present	None (not authenticated)

TACACS+ server redundancy and access sequence

To prevent authentication and authorization interruption, it is common practice to configure more than one TACACS+ server. When identifying TACACS+ servers to the switch, server group order (and server order within the group), determines server access order.



NOTE: When defining the server access sequence for authentication with `aaa authentication login default`, there is an implied `local` included as the last item in the list. If no TACACS+ server can be reached, local authentication will be attempted.



NOTE: When defining the server access sequence for authorization with `aaa authorization commands`, it is recommended to always include either `local` or `none` as the last item in the list.

Single source IP address for consistent source identification to AAA servers



NOTE: If applicable to your installation, it is recommended that you perform the optional configuration mentioned in this section.

If your topology allows the AAA server to be reached through multiple paths, the server interprets the incoming packets to be from different switches even though they are all coming from the same switch. Having a switch associated with multiple IP addresses makes it more difficult to interpret system logs and accounting data.

To ensure that all traffic sent from the switch to the AAA server uses the same source IP address, use `ip source-interface` or `ipv6 source-interface`. These two commands plus the related commands `show ip source-interface` and `show ipv6 source-interface` are described under *Layer 2/3 Interface commands* in the *Command-Line Interface Guide*.

TACACS+ general tasks

General TACACS+ tasks, not specific to authentication or authorization, are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring a TACACS+ server	<u>tacacs-server host</u>	<pre>tacacs-server host 1.1.1.1 vrf default no tacacs-server host 1.1.1.1 vrf default</pre>
Showing global and TACACS+ server configurations	<u>show tacacs-server</u>	<pre>show tacacs-server detail</pre>
Configuring a TACACS+ server group	<u>aaa group server</u>	<pre>aaa group server tacacs sgl no aaa group server tacacs sgl</pre>
Showing server groups	<u>show aaa server-groups</u>	<pre>show aaa server-groups</pre>
Adding a TACACS server to a server-group	<u>server</u>	<pre>aaa group server tacacs sgl server 1.1.1.2 port 32 vrf default</pre>
Deleting a TACACS server from a server-group	<u>server</u>	<pre>aaa group server tacacs sgl no server 1.1.1.2 port 32 vrf default</pre>
Configuring a TACACS+ global passkey	<u>tacacs-server key</u>	<pre>tacacs-server key plaintext mypasskey123</pre>
Configuring PAP or CHAP for TACACS+	<u>tacacs-server auth-type</u>	<pre>tacacs-server auth-type chap no tacacs-server auth-type</pre>
Configuring the TACACS+ global timeout	<u>tacacs-server timeout</u>	<pre>tacacs-server timeout 20 no tacacs-server timeout</pre>

TACACS+ authentication

TACACS+ authentication overview

TACACS+ authentication occurs as follows:

- User credentials are sent from the switch to TACACS+ server using the PAP or CHAP authentication protocol.
- If a user is authenticated, their role is communicated to the switch as Administrator, Operator, or Auditor.
- An unknown user or a user who entered an invalid password is identified as such to the switch, which then rejects user login.

About authentication fail-through

Normally, authentication is performed by the first AAA server reached. A rarely needed feature named "Authentication fail-through" is available. If Authentication fail-through is enabled and authentication fails on the first reachable AAA server, authentication is attempted on the second AAA server, and so on, until successful authentication or the server list is exhausted.

Enabling Authentication fail-through is typically unnecessary because the user credential databases should be consistent across all AAA servers. Authentication fail-through might be helpful if your AAA user credential databases are not quickly synchronized across all AAA servers.

TACACS+ authentication tasks

The TACACS+ authentication-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring the authentication sequence for the default connection type	<u>aaa authentication login</u>	<pre>aaa authentication login default group tg1 tg2 tacacs local</pre>
Configuring the authentication sequence for the console connection type	<u>aaa authentication login</u>	<pre>aaa authentication login console group tg2 tg3 tacacs local</pre>
Configuring the authentication sequence for the ssh connection type	<u>aaa authentication login</u>	<pre>aaa authentication login ssh group tg2 tacacs local</pre>
Removing remote AAA for the default connection type	<u>aaa authentication login</u>	<pre>no aaa authentication login default</pre>
Configuring authentication fail-through	<u>aaa authentication allow-fail-through</u>	<pre>aaa authentication allow-fail-through no aaa authentication allow-fail-through</pre>
Showing the authentication sequence	<u>show aaa authentication</u>	<pre>show aaa authentication</pre>

TACACS+ authorization

TACACS+ authorization overview

Upon successful user authentication, the user is assigned their role by the TACACS+ server. See also [User role assignment using TACACS+ attributes](#).

TACACS+ authorization provides command filtering to allow/disallow individual command or command set execution. Each command is sent to the TACACS+ server for approval, and the switch then allows/disallows command execution according to the server response.



NOTE: TACACS+ authorization applies only to the CLI interface.

Using local authorization as fallback from TACACS+ authorization

Local authorization can be used for the situation in which communication is lost with all TACACS+ servers after a successful authentication. Users that are members of the built-in local user groups (administrators, operators, or auditors) are authorized according to the fixed roles and privilege levels of those groups. Optionally, local user-defined user groups can be configured with specific command execution rules per group. Users that are members of such groups, are authorized according to the command execution rules of the group to which they belong. For configuring local user groups, see [user-group](#).

About authentication fail-through and authorization

For authorization, there is no equivalent of the authentication fail-through feature. Therefore, if the first reachable TACACS+ server responds with "Authorization Denied," no additional TACACS+ servers are interrogated.



NOTE: Rare potential out-of-synchronization situation when using authentication fail-through: Successful authentication on one server can be followed by authorization denial on another. The user is known on the server doing the authentication but unknown on the server attempting the authorization. This situation typically arises only during brief periods in which user credential databases are not synchronized across all TACACS+ servers. See also [TACACS+ server authorization considerations in aaa authorization commands](#).

TACACS+ authorization tasks

The TACACS+ authorization-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring the authorization sequence for the default connection type	aaa authorization commands	<pre>aaa authorization commands default group tg1 tacacs local</pre>
Configuring the authorization sequence for the console connection type	aaa authorization commands	<pre>aaa authorization commands console group tg1 tg2 tacacs none</pre>

Table Continued

Task	Command or procedure	Example
Removing remote AAA for the default connection type	<u>aaa</u> <u>authorization</u> <u>commands</u>	no aaa authorization commands default
Showing the TACACS+ authorization sequence	<u>show aaa</u> <u>authorization</u>	show aaa authorization

TACACS+ accounting

TACACS+ accounting overview

This accounting information is captured and made available for sending to remote accounting servers:

- Exec Accounting: user login/logout events.
- Command accounting: commands executed by users.
- System accounting: remote accounting On/Off events.
- CLI show commands.
- Interactions on the non-CLI interfaces: REST and WebUI.

The following is not captured or made available as accounting information:

- CLI commands that reboot the switch.
- Interactions in the bash shell.



NOTE: Local accounting (always enabled) must be functioning properly for remote Accounting to work.



NOTE: The accounting information is sent to the first reachable remote TACACS+ AAA server (configured for remote accounting). If no remote TACACS+ server is reachable, local accounting remains available.

Sample accounting information on a TACACS+ server

```
Mon May 9 17:52:32 10.10.11.1 UNKNOWN tty 0.0.0.0 start task_id=1525899775430
timezone=UTC start_time=1525913552.428 service=system event=sys_acct
reason="System-accounting-ON" result=success
Mon May 9 17:52:48 10.10.11.1 admin tty 192.168.1.20 start
task_id=1525899775431 timezone=UTC start_time=1525913567.611 service=shell
priv_lvl=15 result=success
Mon May 9 17:52:48 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775432 timezone=UTC stop_time=1525913567.614 service=shell
priv_lvl=15 cmd="enable" result=success
Mon May 9 17:52:51 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775433 timezone=UTC stop_time=1525913570.851 service=shell
priv_lvl=15 cmd="configure" result=success
Mon May 9 17:52:53 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775434 timezone=UTC stop_time=1525913573.427 service=shell
priv_lvl=15 cmd="interface 1/1/3" result=success
Mon May 9 17:52:54 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775435 timezone=UTC stop_time=1525913574.447 service=shell
```

```

priv_lvl=15 cmd="no shutdown" result=success
Mon May 9 17:52:58 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775436 timezone=UTC stop_time=1525913578.131 service=shell
priv_lvl=15 cmd="ip address 10.10.13.1/24" result=success
Mon May 9 17:52:59 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775437 timezone=UTC stop_time=1525913579.468 service=shell
priv_lvl=15 cmd="exit" result=success
Mon May 9 17:53:10 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775442 timezone=UTC stop_time=1525913590.204 service=shell
priv_lvl=15 cmd="exit" result=success
Mon May 9 17:53:10 10.10.11.1 admin tty 192.168.1.20 stop
task_id=1525899775431 timezone=UTC stop_time=1525913590.205 service=shell
priv_lvl=15 result=success
Mon May 9 17:53:44 10.10.11.1 UNKNOWN tty 0.0.0.0 stop task_id=1525899775430
timezone=UTC stop_time=1525913624.473 service=system event=sys_acct
reason="System-accounting-OFF" result=success

```



NOTE: This sample is representative and not from any particular TACACS+ server implementation.

Sample REST accounting information on a TACACS+ server

```

Oct 30 16:31:56 10.10.10.1 admin tty 127.0.0.1 start task_id=1540942055868
timezone=UTC start_time=1540942316.36 service=https-server priv_lvl=15
cmd="http-method=POST http-uri=/rest/v1/login" result=success

```



NOTE: This sample is representative and not from any particular TACACS+ server implementation.

TACACS+ accounting tasks

The TACACS+ accounting-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring the accounting sequence for the default connection type	<u>aaa accounting all-mgmt</u>	aaa accounting all-mgmt default start-stop group tg1 tg2 tacacs local
Configuring the accounting sequence for the console connection type	<u>aaa accounting all-mgmt</u>	aaa accounting all-mgmt console start-stop group tg2 tg3 tacacs local
Configuring the accounting sequence for the ssh connection type	<u>aaa accounting all-mgmt</u>	aaa accounting all-mgmt ssh start-stop group tg2 tacacs local
Removing remote AAA for the default connection type	<u>aaa accounting all-mgmt</u>	no aaa accounting all-mgmt default start-stop
Showing the accounting configuration	<u>show aaa accounting</u>	show aaa accounting

Example: Configuring the switch for Remote AAA with TACACS+

Prerequisites

- TACACS+ servers configured in general according to the information in **Remote AAA TACACS+ server configuration requirements**. The exact settings appropriate to your environment will vary.
- Logged in to the switch with Administrator privilege and in the `config` context.

Procedure

1. Configure the global TACACS+ passkey (shared secret) as "xjKW74932qX3j_\$"

```
switch(config)# tacacs-server key plaintext xjKW74932qX3j_$
switch(config)#
```

2. Add these configuration details for two remote TACACS+ servers:

- Server 1 with IPv4 address 10.0.0.2, on the management interface (belonging to VRF "mgmt"), using the default PAP protocol.
- Server 2 with IPv4 address 4.0.0.2, on the data interface (belonging to VRF "default"), using the CHAP protocol.

```
switch(config)# tacacs-server host 10.0.0.2 vrf mgmt
switch(config)# tacacs-server host 4.0.0.2 auth-type chap
switch(config)#
```

3. Create a TACACS+ group named `tac_grp1`, assign TACACS+ server 10.0.0.2 to the group, show the group information.



NOTE: The default TACACS+ group named `tacacs` includes every TACACS+ server regardless of whether any TACACS+ servers are also assigned to a user-defined TACACS+ group.

```
switch(config)# aaa group server tacacs tac_grp1
switch(config-sg)# server 10.0.0.2 vrf mgmt
switch(config-sg)# exit
switch(config)#
switch(config)# do show aaa server-groups tacacs
```

***** AAA Mechanism TACACS+ *****

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
tac_grp1	10.0.0.2	49	mgmt	1
tacacs (default)	10.0.0.2	49	mgmt	1
tacacs (default)	4.0.0.2	49	default	2

```
switch(config)#
```

4. Define the authentication sequence list so that the new TACACS+ group is first, the default TACACS+ group is second, and local is third. Show the authentication sequence.

```
switch(config)# aaa authentication login default group tac_grp1 tacacs local
switch(config)#
switch(config)# do show aaa authentication
AAA Authentication:
```

```

Fail-through : Disabled
Limit Login Attempts : Not set
Lockout Time : 300
Minimum Password Length : Not set

```

Default Authentication for All Channels:

GROUP NAME	GROUP PRIORITY
tac_grp1	0
tacacs	1
local	2

switch(config)#

5. Define the authorization sequence list with two TACACS+ server groups plus local RBAC. Show the authorization sequence.

```

switch(config)# aaa authorization commands default group tac_grp1 tacacs local
switch(config)#
switch(config)# do show aaa authorization

```

Default command Authorization for All Channels:

GROUP NAME	GROUP PRIORITY
tac_grp1	0
tacacs	1
local	2

switch(config)#

6. Define the accounting sequence list with two TACACS+ server groups. Show the accounting sequence.

```

switch(config)# aaa accounting all default start-stop group tac_grp1 tacacs
switch(config)#
switch(config)# do show aaa accounting
AAA Accounting:
  Accounting Type : all
  Accounting Mode : start-stop

```

Default Accounting for All Channels:

GROUP NAME	GROUP PRIORITY
tac_grp1	0
tacas	1

About remote AAA with RADIUS



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

Remote AAA provides the following for your Aruba switch:

- Authentication using remote RADIUS AAA servers. For added security, two-factor authentication may be used. In two-factor authentication, X.509 certificate-based authentication is combined with RADIUS authentication.
- Command authorization is not supported by RADIUS servers, however, user-defined local user groups can be configured with command-authorization rules, providing locally configured per-command authorization for members of such groups. See [User-defined user groups](#).

In the switch default state (without user-defined local groups), basic role-based authorization is available with the three built-in roles (`administrators`, `operators`, `auditors`).

- Transmission of locally collected accounting information to remote RADIUS servers.



NOTE: For switches that support multiple management modules such as the Aruba 8400, all AAA functionality discussed only applies to the active management module. See also *AAA on switches with multiple management modules* in the *High Availability Guide*.

Default server groups

The switch always has these four default groups:

- `tacacs`: for remote AAA, always contains every configured TACACS+ server.
- `radius`: for remote AAA, always contains every configured RADIUS server.
- `local`: for local authentication.
- `none`: for local (RBAC) authorization.

User-defined AAA servers are always added to the matching default group, either `tacacs` or `radius`. Optionally, each server can also be added to exactly one additional user-defined (custom) group. A maximum of 28 user-defined groups can be created.

The order in which servers are added to a group is important. The server added first is accessed first, and if necessary, the second server is accessed second, and so on.

Remote AAA (RADIUS) defaults and limits

Setting	Default value / limit
Maximum number of RADIUS servers in an AAA group	16
Maximum number of RADIUS servers that can be configured	16
Maximum number of user-defined AAA server groups that can be configured	28
RADIUS authentication	Disabled
RADIUS authentication global timeout	5 seconds
RADIUS authentication passkey (shared secret)	None
RADIUS authentication <code>udp-port</code>	1812
RADIUS global authentication protocol	PAP
RADIUS global retries	1 retry
RADIUS server tracking default interval	300 seconds
RADIUS server access through the default VRF	default ¹

¹ The default value is `default`, unless another VRF is specified during the server configuration.

About global versus per-RADIUS server passkeys (shared secrets)

To communicate with a RADIUS AAA server, the switch must have a passkey (shared secret) configured that matches what is configured on the server. Use one of these commands to achieve the desired configuration:

- For a global passkey common to every RADIUS server, use `radius-server key`.
- For a per-RADIUS server passkey, use `radius-server host` with the `key` parameter.



NOTE: If both passkeys are configured on the switch, the per-RADIUS server passkey is used.

Remote AAA RADIUS server configuration requirements

The user-supplied RADIUS server must:

- Have an IPv4/IPv6 address or fully qualified domain name (FQDN) that is visible to the switch.
- Have a passkey (shared secret) that matches what is configured on the switch.

- Provide username and password definitions for every switch user. Remote users do not require definition on the switch.
- Configure user role assignment using RADIUS attributes.



NOTE: Consult your RADIUS server documentation for installation and general configuration details.



NOTE: If SSH public key authentication is used, the key information is stored locally on the switch, making username and password definition on the RADIUS server unnecessary.

User role assignment using RADIUS attributes

User role assignment is configured on the RADIUS server using VSAs (vendor-specific attributes).

RADIUS servers can return multiple attribute value pairs (AVPs) in response to an authentication request. The attributes are processed in this order of precedence to determine the user role assigned:

- If the `Aruba-Admin-Role` VSA is present, map the user to the matching local user-group name.
 - Else if the `Aruba-Priv-Admin-User` VSA is present, extract the privilege level (1, 15, or 19) and map the user to the local user-group corresponding to this privilege level (1=operators, 15=administrators, 19=auditors). Privilege levels 2 to 14 may also be used with matching local user groups named 2 to 14.
 - Else If `Service-Type` AVP is present, map `Administrative-User (6)` to administrators and map `NAS-Prompt-User (7)` to operators.
 - Otherwise, the user role cannot be determined, and the authentication fails.

This is summarized as follows:

Aruba-Admin-Role	Aruba-Priv-Admin-User	service-type	User role assigned
<GROUP-NAME>	Do not care	Do not care	Matching local user <GROUP-NAME>
Not present	privilege level =1	Do not care	Operators
Not present	privilege level =15	Do not care	Administrators
Not present	privilege level =19	Do not care	Auditors
Not present	privilege level =2 to 14	Do not care	Matching local user groups named 2 to 14
Not present	Not present	Administrative- User (6)	Administrators
Not present	Not present	NAS-Prompt-User (7)	Operators
Not present	Not present	Not present (or = any other value)	None (not authenticated)



NOTE: The `Service-Type` attribute is retained only for backward compatibility. It is recommended that you instead use the `Aruba-Admin-Role` or `Aruba-Priv-Admin-User` VSA.

RADIUS server redundancy and access sequence

To prevent authentication interruption, it is common practice to configure more than one RADIUS server. When identifying RADIUS servers to the switch, server group order (and server order within the group), determines server access order.



NOTE: When defining the server access sequence for authentication with `aaa authentication login default`, there is an implied `local` included as the last item in the list. If no RADIUS server can be reached, local authentication will be attempted.

Single source IP address for consistent source identification to AAA servers



NOTE: If applicable to your installation, it is recommended that you perform the optional configuration mentioned in this section.

If your topology allows the AAA server to be reached through multiple paths, the server interprets the incoming packets to be from different switches even though they are all coming from the same switch. Having a switch associated with multiple IP addresses makes it more difficult to interpret system logs and accounting data.

To ensure that all traffic sent from the switch to the AAA server uses the same source IP address, use `ip source-interface` or `ipv6 source-interface`. These two commands plus the related commands `show ip source-interface` and `show ipv6 source-interface` are described under *Layer 2/3 Interface commands* in the *Command-Line Interface Guide*.

RADIUS general tasks

General RADIUS tasks, not specific to authentication, are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring a RADIUS server	<u>radius-server host</u>	<pre>radius-server host 1.1.1.1 vrf default no radius-server host 1.1.1.1 vrf default</pre>
Showing global and RADIUS server configurations	<u>show radius-server</u>	<pre>show radius-server detail</pre>
Configuring a RADIUS server group	<u>aaa group server</u>	<pre>aaa group server radius sg3 no aaa group server radius sg3</pre>
Showing server groups	<u>show aaa server-groups</u>	<pre>show aaa server-groups</pre>
Adding a RADIUS server to a server-group	<u>server</u>	<pre>aaa group server radius sg3 server 1.1.1.4 port 32 vrf default</pre>
Deleting a RADIUS server from a server-group	<u>server</u>	<pre>aaa group server tacacs sg3 no server 1.1.1.4 port 32 vrf default</pre>

Table Continued

Task	Command or procedure	Example
Configuring a RADIUS global passkey	<u>radius-server key</u>	<code>radius-server key plaintext mypasskey123</code>
Configuring PAP or CHAP for RADIUS	<u>radius-server auth-type</u>	<code>radius-server auth-type chap</code> <code>no radius-server auth-type</code>
Configuring the RADIUS global timeout	<u>radius-server timeout</u>	<code>radius-server timeout 15</code> <code>no radius-server timeout</code>
Configuring the RADIUS global retries	<u>radius-server retries</u>	<code>radius-server retries 3</code> <code>no radius-server retries</code>
Overriding the global retries for a RADIUS server	<u>radius-server host</u>	<code>radius-server host 1.1.1.1 retries 2</code>

RADIUS authentication

RADIUS authentication overview

RADIUS authentication occurs as follows:

- User credentials are sent from the switch to RADIUS server using the PAP or CHAP authentication protocol.
- If a user is authenticated, their role is communicated to the switch as Administrator, Operator, or Auditor.
- An unknown user or a user who entered an invalid password is identified as such to the switch, which then rejects user login.

About authentication fail-through

Normally, authentication is performed by the first AAA server reached. A rarely needed feature named "Authentication fail-through" is available. If Authentication fail-through is enabled and authentication fails on the first reachable AAA server, authentication is attempted on the second AAA server, and so on, until successful authentication or the server list is exhausted.

Enabling Authentication fail-through is typically unnecessary because the user credential databases should be consistent across all AAA servers. Authentication fail-through might be helpful if your AAA user credential databases are not quickly synchronized across all AAA servers.

RADIUS authentication tasks

The RADIUS authentication-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring the authentication sequence for the default connection type	<u>aaa authentication login</u>	aaa authentication login default group rg1 rg2 radius local
Configuring the authentication sequence for the https-server connection type	<u>aaa authentication login</u>	aaa authentication login https-server group rg1 radius local
Removing remote AAA for the default connection type	<u>aaa authentication login</u>	no aaa authentication login default
Configuring authentication fail-through	<u>aaa authentication allow-fail-through</u>	aaa authentication allow-fail-through no aaa authentication allow-fail-through
Showing the authentication sequence	<u>show aaa authentication</u>	show aaa authentication

Configuring two-factor authentication

Two-factor authentication is available for added security. In two-factor authentication, X.509 certificate-based authentication is combined with RADIUS authentication. When a user establishes an SSH connection to the switch, two factor-authentication occurs as follows:

- The username in the user's X.509 certificate is validated against the local user accounts on the switch.
- The username and password are validated against the accounts on the RADIUS server and the configured trust anchors.

Prerequisites

- The switch SSH server is enabled.
- Your switch management computer, though its SSH client, is connected to the switch.
- A remote RADIUS server is available to authenticate switch users and is configured on the switch.
- Every user that will use two-factor authentication is configured both on the RADIUS server and locally on the switch using identical usernames. Users are added locally on the switch with the `user` command. These usernames must precisely match the usernames identified by the X.509 user certificates.
- The X.509 CA certificate is both installed on your switch management computer and is also visible to your computer's SSH client. The X.509 CA certificate is the root of trust for the client certificate being used.
- One X.509 certificate per user is available on your switch management computer and is visible to your computer's SSH client. The usernames identified by these user certificates must be the same as the usernames already defined on the RADIUS server and locally on the switch.

Procedure

1. Create a TA profile with the command `crypto pki ta-profile`. This command switches to the TA configuration context. The TA profile is where the switch stores the root certificate of the CA that is used to validate the certificates of clients communicating with the SSH server.
2. Although optional, it is recommended that you enable certificate revocation checking with the command `revocation-check oosp`.
3. Import the root certificate of the CA with the command `ta-certificate`.
4. Exit the TA configuration context with the command `exit`.
5. For each user that will be using two-factor authentication, import the public key from the individual X.509 user certificate with the command `user <USERNAME> authorized-key <PUBKEY>`. Each user identified by <USERNAME> must exist locally on the switch and on the RADIUS authentication server.
6. Enable two-factor authentication with the command `ssh two-factor-authentication`.

Example

This example installs the root certificate **root-cert** and enables two-factor authentication for user **admin**:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# revocation-check oosp
switch(config-ta-root-cert)# ta-certificate
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIDuTCCAqECCQCuoxeJ2ZNYcjANBgkqhkiG9w0BAQsFADCBq
switch(config-ta-cert)# VVMxEzARBgNVBAGMCkNhbGlmb3JuaWEeEDAOBgNVBACMB1JvY
switch(config-ta-cert)# BAoMA0hQTjEVMBMGA1UECwwMSFBOUm9zZXZpbGxlMSowKAYDV
...
switch(config-ta-cert)# x3Wff3dFZ8o9sd5LVAHneH/ztb9MP34z+le1V346r12L2MDL8
switch(config-ta-cert)# BIzD/ST/HaWI+0S+S80rm93PSscEbb9GWk7vshh5E8DH73nW/
switch(config-ta-cert)# 3LvMLZcSSSe5J2Ca2XIhfDme8UaNZ7syGYoCD/TMsAW0nG7yY
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Issuer: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
        CN=8400/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert)# exit
switch(config)#
switch(config)# user admin authorized-key ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQAC6krLTrFTnzg3YjLiZKTZEYnh4cUiuOK+cjduxFnZUa
...
iAfcGvqvWtWWBSowd011DeEZnKn008uEKeTEcAjfrnRHeOk2QJmw== "sv1@site.net"
switch(config)#
switch(config)# ssh two-factor-authentication
```

RADIUS accounting

RADIUS accounting overview

This accounting information is captured and made available for sending to remote accounting servers:

- Port access accounting
- Exec Accounting: user login/logout events
- Command accounting: commands executed by users. The Vendor-Specific Attribute (VSA) `Aruba_Command_String` with a value of 46 is available.
- System accounting: remote accounting On/Off events.
- CLI show commands.
- Interactions on the non-CLI interfaces: REST and WebUI.



NOTE: With RADIUS, command accounting logs a maximum of 247 characters per command entered by the user.

The following is not captured or made available as accounting information:

- CLI commands that reboot the switch.
- Interactions in the bash shell.



NOTE: Local accounting (always enabled) must be functioning properly for remote Accounting to work.



NOTE: The accounting information is sent to the first reachable remote RADIUS AAA server (configured for remote accounting). If no remote RADIUS server is reachable, local accounting remains available.

Sample general accounting information

```
~~~~~ EXEC ~~~~~
```

```
Mon Jul 16 16:25:27 2018
  User-Name = "admin"
  NAS-Identifier = "switchx"
  NAS-Port = 331
  NAS-Port-Type = Virtual
  Acct-Status-Type = Start
  Acct-Session-Id = "1531769192494"
  Acct-Authentic = Local
  Calling-Station-Id = "0.0.0.0"
  Event-Timestamp = "Jul 16 2018 16:25:22 PDT"
  Acct-Delay-Time = 0
  NAS-IP-Address = 10.10.10.1
  Acct-Unique-Session-Id = "b83e29f4140c17b1"
  Timestamp = 1531783527
```

```
~~~ EXEC stop ~~~
```

```
Mon Jul 16 16:26:42 2018
  User-Name = "admin"
  NAS-Identifier = "switchx"
  NAS-Port = 331
  NAS-Port-Type = Virtual
  Acct-Status-Type = Stop
  Acct-Session-Id = "1531769192494"
  Acct-Authentic = Local
  Calling-Station-Id = "0.0.0.0"
  Event-Timestamp = "Jul 16 2018 16:26:37 PDT"
  Acct-Delay-Time = 0
```

```
Acct-Session-Time = 75
NAS-IP-Address = 10.10.10.1
Acct-Unique-Session-Id = "b83e29f4140c17b1"
Timestamp = 1531783602

~~~~~ CMD ACCOUNTING ~~~~~
Mon Jul 16 16:26:42 2018
  User-Name = "admin"
  NAS-Identifier = "switchx"
  NAS-Port = 331
  NAS-Port-Type = Virtual
  Acct-Status-Type = Stop
  Acct-Session-Id = "1531769192496"
  Acct-Authentic = Local
  Aruba-Command-String = "exit"
  Calling-Station-Id = "0.0.0.0"
  Event-Timestamp = "Jul 16 2018 16:26:37 PDT"
  Acct-Delay-Time = 0
  NAS-IP-Address = 10.10.10.1
  Acct-Unique-Session-Id = "280710992629128c"
  Timestamp = 1531783602
```

```
~~~~~ SYSTEM ACCOUNTING ~~~~~

Mon Jul 16 17:13:02 2018
  User-Name = "UNKNOWN"
  NAS-Identifier = "UNKNOWN"
  NAS-Port = 331
  NAS-Port-Type = Virtual
  Acct-Status-Type = Accounting-On
  Acct-Session-Id = "1531769192506"
  Acct-Authentic = Local
  Calling-Station-Id = "0.0.0.0"
  Event-Timestamp = "Jul 16 2018 17:12:56 PDT"
  Acct-Delay-Time = 0
  NAS-IP-Address = 10.10.10.1
  Acct-Unique-Session-Id = "b478e6402c86933e"
  Timestamp = 1531786382
```

```
Mon Jul 16 17:12:55 2018
  User-Name = "UNKNOWN"
  NAS-Identifier = "UNKNOWN"
  NAS-Port = 331
  NAS-Port-Type = Virtual
  Acct-Status-Type = Accounting-Off
  Acct-Session-Id = "1531769192491"
  Acct-Authentic = Local
  Calling-Station-Id = "0.0.0.0"
  Event-Timestamp = "Jul 16 2018 17:12:49 PDT"
  Acct-Delay-Time = 0
  NAS-IP-Address = 10.10.10.1
  Acct-Unique-Session-Id = "93da1f094121f2ee"
  Timestamp = 1531786375

~~~~~
```



NOTE: This sample is representative and not from any particular RADIUS server implementation.

RADIUS accounting tasks

The RADIUS accounting-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Configuring the accounting sequence for the default connection type	<u>aaa accounting all-mgmt</u>	aaa accounting all-mgmt default start-stop group rg1 rg2 radius local
Configuring the accounting sequence for the https-server connection type	<u>aaa accounting all-mgmt</u>	aaa accounting all-mgmt https-server start-stop group rg1 radius local
Removing remote AAA for the default connection type	<u>aaa accounting all-mgmt</u>	no aaa accounting all-mgmt default start-stop
Showing the accounting configuration	<u>show aaa accounting</u>	show aaa accounting

Example: Configuring the switch for Remote AAA with RADIUS

Prerequisites

- RADIUS servers configured in general according to the information in [**Remote AAA RADIUS server configuration requirements**](#). The exact settings appropriate to your environment will vary.
- Logged in to the switch with Administrator privilege and in the `config` context.

Procedure

1. Configure the global RADIUS passkey (shared secret) as "xjKW74932qX3j_\$"

```
switch(config)# radius-server key plaintext xjKW74932qX3j_$
switch(config)#
```

2. Add these configuration details for two remote RADIUS servers.

- Server 1 with IPv4 address 10.0.0.2, on the management interface (belonging to VRF "mgmt"), using the default PAP protocol.
- Server 2 with IPv4 address 4.0.0.2, on the data interface (belonging to VRF "default"), using the CHAP protocol.

```
switch(config)# radius-server host 10.0.0.2 vrf mgmt
switch(config)# radius-server host 4.0.0.2 auth-type chap
switch(config)#
```

3. Create a RADIUS group named `rad_grp1`, assign RADIUS server 10.0.0.2 to the group, show the group information.



NOTE: The default RADIUS group named `radius` includes every RADIUS server regardless of whether any RADIUS servers are also assigned to a user-defined RADIUS group.

```
switch(config)# aaa group server radius rad_grp1
switch(config-sg)# server 10.0.0.2 vrf mgmt
switch(config-sg)# exit
switch(config)#
switch(config)# do show aaa server-groups radius

***** AAA Mechanism RADIUS *****
```

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
rad_grp1	10.0.0.2	1812	mgmt	1
radius (default)	10.0.0.2	1812	mgmt	1
radius (default)	4.0.0.2	1812	default	2

```
switch(config)#
```

4. Define the authentication sequence list so that the new RADIUS group is first, the default RADIUS group is second, and local is third. Show the authentication sequence.

```
switch(config)# aaa authentication login default group rad_grp1 radius local
switch(config)#
switch(config)# do show aaa authentication
AAA Authentication:
  Fail-through           : Disabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set

Default Authentication for All Channels:
```

GROUP NAME	GROUP PRIORITY
rad_grp1	0
radius	1
local	2

```
switch(config)#
```

5. Define the accounting sequence list with two RADIUS server groups. Show the accounting sequence.

```
switch(config)# aaa accounting all default start-stop group rad_grp1 radius
switch(config)#
switch(config)# do show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop

Default Accounting for All Channels:
```

GROUP NAME	GROUP PRIORITY
rad_grp1	0
radius	1



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

aaa accounting all-mgmt

Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



NOTE: The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

`config`

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console . . .` to define the console accounting list, then this default configuration is used for console.

console

Defines a list of accounting server groups to be used for the `console` connection type.

https-server

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

ssh

Defines a list of accounting server groups to be used for the `ssh` connection type.

start-stop

Selects accounting information capture at both the beginning and end of a process.

local

Selects local-only accounting when used without the `group` parameter.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Local accounting is always active. It cannot be turned off.

Examples

Defining the default accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group tg1 tg2 tacacs local
```

Defining the console accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt console start-stop group tg2 tg3 tacacs local
```

Defining the ssh accounting sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa accounting all-mgmt ssh start-stop group tg2 tacacs
```

Defining the default accounting sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group rg1 rg2 radius local
```

Defining the https-server accounting sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa accounting all-mgmt https-server start-stop group rg1 radius
```

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

aaa authentication allow-fail-through

Syntax

```
aaa authentication allow-fail-through
```

```
no aaa authentication allow-fail-through
```

Description

Enables authentication fail-through. When this option is enabled, the next server/authentication method is tried after an authentication failure.

The `no` form of this command disables authentication fail-through. If the system fails to authenticate with a reachable TACACS+ or RADIUS server, the system does not attempt to authenticate with the next TACACS+/RADIUS server.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling authentication fail-through:

```
switch(config)# aaa authentication allow-fail-through
```

aaa authentication login

Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}
```

```
no aaa authentication login <CONNECTION-TYPE>
```

Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.



NOTE: If you do not want local authentication to occur in cases where all AAA servers contacted reject the user's credentials, do not enable authentication fail-through (command `aaa authentication allow-fail-through`).

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for console.

console

Defines a list of authentication server groups to be used for the `console` connection type.

https-server

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

ssh

Defines a list of authentication server groups to be used for the `ssh` connection type.

local

Selects local-only authentication when used without the `group` parameter.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining the default authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group tg1 tg2 tacacs local
```

Defining the console authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login console group tg2 tg3 tacacs local
```

Defining the ssh authentication sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa authentication login ssh group tg2 tacacs
```

Defining the default authentication sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group rg1 rg2 radius local
```

Defining the https-server authentication sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa authentication login https-server group rg1 radius
```

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

aaa authorization commands

Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}  
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>
```

```
no aaa authorization commands <CONNECTION-TYPE>
```

Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



NOTE: Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

default

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

console

Selects the `console` connection type for configuration.

ssh

Selects the `ssh` connection type for configuration.

local (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



NOTE: If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

none

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (administrators, operators, auditors).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



NOTE: With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

group <GROUP-LIST>

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

Authority

Administrators or local user group members with execution rights for this command.

Usage

TACACS+ server authorization considerations



NOTE: Use caution when configuring authorization, as it has no fail through. If the switch is not configured properly, the switch might get into an unusable state in which all command execution is prohibited.

To prevent authorization difficulties:

- Make sure that all listed TACACS+ servers can authorize users for command execution.
- Make sure that credential database changes are promptly synchronized across all TACACS+ servers.

- Make sure either `local` or `none` is included as the last name in the group list. If both `local` and `none` are omitted, and no remote TACACS+ server is reachable (or the first reachable server cannot authorize), authorization will not be possible.
- Although not recommended, if you choose to omit both `local` and `none` from the list, and are manipulating configuration files, special caution is necessary. If the source configuration includes TACACS+ authorization and you are copying configuration from an existing switch into the running configuration of a new switch, and you have not yet configured the interface or routing information to reach the TACACS+ server, the switch will enter an unusable state, requiring hard reboot.

To avoid getting into this situation that can occur when `local` and `none` have been omitted, do either of the following:

- In the configuration source, delete or comment-out the line configuring remote authorization. Then, after the configuration copy and paste, manually configure authorization.
- Move the line configuring the authorization to the end of the source configuration before copying and pasting.

Examples

Defining the default authorization sequence based on a user-defined TACACS+ server group, then the default TACACS+ server group, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands default group tg1 tacacs local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Defining the console authorization sequence based on two user-defined TACACS+ server groups, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands console group tg1 tg2 local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```

aaa group server

Syntax

```
aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

```
no aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

Description

Creates an AAA server group that is either empty or contains preconfigured RADIUS/TACACS+ servers. You can create a maximum of 28 server groups.

The `no` form of this command deletes a server group. Only a preconfigured user-defined RADIUS/TACACS+ server group can be deleted. RADIUS or TACACS+ servers that were in a deleted server group remain a part

of their default server group. The default server group for TACACS+ servers is `tacacs`. The default server group for RADIUS servers is `radius`.

Command context

`config`

Parameters

server {tacacs | radius}

Select either `tacacs` or `radius` for the server type.

<SERVER-GROUP-NAME>

Specifies the name of the server group to be created. The name of the server group can have a maximum of 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating TACACS+ server group `sg1`:

```
switch(config)# aaa group server tacacs sg1
```

Creating RADIUS server group `sg3`:

```
switch(config)# aaa group server radius sg3
```

Deleting TACACS+ server group `sg1`:

```
switch(config)# no aaa group server tacacs sg1
```

Deleting RADIUS server group `sg3`:

```
switch(config)# no aaa group server radius sg3
```

radius-server auth-type

Syntax

`radius-server auth-type {pap | chap}`

`no radius-server auth-type`

Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the RADIUS servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for RADIUS to PAP, which is the default authentication mechanism for RADIUS.

Command context

`config`

Parameters

auth-type {pap | chap}

Selects either the PAP or CHAP authentication protocol.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Authenticating CHAP:

```
switch(config)# radius-server auth-type chap
```

Authenticating PAP:

```
switch(config)# radius-server auth-type pap
```

radius-server host

Syntax

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |  
    ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
    [auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]  
    [tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]  
  
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a RADIUS server. By default, the RADIUS server is associated with the server group named `radius`.

The `no` form of this command removes a previously added RADIUS server.



NOTE: For enhanced security with IPsec, the alternative command `radius-server host secure ipsec` is available. The standard non-IPsec `radius-server host` command does not modify any existing IPsec configuration. If IPsec is already configured for the RADIUS server, then IPsec will remain enabled for the server.

Command context

config

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

key {plaintext <PASSKEY> | ciphertext <PASSKEY>}

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, the shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

timeout <TIMEOUT-SECONDS>

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default RADIUS: 1812.

auth-type {pap | chap}

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the RADIUS global default is used.

acct-port <ACCT-PORT>

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

retries <RETRY-COUNT>

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

tracking {enable | disable}

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



NOTE: Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

tracking-mode {any | dead-only}

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

vrf <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see [`radius-server key`](#).

If the fully qualified domain name is provided for the RADIUS server, a DNS server must be configured and accessible through the same VRF which is configured for the RADIUS server. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a RADIUS server with an IPv4 address and a named VRF:

```
switch(config)# radius-server host 1.1.1.1 vrf mgmt
```

Adding a RADIUS server with an IPv4 address, a port, and a named VRF:

```
switch(config)# radius-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a RADIUS server with an IPv6 address:

```
switch(config)# radius-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Adding a RADIUS server with tracking enabled and tracking mode is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only
```

Adding a RADIUS server with tracking disabled:

```
switch(config)# radius-server host 1.1.1.1 tracking disable
```

Deleting a RADIUS server with an IPv4 address and specified VRF:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt
```

radius-server host secure ipsec

Syntax

Syntax for a RADIUS server that uses IPsec for authentication:

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |  
  ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]  
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]  
secure ipsec authentication spi <SPI-INDEX> <AUTH-TYPE> <AUTH_KEY-TYPE> <AUTH-KEY>  
  
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]  
secure ipsec authentication
```

Syntax for a RADIUS server that uses IPsec for both authentication and encryption:

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |  
  ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]  
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]  
secure ipsec encryption spi <SPI-INDEX> <AUTH-TYPE> <AUTH_KEY-TYPE> <AUTH-KEY>  
  <ENCRYPT-TYPE> <ENCRYPT-KEY-TYPE> <ENCRYPT-KEY>  
  
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]  
secure ipsec encryption
```

Description

Adds a RADIUS server that uses IPsec for enhanced security (authentication and possibly encryption). By default, the RADIUS server is associated with the server group named `radius`.

The `no` form of this command removes a previously added RADIUS (with IPsec) server.



NOTE: Unless enhanced security with IPsec is required, use the `radius-server host` command instead.

Command context

`config`

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the RADIUS server as:

- `<FQDN>`: a fully qualified domain name.
- `<IPv4>`: an IPv4 address.
- `<IPv6>`: an IPv6 address.

key {plaintext <PASSKEY> | ciphertext <PASSKEY>}

Selects either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

timeout <TIMEOUT-SECONDS>

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

auth-type {pap | chap}

Selects either the PAP (the default) or CHAP authentication types. If this parameter is not specified, the RADIUS global default is used.

acct-port <ACCT-PORT>

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

retries <RETRY-COUNT>

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

tracking {enable | disable}

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



NOTE: Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

tracking-mode {any | dead-only}

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

vrf <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

spi <SPI-INDEX>

Specifies the Security Parameters Index. The SPI is an identification tag carried in the IPsec AH header. The SPI must be unique on the switch. Range: 256 to 4294967295.

<AUTH-TYPE>

Specifies the authentication algorithm: `md5`, `sha1`, or `sha256`.

<AUTH_KEY-TYPE>

Specifies the authentication key type: `plaintext`, `hex-string`, or `ciphertext`.

<AUTH-KEY>

Specifies the authentication key. For **<AUTH-TYPE>** of `ciphertext`, this is the ciphertext string.

For **<AUTH-TYPE>** of `plaintext` or `hex-string`:

- `md5` (`plaintext`): 1 to 16 characters, (`hex-string`): 2 to 32 hexadecimal digits.
- `sha1` (`plaintext`): 1 to 20 characters, (`hex-string`): 2 to 40 hexadecimal digits.
- `sha256` (`plaintext`): 1 to 32 characters, (`hex-string`): 2 to 64 hexadecimal digits.

<ENCRYPT-TYPE>

Specifies the encryption algorithm: `3des`, `aes`, `des`, or `null`.

<ENCRYPT-KEY-TYPE>

Specifies the encryption key type: `plaintext`, `hex-string`, or `ciphertext`.

<ENCRYPT-KEY>

Specifies the encryption key. For **<ENCRYPT-TYPE>** of `ciphertext`, this is the ciphertext string.

For **<ENCRYPT-TYPE>** of `plaintext` or `hex-string`:

- `3des` (`plaintext`): 24 characters, (`hex-string`): 48 hexadecimal digits.
- `aes` (`plaintext`): 16, 24, or 32 characters, (`hex-string`): 32, 48, or 64 hexadecimal digits.
- `des` (`plaintext`): 8 characters, (`hex-string`): 16 hexadecimal digits.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see [radius-server key](#).

If the fully qualified domain name is provided for the RADIUS server host, a DNS server must be configured and accessible through the same VRF as mentioned for the server host. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a RADIUS server with an IPv4 address, a plaintext passkey, and IPsec authentication (md5 plaintext).

```
switch(config)# radius-server host 1.1.1.1 key plaintext 98ab vrf mgmt secure  
ipsec authentication spi 261 md5 plaintext 1abc
```

Adding a RADIUS server with an IPv4 address, IPsec authentication (MD5 plaintext), and IPsec encryption (AES plaintext):

```
switch(config)# radius-server host 1.1.1.2 vrf mgmt secure  
ipsec encryption spi 262 md5 plaintext 9xyz aes plaintext 1234567890abcdef
```

Adding a RADIUS server with an IPv4 address, tracking enabled, tracking mode, IPsec authentication (MD5 plaintext), IPsec encryption (AES plaintext) is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only  
vrf mgmt secure ipsec encryption spi 262 md5 plaintext 9xyz aes plaintext 1234567890abcdef
```

Removing a RADIUS server with an IPv4 address and IPsec authentication:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt secure ipsec authentication
```

Removing a RADIUS server with an IPv4 address and IPsec authentication and IPsec encryption:

```
switch(config)# no radius-server host 1.1.1.2 vrf mgmt secure ipsec encryption
```

radius-server key

Syntax

```
radius-server key {plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>}
```

```
no radius-server key
```

Description

Creates or modifies a RADIUS global passkey. The RADIUS global passkey is used as a shared-secret for encrypting the communication between all RADIUS servers and the switch. The RADIUS global passkey is required for authentication unless local passkeys have been set. By default, the RADIUS global passkey is empty. If the administrator has not set this key, the switch will not be able to perform RADIUS authentication. The switch will instead rely on the authentication mechanism configured with [aaa authentication login](#).

The `no` form of the command removes the global passkey.

Authority

Administrators or local user group members with execution rights for this command.

Parameters

plaintext <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, a shared-secret can be a mix of alphanumeric and special characters.

ciphertext <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in encrypted format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding the global passkey:

```
switch(config)# radius-server key plaintext mypasskey123
```

Removing the global passkey:

```
switch(config)# no radius-server key
```

radius-server retries

Syntax

```
radius-server retries <0-5>
```

```
no radius-server retries
```

Description

Sets at the global level the number of retries the switch makes before concluding that the RADIUS server is unreachable.

You can override this setting with a fine-grained per RADIUS server retries configuration. See [radius-server host](#).

The **no** form of this command resets the RADIUS global retries to the default retries value of 1.

Command context

```
config
```

Parameters

retries <0-5>

Specifies the number of retry attempts for contacting RADIUS servers. Range is 0 to 5 retries.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch(config)# radius-server retries 3
```

radius-server timeout

Syntax

```
radius-server timeout [<1-60>]
```

```
no radius-server timeout
```

Description

Specifies the number of seconds to wait for a response from the RADIUS server before trying the next RADIUS server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The `no` form of this command resets the RADIUS global authentication timeout to the default of 5 seconds.

Command context

```
config
```

Parameters

timeout <1-60>

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the RADIUS server timeout:

```
switch(config)# radius-server timeout 10
```

Resetting the timeout for the RADIUS server to the default:

```
switch(config)# no radius-server timeout
```

radius-server tracking

Syntax

```
radius-server tracking {interval <INTERVAL> | retries <RETRIES> | user-name <NAME>  
                        [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]}
```

```
no radius-server tracking {interval | retries | user-name}
```

Description

Configures RADIUS server tracking settings globally for all configured RADIUS servers that have tracking enabled with the `radius-server host` command on individual servers.

The `no` form of the command removes the specified configuration, reverting it to its default. The `no` form with `user-name` also clears the password (resets it to empty).

Authority

Administrators or local user group members with execution rights for this command.

Parameters

interval <INTERVAL>

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

retries <RETRIES>

Specifies the number of server retries. Default: Global RADIUS retries. Range: 0 to 5.

user-name <NAME> [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]

Specifies the user name to be used for server checking. The default user name is `radius-tracking-user` with an empty password.



NOTE: The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.

The password for the server tracking user name is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



NOTE: Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# radius-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no radius-server tracking interval
```

Configuring three retries:

```
switch(config)# radius-server tracking retries 3
```

Configuring user `radius-tracker` with a plaintext password.

```
switch(config)# radius-server tracking user-name radius-tracker password plaintext track$1
```

Reverting the tracking user name to its default of `radius-tracking-user`:

```
switch(config)# no radius-server tracking user-name
```


server

Syntax

```
server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

```
no server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a TACACS+/RADIUS server to a server-group. Only the configured TACACS+/RADIUS servers are allowed to be added within the server group. If the same server name exists with multiple ports or multiple VRFs, specify the server name, port, and VRF when adding the server to the server-group.

The `no` form of this command removes a TACACS+/RADIUS server from a server-group.

Command context

config-sg

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default TACACS+ (TCP): 49, RADIUS (UDP): 1812.

If a port number is not provided, the system searches the TACACS+/RADIUS server by host name and sets the default authentication port. Group server priority is assigned based on the sequence in which the servers are added.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a server to TACACS+ server group sg1 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server tacacs sg1  
switch(config-sg)# server 1.1.1.2 port 32 vrf default
```

Adding a server to TACACS+ server group sg2 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server tacacs sg2  
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

Adding a server to RADIUS server group sg3 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg3
switch(config-sg)# server 1.1.1.5 port 12 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0371:7334 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 1.1.1.6 port 32 vrf vrf_red
```

Specifying an IPv4 address when removing a TACACS+ server from server group sg1:

```
switch(config)# aaa group server tacacs sg1
switch(config-sg)# no server 1.1.1.2 port 12 vrf default
```

Specifying an IPv6 address when removing a TACACS+ server from server group sg2 with the default VRF:

```
switch(config)# aaa group server tacacs sg2
switch(config-sg)# no server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

show aaa accounting

Syntax

```
show aaa accounting
```

Description

Shows the accounting configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing the accounting sequence for TACACS+ groups and local:

```
switch(config)# aaa accounting all default start-stop group tg1 tg2 tacacs local
switch(config)# aaa accounting all ssh start-stop group tg1 tg2
switch(config)# aaa accounting all console start-stop group tg4 tacacs local
switch(config)# aaa accounting all https-server start-stop local group tacacs tg3
switch(config)# exit
switch# show aaa accounting
```

```
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop
```

Accounting for default channel:

```
-----
GROUP NAME                  | GROUP PRIORITY
-----
tg1                          | 0
```

tg2	1
tacacs	2
local	3

Accounting for ssh channel:

GROUP NAME	GROUP PRIORITY
tg1	0
tg2	1

Accounting for console channel:

GROUP NAME	GROUP PRIORITY
tg4	0
tacacs	1
local	2

Accounting for https-server channel:

GROUP NAME	GROUP PRIORITY
local	0
tacacs	1
tg3	2

Configuring and then showing the accounting sequence for RADIUS groups and local:

```
switch(config)# aaa accounting all default start-stop group rg1 rg2 radius local
switch(config)# aaa accounting all console start-stop group rg4 radius local
switch(config)# exit
switch# show aaa accounting
```

AAA Accounting:

Accounting Type	: all
Accounting Mode	: start-stop

Accounting for default channel:

GROUP NAME	GROUP PRIORITY
rg1	0
rg2	1
radius	2
local	3

Accounting for console channel:

GROUP NAME	GROUP PRIORITY
tg4	0
radius	1
local	2

Configuring and then showing only local accounting for default:

```

switch(config)# aaa accounting all default start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop

```

Accounting for default channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authentication

Syntax

```
show aaa authentication
```

Description

Shows the authentication configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configuring TACACS+ authentication sequences and then showing the configuration per connection type (channel):

```

switch(config)# aaa authentication login default group tg1 tg2 tg3 tg4 tacacs local
switch(config)# aaa authentication login ssh group tg1 tg2
switch(config)# aaa authentication login console group tg4 tacacs local
switch(config)# aaa authentication login https-server local group tacacs tg3
switch(config)# exit
switch# show aaa authentication

```

```

AAA Authentication:
  Fail-through           : Enabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set

```

Authentication for default channel:

GROUP NAME	GROUP PRIORITY
tg1	0
tg2	1
tg3	2
tg4	3
tacacs	4
local	5

Authentication for ssh channel:

GROUP NAME	GROUP PRIORITY
tg1	0
tg2	1

Authentication for console channel:

GROUP NAME	GROUP PRIORITY
tg4	0
tacacs	1
local	2

Authentication for https-server channel:

GROUP NAME	GROUP PRIORITY
local	0
tacacs	1
tg3	2

Configuring RADIUS authentication sequences and then showing the configuration per connection type (channel):

```
switch(config)# aaa authentication login default group rg1 rg2 rg3 rg4 radius local
switch(config)# aaa authentication login console group rg4 radius local
switch(config)# exit
switch# show aaa authentication
```

AAA Authentication:

Fail-through	: Enabled
Limit Login Attempts	: Not set
Lockout Time	: 300
Minimum Password Length	: Not set

Authentication for default channel:

GROUP NAME	GROUP PRIORITY
rg1	0
rg2	1
rg3	2
rg4	3
radius	4
local	5

Authentication for console channel:

GROUP NAME	GROUP PRIORITY
rg4	0
radius	1
local	2

Configuring only default authentication and then showing the default connection type (channel):

```
switch(config)# aaa authentication login default local
switch(config)# exit
switch# show aaa authentication
```

```
AAA Authentication:
  Fail-through           : Disabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set
```

Authentication for default channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authorization

Syntax

```
show aaa authorization
```

Description

Shows the authorization configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing the authorization sequence for default and console connection types (channels):

```
switch(config)# aaa authorization commands default group tg1 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)# aaa authorization commands console group tg1 tg2 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)# exit
switch# show aaa authorization
```

Authorization for default channel:

GROUP NAME	GROUP PRIORITY
tg1	0
tacacs	1
none	2

Authorization for console channel:

GROUP NAME	GROUP PRIORITY
tg1	0
tg2	1
tacacs	2
none	3

show aaa server-groups

Syntax

```
show aaa server-groups [tacacs | radius]
```

Description

Shows TACACS+ and RADIUS AAA server group information for all server types or for the specified server type.

Command context

Operator (>) or Manager (#)

Parameters

tacacs

Narrows the command output to only TACACS+ servers.

radius

Narrows the command output to only RADIUS servers.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing all AAA server group information:

```
switch# show aaa server-groups

***** AAA Mechanism TACACS+ *****
```

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
sg2	2001:0db8:85a3:0000:0000:8a2e:0370:7334	49	default	1
sg1	1.1.1.2	12	mgmt	1

```

tacacs (default) | FQDN.com | 32 | mgmt | 1
tacacs (default) | 1.1.1.1 | 49 | mgmt | 2
tacacs (default) | 1.1.1.2 | 12 | mgmt | 3
tacacs (default) | abc.com | 32 | vrf_red | 4
tacacs (default) | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49 | default | 5
tacacs (default) | 1.1.1.3 | 32 | vrf_blue | 6
***** AAA Mechanism RADIUS *****
```

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
sg4	2001:0db8:85a3:0000:0000:8a2e:0370:7334	1812	default	1
sg3	1.1.1.5	12	mgmt	1
radius (default)	1.1.1.4	1812	mgmt	1
radius (default)	1.1.1.5	12	mgmt	2
radius (default)	abc1.com	32	mgmt	3
radius (default)	2001:0db8:85a3:0000:0000:8a2e:0370:7334	1812	default	4
radius (default)	1.1.1.6	32	vrf_red	5
radius (default)	1.1.1.7	32	vrf_blue	6

Showing TACACS+ server group information:

```
switch# show aaa server-groups tacacs
```

***** AAA Mechanism TACACS+ *****

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
sg2	2001:0db8:85a3:0000:0000:8a2e:0370:7334	49	default	1
sg1	1.1.1.2	12	mgmt	1
tacacs (default)	FQDN.com	32	mgmt	1
tacacs (default)	1.1.1.1	49	mgmt	2
tacacs (default)	1.1.1.2	12	mgmt	3
tacacs (default)	abc.com	32	vrf_red	4
tacacs (default)	2001:0db8:85a3:0000:0000:8a2e:0370:7334	49	default	5
tacacs (default)	1.1.1.3	32	vrf_blue	6

Showing RADIUS server group information:

```
switch# show aaa server-groups radius
```

***** AAA Mechanism RADIUS *****

GROUP NAME	SERVER NAME	PORT	VRF	PRIORITY
sg4	2001:0db8:85a3:0000:0000:8a2e:0370:7334	1812	default	1
sg3	1.1.1.5	12	mgmt	1
radius (default)	1.1.1.4	1812	mgmt	1
radius (default)	1.1.1.5	12	mgmt	2
radius (default)	abc1.com	32	mgmt	3
radius (default)	2001:0db8:85a3:0000:0000:8a2e:0370:7334	1812	default	4
radius (default)	1.1.1.6	32	vrf_red	5
radius (default)	1.1.1.7	32	vrf_blue	6

show accounting log

Syntax

```
show accounting log [last <QTY-TO-SHOW> | all]
```

Description

Entered without optional parameters, this command shows all accounting log records for the current boot. Sensitive information is masked from the log, by being represented as asterisks.



NOTE: This `show accounting log` command replaces the `show audit-log` command that is supported only in 10.00 releases.

Command context

Manager (#) or Auditor (auditor>)

Parameters

last <QTY-TO-SHOW>

Specifies how many most-recent accounting log records to show for the current boot. Range: 1 to 1000.

all

Selects for showing, all accounting records from the current boot and the previous boot.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

The log message starts with the record type, which is specific to AOS-CX. Values are the following:

USER_START

Record of a user login action.

USER_END

Record of a user logout action.

USYS_CONFIG

Record of a command executed by the user.

The three types of accounting log information are identified by the `msg=` element starting with the `rec=` item as follows:

- Exec is identified with: `msg='rec=ACCT_EXEC'`
- Command is identified with: `msg='rec=ACCT_CMD'`
- System is identified with: `msg='rec=ACCT_SYSTEM'`

The user group is indicated by `priv-lvl`, which is specific to AOS-CX. Values are the following:

Privilege level	User group
1	operators
15	administrators
19	auditors

The value of `service` indicates which user interface was used:

service=shell

Indicates that the log entry is a result of a CLI command.

service=https-server

Indicates that the log entry is a result of a REST API request or a Web UI action.

The string value of `data` identifies the CLI command or REST API request that was executed.

These elements are shown in context under *Examples*.

Examples

Showing the accounting log for the previous and current boot. Line breaks have been added for readability.

```
switch# show accounting log all
```

```
-----
Local accounting logs from previous boot
-----
```

```
-----
type=DAEMON_START msg=audit(Nov 05 2018 23:00:58.607:9057) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success
-----
type=USER_START msg=audit(Nov 05 2018 23:06:42.398:42) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'
-----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:06:42.399:43) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
data="enable" hostname=8xxx addr=0.0.0.0 res=success'
-----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:24.693:51) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=1
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
data="configure terminal" hostname=8xxx addr=0.0.0.0 res=success'
-----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:39.108:52) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="https-server rest access-mode read-write"
hostname=8xxx addr=0.0.0.0 res=success'
-----
type=USER_START msg=audit(Nov 05 2018 23:10:57.238:58) :
msg='rec=ACCT_EXEC op=start session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/login"
hostname=8xxx addr=127.0.0.1 res=success'
-----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:11.958:75) :
```

```

msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="tacacs-server host 2.2.2.2" hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:37.090:76) :
msg='rec=ACCT_CMD op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=GET http-uri=/rest/v1/system/vrfs/mgmt/tacacs_servers"
hostname=8xxx addr=127.0.0.1 res=success'
----
type=USER_END msg=audit(Nov 05 2018 23:26:59.207:90) :
msg='rec=ACCT_EXEC op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/logout"
hostname=8xxx addr=127.0.0.1 res=success'
----
type=USER_END msg=audit(Nov 05 2018 23:27:49.164:93) :
msg='rec=ACCT_EXEC op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'

-----
Local accounting logs from current boot
-----
----
type=DAEMON_START msg=audit(Nov 05 2018 23:32:05.642:626) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success
----
type=USER_START msg=audit(Nov 05 2018 23:35:52.915:11) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:35:52.917:12) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no data="enable"
hostname=8xxx addr=0.0.0.0 res=success'

```

show radius-server

Syntax

```
show radius-server [detail]
```

Description

Shows configured RADIUS servers information.

Command context

Operator (>) or Manager (#)

Parameters

detail

Selects additional RADIUS server details and global parameters for showing.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

When the `show radius-server` command shows `None` for the shared-secret, the passkey is missing.

Examples

Showing a summary of the global RADIUS configuration:

```
switch# show radius-server
***** Global RADIUS Configuration *****
```

```
Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

SERVER NAME	PORT	VRF
20.1.1.129	1812	default
1.1.1.4	1812	default
1.1.1.5	12	default
abc1.com	32	default
2001:0db8:85a3:0000:0000:8a2e:0371:7334	1812	default

Showing a summary of a RADIUS server when the per-server shared key or the global RADIUS shared key is not set:

```
switch# show radius-server
***** Global RADIUS Configuration *****
```

```
Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

SERVER NAME	PORT	VRF
20.1.1.129	1812	default

Showing details of a global RADIUS configuration:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****
```

```
Shared-Secret: AQBapb+HsdpqV1QcA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
```

```
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

```
***** RADIUS Server Information *****
Server-Name           : 20.1.1.129
Auth-Port              : 1812
Accounting-Port        : 1813
VRF                    : default
Shared-Secret          : None
Timeout                : 60
Retries                : 5
Auth-Type              : pap
Server-Group           : radius
Default-Priority        : 4
Tracking               : disabled
Tracking-Mode           : any
Reachability-Status    : N/A
ClearPass-Username     :
ClearPass-Password     : None
```

Showing details of a RADIUS server when the per-server shared key and the global RADIUS shared key are not set:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 1

***** RADIUS Server Information *****
Server-Name           : 1.1.1.1
Auth-Port              : 2083
VRF                    : default
Shared-Secret (default) : None
Timeout (default)      : 5
Retries (default)      : 1
Auth-Type (default)    : pap
Server-Group (default) : radius
Default-Priority        : 1
```

show radius-server secure ipsec

Syntax

```
show radius-server secure ipsec { server-list | host {<FQDN> | <IPV4> | <IPV6>}
                                   [port <PORT-NUMBER>] [vrf <VRF-NAME>]}
```

Description

Shows information for one or all RADIUS servers configured with IPsec.

Command context

Operator (>) or Manager (#)

Parameters

server-list

Selects all servers for showing.

host {<FQDN> | <IPv4> | <IPv6>}

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

vrf <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The IPsec key is shown in an exportable ciphertext format.

Examples

Showing information for RADIUS server 1.1.1.1 secured with IPsec:

```
switch# show radius-server secure ipsec host 1.1.1.1
IPsec           : enabled
Protocol        : ESP
Authentication   : MD5
Encryption      : AES
SPI             : 1234
```

Showing information for all RADIUS servers secured with IPsec:

```
switch# show radius-server secure ipsec server-list
Server          : 1.1.1.1
IPsec           : enabled
Protocol        : ESP
Authentication   : MD5
Encryption      : AES
SPI             : 1234

Server          : 1.1.1.2
IPsec           : enabled
Protocol        : ESP
Authentication   : MD5
Encryption      : AES
SPI             : 12341
```

show tacacs-server

Syntax

```
show tacacs-server [detail]
```

Description

Shows the configured TACACS+ servers.

Command context

Operator (>) or Manager (#)

Parameters

detail

Selects additional TACACS+ server details and global parameters for showing.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing a summary of a global TACACS+ configuration with a shared-secret:

```
switch# show tacacs-server
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqV1Q3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

-----
SERVER NAME                               | PORT | VRF
-----
1.1.1.1                                   | 49    | mgmt
1.1.1.2                                   | 12    | mgmt
abc.com                                   | 32    | vrf_blue
2001:0db8:85a3:0000:0000:8a2e:0370:7334  | 49    | default
1.1.1.3                                   | 32    | vrf_red
-----
```

Showing details of a global TACACS+ configuration:

```
switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqV1Q3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

***** TACACS+ Server Information *****
Server-Name       : 1.1.1.2
Auth-Port         : 12
VRF               : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)  : 5
```

```

Auth-Type (default)      : pap
Server-Group             : sg1
Group-Priority            : 1

Server-Name              : 2001:0db8:85a3:0000:0000:8a2e:0370:7334
Auth-Port                : 49
VRF                     : default
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)       : 5
Auth-Type (default)     : pap
Server-Group            : sg2
Group-Priority          : 1

Server-Name              : 1.1.1.1
Auth-Port                : 49
VRF                     : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)       : 5
Auth-Type (default)     : pap
Server-Group (default)  : tacacs
Default-Priority        : 1

Server-Name              : abc.com
Auth-Port                : 32
VRF                     : vrf_red
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout                 : 15
Auth-Type (default)     : pap
Server-Group (default)  : tacacs
Default-Priority        : 3

Server-Name              : 1.1.1.3
Auth-Port                : 32
VRF                     : vrf_blue
Shared-Secret           : AQBapfnqbSswqKC476tdUFZ+AncIRY92hDTYkQCAAAAFEaHn43vNC
Timeout                 : 15
Auth-Type               : chap
Server-Group (default)  : tacacs
Default-Priority        : 5

```

Showing TACACS+ server when per-server shared key and global TACACS+ shared key is not set:

```

switch# show tacacs-server
***** Global TACACS+ Configuration *****

```

```

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Number of Servers: 1

```

SERVER NAME	PORT	VRF
1.1.1.1	49	default

Showing TACACS+ server details when per-server shared key and global TACACS+ shared key is not set:

```

switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

```

```

Shared-Secret: None
Timeout: 5
Auth-Type: pap

```



```
Number of Servers: 1
```

```
***** TACACS+ Server Information *****
Server-Name           : 1.1.1.1
Auth-Port             : 49
VRF                   : default
Shared-Secret (default) : None
Timeout (default)     : 5
Auth-Type (default)   : pap
Server-Group (default) : tacacs
Default-Priority      : 1
```

show tacacs-server statistics

Syntax

```
show tacacs-server statistics
```

Description

Shows authentication statistics for all configured TACACS+ servers.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing TACACS+ server authentication statistics:

```
switch# show tacacs-server statistics
Server Name       : tac1
Auth-Port         : 49
VRF               : mgmt
Authentication Statistics
-----
Round Trip Time   : 1
Pending Requests  : 0
Timeout           : 0
Unknown Types     : 0
Packet Dropped    : 0
Auth Start        : 8
Auth challenge     : 0
Auth Accepts      : 4
Auth Rejects      : 4
Auth reply malformed : 0
Tracking Requests : 0
Tracking Responses : 0
```

show tech aaa

Syntax

```
show tech aaa
```

Description

Shows the AAA configuration settings.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

Showing the AAA configuration settings:

```
switch# show tech aaa
=====
Show Tech executed on Tue Feb 14 02:19:11 2017
=====
[Begin] Feature aaa
=====

*****
Command : show aaa authentication
*****
AAA Authentication:
  Fail-through           : Enabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set

Authentication for ssh channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
local                      | 0
-----

Authentication for https-server channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
local                      | 0
-----

Authentication for console channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
local                      | 0
-----

Authentication for default channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
tacacs                    | 0
local                     | 1
-----

*****
Command : show aaa accounting
```

AAA Accounting:

Accounting Type : all
Accounting Mode : start-stop

Accounting for default channel:

GROUP NAME	GROUP PRIORITY
local	0

Accounting for ssh channel:

GROUP NAME	GROUP PRIORITY
tacacs	0
local	1

Accounting for https-server channel:

GROUP NAME	GROUP PRIORITY
tacacs	0

Command : show aaa authorization

Authorization for default channel:

GROUP NAME	GROUP PRIORITY
none	0

Authorization for console channel:

GROUP NAME	GROUP PRIORITY
none	0

Authorization for ssh channel:

GROUP NAME	GROUP PRIORITY
tacacs	0
none	1

Command : show aaa server-groups

***** AAA Mechanism TACACS+ *****

GROUP NAME	SERVER NAME	PORT	PRIORITY	VRF
tacacs (default)	1.1.1.1	49	1	mgmt

```

***** AAA Mechanism RADIUS *****
-----
GROUP NAME | SERVER NAME | PORT | PRIORITY | VRF
-----

*****
Command : show tacacs-server detail
*****
***** Global TACACS+ Configuration *****

Shared-Secret: G8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Tracking: disabled
Tracking Time Interval (seconds): 300
Tracking User-name: tacacs-tracking-user
Tracking Password: None
Number of Servers: 1

***** TACACS+ Server Information *****
Server-Name : 1.1.1.1
Auth-Port : 49
VRF : mgmt
Shared-Secret : KCdmOMxMD26T0fQoXfJbtj9j2AUxlGn6eCAAAAF2MkfMTtojQX
Timeout (default) : 5
Auth-Type (default) : pap
Server-Group (default) : tacacs
Default-Priority : 1
Tracking : disabled
Reachability-Status : N/A

*****
Command : show radius-server detail
*****
***** Global RADIUS Configuration *****

Shared-Secret: CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 0

=====
[End] Feature aaa
=====

=====
Show Tech commands executed successfully
=====

```

tacacs-server auth-type

Syntax

```
tacacs-server auth-type {pap | chap}
```

```
no tacacs-server auth-type
```

Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the TACACS+ servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for TACACS+ to PAP, which is the default authentication mechanism for TACACS+.

Command context

`config`

Parameters

`auth-type {pap | chap}`

Selects either the PAP or CHAP authentication protocol.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling command for CHAP authentication:

```
switch(config)# tacacs-server auth-type chap
```

Enabling command for PAP authentication:

```
switch(config)# tacacs-server auth-type pap
```

tacacs-server host

Syntax

```
tacacs-server host {<FQDN> | <IPV4> | <IPV6>} [key {plaintext <PASSKEY> |  
    ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
    [auth-type {pap | chap}] [tracking {enable | disable}] [vrf <VRF-NAME>]
```

```
no tacacs-server host {<FQDN> | <IPV4> | <IPV6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a TACACS+ server. By default, the TACACS+ server is associated with the server group named `tacacs`.

The `no` form of this command removes a previously added TACACS+ server.

Command context

`config`

Parameters

`{<FQDN> | <IPV4> | <IPV6>}`

Specifies the TACACS+ server as:

- `<FQDN>`: a fully qualified domain name.
- `<IPv4>`: an IPv4 address.
- `<IPv6>`: an IPv6 address.

key {plaintext `<PASSKEY>` | ciphertext `<PASSKEY>`}

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. The length of shared-secret in plaintext format is fewer than 32 characters.

timeout `<TIMEOUT-SECONDS>`

Specifies the timeout. The range is 1 to 60 seconds. The default timeout is 5 seconds.

port `<PORT-NUMBER>`

Specifies the TCP authentication port number. Range: 1 to 65535. Default: 49.

auth-type {`pap` | `chap`}

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the TACACS+ global default is used.

tracking {`enable` | `disable`}

Enables or disables server tracking for the server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable. Unreachable servers are skipped in favor of servers that are proven to be reachable. Use command `tacacs-server tracking` to configure TACACS+ server tracking.

vrf `<VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see `tacacs-server key`.

If the fully qualified domain name is provided for the TACACS+ server, a DNS server must be configured and accessible through the same VRF which is configured for the TACACS+ server. This configuration is required for the resolution of the TACACS+ server hostname to its IP address. If a DNS server is not available for this VRF, the TACACS+ servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a TACACS+ server with an IPv4 address and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.1 vrf mgmt
```

Adding a TACACS+ server with an IPv4 address, a port, and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a TACACS+ server with an FQDN, a timeout, port number, and a named VRF:

```
switch(config)# tacacs-server host abc.com timeout 15 port 32 vrf vrf_blue
```

Adding a TACACS+ server with an IPv6 address:

```
switch(config)# tacacs-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Adding a TACACS+ server with an IPv4 address, plaintext passkey, timeout, port, authentication type, and VRF name:

```
switch(config)# tacacs-server host 1.1.1.3 key plaintext test-123 timeout 15 port 32 auth-type chap vrf vrf_red
```

Deleting a TACACS+ server with an IPv4 address and specified VRF:

```
switch(config)# no tacacs-server host 1.1.1.1 vrf mgmt
```

Deleting a TACACS+ server with an FQDN, port, and specified VRF:

```
switch(config)# no tacacs-server host abc.com port 32 vrf vrf_blue
```

tacacs-server key

Syntax

```
tacacs-server key {plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>}
```

```
no tacacs-server key
```

Description

Creates or modifies a TACACS+ global passkey. The TACACS+ global passkey is used as a shared-secret for encrypting the communication between all TACACS+ servers and the switch. The TACACS+ global passkey is required for authentication unless local passkeys have been set. By default, the TACACS+ global passkey is empty. If the administrator has not set this key, the switch will not be able to perform TACACS+ authentication. The switch will instead rely on the authentication mechanism configured with aaa authentication login.

The `no` form of the command removes the global passkey.

Command context

```
config
```

Parameters

plaintext <GLOBAL-PASSKEY>

Specifies the TACACS+ global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters.

ciphertext <GLOBAL-PASSKEY>

Specifies the TACACS+ global passkey in encrypted format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding the global passkey:

```
switch(config)# tacacs-server key plaintext mypasskey123
```

Removing the global passkey:

```
switch(config)# no tacacs-server key
```

tacacs-server timeout

Syntax

```
tacacs-server timeout [<1-60>]
```

```
no tacacs-server timeout
```

Description

Specifies the number of seconds to wait for a response from the TACACS+ server before trying the next TACACS+ server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The `no` form of this command resets the TACACS+ global authentication timeout to the default of 5 seconds.

Command context

```
config
```

Parameters

timeout *<1-60>*

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Specifying the TACACS+ server timeout:

```
switch(config)# tacacs-server timeout 10
```

Resetting the timeout for the TACACS+ server to the default:

```
switch(config)# no tacacs-server timeout
```

tacacs-server tracking

Syntax

```
tacacs-server tracking {interval <INTERVAL> | user-name <NAME> [password  
                        {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]}
```

```
no tacacs-server tracking {interval | user-name}
```

Description

Configures TACACS+ server tracking settings globally for all configured TACACS+ servers that have tracking enabled with the `tacacs-server host` command.

The `no` form of the command removes the specified configuration, reverting it to its default. The `no` form with `user-name` also clears the password (resets it to empty).

Authority

Administrators or local user group members with execution rights for this command.

Parameters

interval <INTERVAL>

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

user-name <NAME> [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]

Specifies the user name to be used for server tracking requests. The default user name is `tacacs-tracking-user` with an empty password.



NOTE: The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.

The password for the server tracking user name is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



NOTE: Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# tacacs-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no tacacs-server tracking interval
```

Configuring user `tacacs-tracker` with a plaintext password.

```
switch(config)# tacacs-server tracking user-name tacacs-tracker password plaintext track$1
```

Reverting the tracking user name to its default of `tacacs-tracking-user`:

```
switch(config)# no tacacs-server tracking user-name
```

About RADIUS dynamic authorization

RADIUS dynamic authorization provides the ability to make changes to a user account session while it is in progress. This ability includes disconnecting a session or updating some aspect of the authorization for the session. It also includes "pot bounce" in which the interface on which a client is connected is brought down and then back up (using COA (change of authorization)).

RADIUS dynamic authorization enables or disables the processing of "Disconnect" and "Change of Authorization (CoA)" messages from the RADIUS server. When enabled, the RADIUS server can dynamically terminate or change the authorization parameters (such as VLAN/user-role assignment) used in an active client session on the switch.



NOTE: See also *RFC 3576* available at <http://www.ietf.org/rfc/rfc3576.txt> for general information on the dynamic authorization extensions to RADIUS.

Requirements and tips

The switch validates these mandatory attributes that must be present in the CoA/Disconnect Message:

- Username
- NAS IP or NAS IPV6 or NAS Identifier
- Any one of the following combinations is used to identify the client session:
 - NAS-Port and Calling-Station-ID
 - NAS-Port-ID and Calling-Station-ID
 - Accounting-Session-ID

RADIUS server requirements:

- For ClearPass to provide CoA capabilities, in the case where the switch sends the NAS-IP address as a routable IP address, the CLI command `ip source interface` must be executed with the `radius` parameter.
- In CISCO ISE, to send the CoA request with the same username as in the RADIUS Accept, the Identity rewrite option has to be configured.

RADIUS dynamic authorization commands



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

radius dyn-authorization enable

Syntax

```
radius dyn-authorization enable

no radius dyn-authorization enable
```

Description

Enables RADIUS dynamic authorization. This command must be issued before the configuration set with other `radius dyn-authorization` commands takes effect.

The `no` form of this command disables RADIUS dynamic authorization.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RADIUS dynamic authorization:

```
switch(config)# radius dyn-authorization enable
```

radius dyn-authorization client

Syntax

```
radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>}
                                [secret-key {plaintext | ciphertext} <PASSKEY>]
                                [time-window <WIDTH>] [replay-protection {enable|disable}]
                                [vrf <VRF-NAME>]

no radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>} [vrf <VRF-NAME>]
```

Description

Configures RADIUS dynamic authorization for the specified client on the specified (or default) VRF.

The `no` form of this command unconfigures RADIUS dynamic authorization for the specified client on the specified (or default) VRF.

Command context

```
config
```

Parameters

<IPV4> | <IPV6> | <HOSTNAME>

Specifies the client IPv4 address, IPv6 address, or host name.

secret-key {plaintext | ciphertext} <PASSKEY>

Specifies the dynamic authorization server (RADIUS server) shared secret key required for client access. Provide either a plaintext or an encrypted shared-secret passkey. As per RFC 2865, the shared-secret can

be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

time-window <WIDTH>

Specifies the width of the synchronization window (in seconds) between the RADIUS dynamic authorization client and the RADIUS dynamic authorization server. Default 300. Range: 1 to 65535.

replay-protection {enable|disable}

Enables or disables RADIUS dynamic authorization replay protection for the specified client on the specified (or default) VRF.

vrf <VRF-NAME>

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RADIUS dynamic authorization with replay protection for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.5 replay-protection enable
```

Configuring RADIUS dynamic authorization with time window and shared secret for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.7 time-window 8 secret-key sk
```

radius dyn-authorization port

Syntax

`radius dyn-authorization port <PORT-NUMBER>`

Description

Sets the RADIUS dynamic authorization server UDP/TCP port.

Command context

`config`

Parameters

<PORT-NUMBER>

Specifies the UDP or TCP port. Default UDP: 3799 and TCP:2083

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the RADIUS dynamic authorization server UDP port back to its default 3799:

```
switch(config)# radius dyn-authorization port 3799
```

Setting the RADIUS dynamic authorization server TCP port back to its default 2083:

```
switch(config)# radius dyn-authorization port 2083
```

show radius dyn-authorization

Syntax

```
show radius dyn-authorization
```

Description

Shows RADIUS dynamic authorization configuration and summarized statistics for all clients configured for dynamic authorization.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Show command output item identification:

- Radius Dynamic Authorization: Enabled or Disabled status, system wide.
- Radius Dynamic Authorization Port: The UDP or TCP port used for dynamic authorization (default 3799).
- Invalid Client Address in CoA Requests: The number of CoA (change of authorization) requests received with an incorrect DAC (dynamic authorization client) address.
- Invalid Client Address in Disconnect Requests: The number of disconnect requests received with incorrect DAC address.
- Disconnect Requests: The number of disconnect requests received from the DAC.
- Disconnect ACKs: The number of Disconnect-ACKs sent to the DAC.
- Disconnect NAKs: The number of Disconnect-NAKs sent to the DAC.
- CoA Requests: The number of CoA-requests received from the DAC.
- CoA ACKs: The number of CoA-ACKs sent to the DAC.
- CoA NAKs: The number of CoA-NAKs sent to the DAC.

Example

Showing RADIUS dynamic authorization summarized statistics for all clients configured for dynamic authorization:

```
switch# show radius dyn-authorization
Status and Counters - RADIUS Dynamic Authorization Information

RADIUS Dynamic Authorization           : Enabled
RADIUS Dynamic Authorization UDP Port  : 3799
Invalid Client Addresses in CoA Requests : 0
Invalid Client Addresses in Disconnect Requests: 0

Dynamic Authorization Client Information
=====
```

```

IP Address      : 1.1.2.1
VRF             : adm2
Replay Protection : Disabled
Time Window     : 20
Disconnect Requests : 1
Disconnect ACKs  : 1
Disconnect NAKs  : 0
CoA Requests    : 7
CoA ACKs        : 2
CoA-NAKs        : 5
Shared-Secret   : AQBapb+HsdpqVlQ3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address      : 1.1.2.5
VRF             : default
Replay Protection : Enabled
Time Window     : 20
Disconnect Requests : 6
Disconnect ACKs  : 6
Disconnect NAKs  : 0
CoA Requests    : 9
CoA ACKs        : 5
CoA-NAKs        : 4
Shared-Secret   : AQBapb+HsdpqVlQ3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address      : 1.1.2.7
VRF             : default
Replay Protection : Disabled
Time Window     : 8
Disconnect Requests : 6
Disconnect ACKs  : 6
Disconnect NAKs  : 0
CoA Requests    : 9
CoA ACKs        : 5
CoA-NAKs        : 4
Shared-Secret   : AQBapb+HsdpqVlQ3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

```

show radius dyn-authorization client

Syntax

```
show radius dyn-authorization client <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Shows RADIUS dynamic authorization statistics for the specified client on the specified VRF.

Command context

Operator (>) or Manager (#)

Parameters

client <IP-ADDR>

Specifies the client IPv4 or IPv6 address.

vrf <VRF-NAME>

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Show command output item identification:

- **Total Requests:** The number of Disconnect and CoA (change of authorization) requests received from the DAC (dynamic authorization client).
- **Authorize Only Requests:** The number of Disconnect and CoA requests received from the DAC with an "Authorize only" Service-Type attribute.
- **Malformed Requests:** The number of malformed Disconnect and CoA requests received from the DAC.
- **Bad Authenticator Requests:** The number of Disconnect and CoA requests received from this DAC with an invalid authenticator field.
- **Dropped Requests:** The number of Disconnect and CoA requests from this DAC that have been silently discarded for reasons other than malformed, bad authenticators, or unknown type.
- **Total ACK Responses:** The number of Disconnect-ACKs sent to the DAC.
- **Total NAK Responses:** The number of Disconnect-NAKs sent to the DAC.
- **Session Not Found Responses:** The number of Disconnect-NAKs sent to the DAC because no session context could be found.
- **User Sessions Modified:** The number of user sessions for which authorization changed due to Disconnect and CoA requests received from the DAC.

Example

Showing RADIUS dynamic authorization statistics for client 1.1.2.1 on VRF default:

```
switch# show radius dyn-authorization client 1.1.2.1 vrf default
Status and Counters - RADIUS Dynamic Authorization Client Information
```

VRF Name	: default	
Authorization Client	: 1.1.2.1	
Unknown Packets	: 55	
Message-Type	Disconnect	CoA

Total Requests	2147483647	10
Authorize Only Requests	10	10
Malformed Requests	10	10
Bad Authenticator Requests	2147483647	2147483647
Dropped Requests	10	10
Total ACK Responses	10	10
Total NAK Responses	10	10
Session Not Found Responses	10	10
User Sessions Modified	20	20

The public key infrastructure (PKI) feature enables administrators to manage digital certificates on the switch. The switch uses certificates to validate SSH clients when acting as an SSH server, and when communicating with syslog servers when TLS encryption is used.



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

PKI concepts

Digital certificate

A digital certificate is an electronic form of identification that stores important information about an entity (such as a computer, program, or website). Certificates help secure digital transactions by enabling the end parties to validate each other's identity. Digital certificates are issued by a certificate authority (CA) and are composed of an encoded string of characters (usually stored in a file). For example:

```
-----BEGIN CERTIFICATE-----
MIIDSDCCApGCCQDJotuPPj9GCDANBgkqhkiG9w0BAQsAAADCBQzELMAkGA1UEBh
VVMxEzARBgNVBAgMCkNhbm3JuaWExEDAObgNVBACBM1JvY2tsaW4xDDAKBg
BAoMA0hQTjEVMBMGA1UECwwMSFBOUm9zZXZpbGxlMSokwAYDVQQDDCFocG5zd
...
MioDy0096DvSMPsnOaI+jnZ3AozN8y+nLgotXUsg36pO/Ncc51oQhyUdcAbgA1
rzSLgyTnpXZKumvlaoTk3pziIf7m5V103GTbgHGSFCzgO6QWxVxu9d7ju1o59S
aOIT7JSsYI5LsLpVz9ZqS599rj/1LoH+rLN1RDVXpS+J51
-----END CERTIFICATE-----
```

The switch can import PEM encoded ITU-T X.509 v3 certificates. (Certificates can be converted to human-readable form using a software decoder.)

An X.509 digital certificate typically includes the following information:

- Signature algorithm: The cryptographic algorithm used to generate the digital signature.
- Signature value: Digital signature of the certificate generated using the CA's private key.
- Version number: X.509 version number.
- Serial number: Certificate serial number.
- Issuer name: Name of the certificate authority (CA) that issued the certificate.
- Validity period: Beginning and ending dates.
- Subject name: Name of the entity to which the certificate is issued.
- Subject public key and key algorithm.
- Key usage extension: Purpose of the certificate.

Certificate authority

A certificate authority (CA) is an entity that can issue and sign digital certificates. A CA can be a well-known, trusted commercial company, or a private entity controlled by your organization. For a commercial CA, the

CA validates the credentials of a user before issuing a certificate and signing it, guaranteeing a certificate holder's identity. For a private CA, self-signed certificates can be generated as needed for devices on your network without paying a commercial company.

Root certificate

A root certificate is a self-signed certificate that is deemed the root of trust for a certificate chain. This is the certificate that identifies a CA, and is used by the CA to sign any certificates that it issues. When two peers attempt to establish a secure connection, they use the CA's public key to verify that each other's certificates were indeed signed by a trusted certificate authority.

Each root CA certificate has a unique fingerprint, which is the hash value of the certificate content. The fingerprint of a root CA certificate can be used to authenticate the validity of the root CA.

In a certificate chain, the root CA generates a self-signed certificate, and each lower level CA holds a CA certificate (intermediate certificate) issued by the CA immediately above it. The hierarchy of these certificates forms a *chain of trust*.

Leaf certificate

This is the certificate used by a software entity, such as a syslog client, to identify itself to a peer when establishing a secure connection.

Intermediate certificate

An intermediate certificate is a CA which has been issued by the root certificate or by another intermediate certificate. Intermediate CAs can issue leaf certificates and sit in between the root and leaf certificates. The use of an intermediate CA allows administrators to segregate their PKI groups.

Trust anchor

This is the certificate that acts as the base of trust for the validation of other certificates. A trust anchor can be a root or intermediate certificate issued by a CA.

OCSP

The online certificate status protocol (OCSP) is a real-time method for determining the revocation status of a certificate. When two peers attempt to establish a secure connection, they can query an OCSP responder to determine the status (valid or revoked) of each other's certificates. The OCSP responder for a certificate is typically provided by a server managed by the CA that issued the certificate.

PKI on the switch

The AOS-CX Switch Series switches provides for installation of certificate authority (CA) certificates and the generation and installation of leaf certificates.

Trust anchor profiles

The switch supports 64 trust anchor (TA) profiles. Each TA profile stores a trusted CA certificate. The certificate can be either a root CA certificate, which must be self-signed, or an intermediate CA certificate that is issued by another CA.



NOTE: The certificate must have its `BasicConstraints` field with `CA` key set to `true`, and its `KeyUsage` extension field set with `keyCertSign` and/or `cRLSign`.

CA certificates are used to:

- Validate the certificates that remote peers present when attempting to establish a secure connection with a service on the switch, for example, the SSH server.
- Validate leaf certificates installed on the switch that are used, for example, by the syslog client, the Web UI, or REST API.

The TA profile also enables configuration of real-time checking of certificate revocation (through OCSP).

Leaf certificates

Leaf certificates can be installed on the switch for use by features such as the syslog client, the Web UI, or REST API. If you are purchasing a certificate from a trusted CA, the switch can generate the certificate signing request (CSR) that is used to obtain the certificate. The switch can also directly generate self-signed certificates. Alternatively, the certificate and private key can be generated outside the switch and then imported. X509 certificate management software such as OpenSSL can be used to generate the private key and CSR and then combine the certificate and private key into one PEM or PKCS#12 file suitable for importation into the switch.

Mandatory matching of peer device hostname

While validating the peer device certificates, the switch checks that the peer device configured hostname matches either the Subject Alternative Name (SAN) field or the Common Name (CN) within the certificate Subject field. If the SAN field is present and matches the hostname, validation succeeds, otherwise it fails. If the SAN field is not present, and the CN matches the hostname, validation succeeds, otherwise it fails.

PKI EST

EST (Enrollment over Secure Transport) (RFC 7030) defines the protocol that devices use to request trusted certificate authority (CA) certificates and to enroll / re-enroll device certificates from CA services using secure channels, specifically HTTP over TLS.

Devices can be configured to request the trusted CA certificates and to request enrollment, and re-enrollment of device certificates automatically, without the need for administrator intervention, while maintaining the security and integrity of the whole enrollment process.

The switch includes an EST client implemented as a part of the PKI infrastructure.



NOTE: For detailed CLI command descriptions see:

- [PKI commands](#)
- [PKI EST commands](#)

EST usage overview

- The EST client on the switch requires EST profile configuration, including EST server URL and the VRF providing HTTP connection to the EST server.
- At the time the URL is set in the EST profile, the switch connects to the EST server and downloads the trusted CA certificate chain. To accommodate CA certificate updates, the certificate chain is also downloaded before a certificate enrollment or re-enrollment is attempted.
- EST supports up to:
 - 16 EST profiles
 - 63 trusted CA certificates downloaded from EST servers.
 - 18 device certificates enrolled through EST services.

- EST profile configuration is supported through the CLI and the REST API `PKI_EST_Profile`.
- CA certificate request and device certificate enrollment is supported through the CLI and the REST custom API `CertificateManager /certificate`.

Prerequisites for using EST for certificate enrollment

- Establish the PKI infrastructure for your organization, with the CA chain and service ready to issue certificates. Issue a service certificate for the EST server.
- Install the root CA certificate in a TA profile on the switch that will validate the EST server certificate using CLI commands `crypto pki ta-profile` and `ta-certificate`.
- Optionally, preconfigure an EST client certificate on the switch.
- Make the EST server reachable from the switch. Connect the CA service(s) to the EST server. If there is a client certificate for the EST client, install the root CA certificate on the server that will validate the client certificate.

EST profile configuration

In the global configuration context, create an EST profile and enter its context:

```
crypto pki est-profile <EST-NAME>
```

In an EST profile context, configure the EST profile parameters using these commands:

```
url <URL>
vrf <VRF-NAME>
username <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> |
                                plaintext <PLAINTEXT-PASSWORD>]
retry-interval <INTERVAL>
retry-count <RETRIES>
arbitrary-label <LABEL>
arbitrary-label-enrollment <LABEL>
arbitrary-label-reenrollment <LABEL>
reenrollment-lead-time <LEAD-TIME>
```

Certificate enrollment

In the global configuration context, create a certificate and enter its context:

```
crypto pki certificate <CERT-NAME>
```

In a certificate configuration context, configure the certificate parameters:

```
key-type {rsa [key-size <K-SIZE>] | ecdsa [curve-size <C-SIZE>]}
subject [common-name <COMMON-NAME>]
        [country <COUNTRY>]
        [locality <LOCALITY>]
        [org <ORG-NAME>]
        [org-unit <ORG-UNIT>]
        [state <STATE>]
```

In a certificate configuration context, enroll the certificate using an EST service:

```
enroll est-profile <EST-NAME>
```

Certificate re-enrollment

- The re-enrollment request is sent automatically to the same EST server that was used for the original enrollment.
- The switch presents the enrolled certificate being re-enrolled to the EST server for authentication. If the certificate has expired or authentication fails for any reason, the switch falls back to using the EST client certificate or the username and password in the EST profile, whichever is configured, and performs a new certificate enrollment.
- Re-enrollment lead-time is configurable in the EST profile using CLI command `reenrollment-lead-time`. It sets the number of days before certificate expiry date that certificate re-enrollment will be initiated.

Checking EST profile and certificate configuration

Show the list of EST profiles or details of a specific EST profile:

```
show crypto pki est-profile [<EST-NAME>]
```

Show a list of TA profiles whether directly configured or EST-enrolled, or details of a specific TA profile:

```
show crypto pki ta-profile [<TA-NAME>]
```

Show the list of certificates whether directly configured or EST-enrolled, or details of a specific certificate:

```
show crypto pki certificate [<CERT-NAME> [plaintext | pem]]
```

Show all certificates assigned to the switch EST client as well as certificates that are assigned to other applications on the switch.:

```
show crypto pki application
```

EST best practices

Ensure the following:

- A time synchronization service is used on both the switch (the EST client) and the EST server.
- In all CA certificates, the `Basic Constraints` field has `CA` set to `true`, `pathlen` is set appropriately, and `Key Usage` is set with `keyCertSign`.
- In all leaf certificates, the `Extended Key Usage` field is set with the appropriate purpose as follows:
 - For server certificates, set with `serverAuth`. The `Key Usage` field has at least one of `digitalSignature`, `keyEncipherment`, or `keyAgreement`.
 - For client certificates, set with `clientAuth`. The `Key Usage` field has at least one of `digitalSignature`, or `keyAgreement`.
- The EST server is configured to include the intermediate issuer CA certificates in the trusted CA certificate chain that the EST server sends to the switch (the EST client) upon request.

Example using EST for certificate enrollment

This example illustrates the configuration of an EST profile and enrolling application certificates using an EST server.

Prerequisites:

- An EST server is reachable from the switch management port.
- Availability of the root CA certificate used to validate the server certificate.

This example shows the following:

- Installing the root CA certificate as a TA profile for validation of the EST server certificate.
- Configuring an EST profile with the EST server information, including the username and password for client authentication and the EST server URL.
- Issuing a request to enroll a leaf certificate using the EST server.
- Assigning the enrolled certificate to the EST client and syslog client on the switch.

Each section in the below example is preceded by descriptive text.

Example

=====

The switch in its default configuration state.

=====

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX FL.10.06.0001CM
!export-password: default
user admin group administrators password ciphertext AQBapTLgcT+DNrtd0bmdXIP2L0AY
NUPwwyQEIZX4oMKtwlXcYgAAAOmKlfxH+ugf3Fe2JuWar2uKG7A/R6bqMO/ZHS364NompXV/Ko37ZhCq
cFpaOJsk01+IJPRUkbpigCeEObM67Od8/vrASJaO6EAj+RBnWCrifwdChcUUS3XpbCUl7dmxYHNg
!
!
ssh server vrf default
ssh server vrf mgmt
vsf member 1
    type jl668a
vlan 1
spanning-tree
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    no routing
    vlan access 1
interface 1/1/2
    no shutdown
    no routing
    vlan access 1
interface 1/1/3
    no shutdown
    no routing
    vlan access 1
...
interface 1/1/26
    no shutdown
    no routing
    vlan access 1
interface 1/1/27
    no shutdown
    no routing
    vlan access 1
interface 1/1/28
```

```

no shutdown
no routing
vlan access 1
interface vlan 1
ip dhcp
!
!
https-server vrf default
https-server vrf mgmt
switch#

```

=====

The mgmt port is connected to a network with DNS available and the EST server reachable.

=====

```

switch# show interface mgmt
Address Mode           : dhcp
Admin State            : up
Mac Address             : 38:21:c7:59:cd:81
IPv4 address/subnet-mask : 999.100.205.146/24
Default gateway IPv4    : 999.100.205.1
IPv6 address/prefix     :
IPv6 link local address/prefix: fe80::3a21:c7ff:fe59:cd81/64
Default gateway IPv6    :
Primary Nameserver      :
Secondary Nameserver    :
switch#

```

=====

Configure the root CA cert as a TA profile that will validate the server cert.

=====

```

switch# config
switch(config)#
switch(config)# crypto pki ta-profile root-ca-for-est-server
switch(config-ta-root-ca-for-est-server)#
switch(config-ta-root-ca-for-est-server)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
NVBAYTonfig-ta-cert)# MIIB2DCCAX6gAwIBAgIJAKtmJvZZy9RdMAoGCCqGSM49BAMCMGIxCzAJBg
QKEwNIonfig-ta-cert)# AlVTMQswCQYDVQQLIEwJDQTESMBAGA1UEBxMJUm9zZXZpbGxlMQwwCgYDVQ
0yMDAlonfig-ta-cert)# UEUxDjAMBgNVBAsTBUFydwJhMRQwEgYDVQQDEwtkYW5lc3Qtcml9vdDAeFw
...
YDVR0Ponfig-ta-cert)# VCnKtlhxfmV72nfxYpI979UsopuP5nCjHTAbMAwGA1UdEwQFMAMBAf8wCw
eo6yN0onfig-ta-cert)# BAQDAgEGMAoGCCqGSM49BAMCA0gAMEUCIQDb/uHvU8DFRTyfnP9wkli6sd
c=00 (config-ta-cert)# UvUO5t7/rrVxRQIgMHGjHhaNlnkjYBG8Ei3C1UDILiKlO7McMTCWVo4Ik5
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Subject: C = US, ST = CA, L = Roseville, O = HPE, OU = Aruba, CN = danest-root
Issuer: C = US, ST = CA, L = Roseville, O = HPE, OU = Aruba, CN = danest-root
Serial Number: 0xAB6626FXXXXD45D
TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
switch(config-ta-root-ca-for-est-server)#
switch(config-ta-root-ca-for-est-server)# exit
switch(config)#
switch(config)# show crypto pki ta-profile

```

TA Profile Name	TA Certificate	Revocation Check
root-ca-for-est-server	Installed, valid	disabled

```
switch(config)#
```

```
=====
Configure the EST profile with the EST server URL, username/password.
=====
```

```
switch(config)# crypto pki est-profile test-est-server
switch(config-est-test-est-server)#
switch(config-est-test-est-server)# user fred password plaintext barney
switch(config-est-test-est-server)#
switch(config-est-test-est-server)# url https://999.0.10.229:8443/.well-known/est
switch(config-est-test-est-server)#
switch(config-est-test-est-server)# exit
switch(config)#
```

```
=====
At the time the EST URL is set, the switch sends a request to the EST server to
get the set of trusted CA certs. If that is successful, TA profiles will be
auto-created for those CA certs.
=====
```

```
Display the list of TA profiles and EST profile details.
=====
```

```
switch(config)# show crypto pki ta-profile
```

TA Profile Name	TA Certificate	Revocation Check
test-est-server-est-ta00	Installed, valid	OCSP
test-est-server-est-ta02	Installed, valid	OCSP
test-est-server-est-ta05	Installed, valid	OCSP
test-est-server-est-ta01	Installed, valid	OCSP
root-ca-for-est-server	Installed, valid	disabled
test-est-server-est-ta04	Installed, valid	OCSP
test-est-server-est-ta03	Installed, valid	OCSP

```
switch(config)# show crypto pki est-profile
```

Profile Name	Downloaded TA Profiles	Enrolled Certificates
test-est-server	6	1

```
switch(config)# show crypto pki est-profile test-est-server
```

```
Profile Name      : test-est-server
Service VRF       : mgmt
Service URL       : https://999.0.10.229:8443/.well-known/est
Arbitrary Label   : not configured
Arbitrary Label Enrollment : not configured
Arbitrary Label Reenrollment : not configured
Authentication Username : fred
Authentication Password :
  AQBapR7ndgoxkMlWQUQvK+Dvd3S6m+s9fdaPQwdkMbIYEMnMBgAAHRhhlYwA==
Retry Interval    : 30 seconds
Retry Count       : 3 times
Reenrollment Lead Time : 2 days
Downloaded TA Profiles : 6
Enrolled Certificates :
  cert-for-app
switch(config)#
```

```
=====
Originally, the switch only has two built-in certificates.
=====
```

```
switch(config)# show crypto pki certificate
```

Certificate Name	Cert Status	EST Status	Associated Applications
local-cert	installed	n/a	est-client, https-server, syslog-client
device-identity	installed	n/a	none

Create a new certificate, configure its key type, key size, and subject fields.

```
switch(config)# crypto pki certificate cert-for-app
switch(config-cert-cert-for-app)#
switch(config-cert-cert-for-app)# key-type ecdsa curve-size 521
switch(config-cert-cert-for-app)# subject
Do you want to use the switch serial number as the common name (y/n)? n
Common Name: 999.100.205.146
Org Unit: Aruba-Roseville
Org Name: HPE
Locality: Roseville
State: CA
Country: US
switch(config-cert-cert-for-app)#
```

Request to enroll the certificate through the EST server.

```
switch(config-cert-cert-for-app)# enroll est-profile test-est-server
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Roseville, OU=Aruba-Roseville, O=HPE,
        CN=999.100.205.146
Key Type: ECDSA (521)

Continue (y/n)? y
Certificate enrollment via test-est-server has been initiated. Please use
'show crypto pki certificate cert-for-app' to check its status.
switch(config-cert-cert-for-app)#
```

Check the cert status to see if enrollment is successful. It is.

```
switch(config-cert-cert-for-app)# show crypto pki certificate
```

Certificate Name	Cert Status	EST Status	Associated Applications
local-cert	installed	n/a	est-client, https-server, syslog-client
device-identity	installed	n/a	none
cert-for-app	installed	enroll success	none

```
switch(config-cert-cert-for-app)#
switch(config-cert-cert-for-app)# exit
switch(config)#
switch(config)# show crypto pki certificate cert-for-app pem
Certificate Name: cert-for-app
```



```

Associated Applications:
  est-client
Certificate Status: installed
EST Status: enroll success
Certificate Type: regular
Intermediates:
  Subject: C = US, ST = CA, O = HPE, OU = Aruba, CN = danest-int2
  Issuer: C = US, ST = CA, O = HPE, OU = Aruba, CN = danest-int1
  Serial Number: 0x02
  Subject: C = US, ST = CA, O = HPE, OU = Aruba, CN = danest-int1
  Issuer: C = US, ST = CA, L = Roseville, O = HPE, OU = Aruba, CN = danest-root
  Serial Number: 0x01
  Subject: C = US, ST = CA, L = Roseville, O = HPE, OU = Aruba, CN = danest-root
  Issuer: C = US, ST = CA, L = Roseville, O = HPE, OU = Aruba, CN = danest-root
  Serial Number: 0xAB6626FXXXXD45D
-----BEGIN CERTIFICATE-----
MIICizCCAjKgAwIBAgICAIGwCQYHKoZiZjOEATBOMQswCQYDVQGEwJVUzELMAkG
A1UECBMCQ0ExDDAKBgNVBAoTA0hQRTEOMAwGA1UECxmFQXJlYmExFDASBgNVBAMT
C2RhbmVzdC1pbmQyMB4XDTIwMTAyODE5NTczOVVoXDTIwMTEyNTE5NTczOVowbzEL
...
RTEOMAwGA1UECxmFQXJlYmExFDASBgNVBAMTC2RhbmVzdC1pbmQxggECMAkGByqG
SM49BAEDSAAwRQIgVC1kVieWxhpBSQVqVsQ36MbZrhR4XsaGbQeu7+08gbUCIQCH
cS17gcLbNxJlWVr2jnZpPBxy9vID38FjirJiGZ5cZw==
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
MIIBPzCCAUA2gAwIBAgIBAJBgqhkJOPQQBME4xCzAJBgNVBAYTA1VTMQswCQYD
VQQIEwJDQTEEMMAoGA1UEChMDSFBFMQ4wDAYDVQQLEwVBcnViYTEUMBIGA1UEAxML
ZGFuZlZlLWludDEwHhcNMjAwNTIwMDUyNDE5NTczOVVoXDTIwMTEyNTE5NTczOVowbzEL
...
7ovbXodgN8lqDvB1lVTJYlLBSzl9FKMdMBswDAYDVR0TBAUwAwEB/zALBgNVHQ8E
BAMCAQYwCQYHKoZiZjOEANJADBGAIeA+i3x7KEZsxObVruM1kwqWe+QXiLKbgNL
fL077jsSMhYCIQD/dFBkH/yN0NFzb3wi7Oaoo083HY2p/47t2pIBk/JNfg==
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
MIIBuTCCAWGgAwIBAgIBATAJBgcqhkJOPQQBMIxGzAJBgNVBAYTA1VTMQswCQYD
VQQIEwJDQTESMBAGA1UEBxMjU09zZXZpbGx1MQwwCgYDVQQKEwNIUEUxDjAMBGNV
BASTBUFYdWJhMRQwEgYDVQQDEWtkYW5lc3Qtcm9vdDAeFw0yMDA1MjAwNTIwMDUyNDE5
...
BgNVHRMEBTADAQH/MASGA1UdDwQEAwIBBjAJBgqhkJOPQQBA0cAMEQCIGrlZmBX
SmbhDvG9pRiXG0YmQVbvZd37jRQdE+mEk2jFAiBFghzMjUadhQbuPUTNs9A7bdYk
wej0mJe5bRpd7sqwRQ==
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
MIIB2DCCAX6gAwIBAgIJAKtmJvZzy9RdMAoGCCqGSM49BAMCMGIXCzAJBgNVBAYT
A1VTMQswCQYDVQQIEwJDQTESMBAGA1UEBxMjU09zZXZpbGx1MQwwCgYDVQQKEwNI
UEUxDjAMBGNVBAsTBUFFYdWJhMRQwEgYDVQQDEWtkYW5lc3Qtcm9vdDAeFw0yMDA1
...
VCnKtlhxfmV72nfxYpI979UsopuP5nCjHTAbMAwGA1UdEwQFMAMBAf8wCwYDVROp
BAQDAgEGMAoGCCqGSM49BAMCA0gAMEUCIQDb/uHvU8DFRTyfnP9wkli6sdeo6yN0
UvU05t7/rrVxRQIgMHGjHhaN1nkjYBG8Ei3C1UDILiKl07McMTCWVo4Ik5c=
-----END CERTIFICATE-----
switch(config)#

```

```

=====
Initially, all applications use the default local-cert.
=====

```

```
switch(config)# show crypto pki application
```

Associated Applications	Certificate Name	Cert Status
est-client		not configured, using local-cert
https-server		not configured, using local-cert

```
syslog-client                               not configured, using local-cert
switch(config)#
```

```
=====
Assign the newly enrolled cert to applications as desired.
In this example, the cert is assigned to the est-client and syslog.
=====
```

```
switch(config)# crypto pki application est-client certificate cert-for-app
switch(config)# crypto pki application syslog-client certificate cert-for-app
switch(config)#
switch(config)# show crypto pki application
```

Associated Applications	Certificate Name	Cert Status
est-client	cert-for-app	valid
https-server		not configured, using local-cert
syslog-client	cert-for-app	valid

```
switch(config)#
```

Example including the use of an intermediate certificate

This example shows the following:

- Installing a root CA as a TA profile.
- Creating a CSR for a leaf certificate.
- Installing the signed leaf certificate issued by an intermediate CA. The intermediate CA certificate is included after the signed leaf certificate.

Each section in the below example is preceded by descriptive text.

Example

```
=====
Install root CA as a TA profile
=====
```

```
switch(config)# crypto pki ta-profile root
switch(config-ta-root)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIGATCCA+mgAwIBAgIJAL/JIZfJ0GpcMA0GCSqGSIUAMIGOMQswCQYD
switch(config-ta-cert)# VQQGEwJVUzETMBEGA1UECAwKQ2FsaWZvcn5pYTESBwwJUm9zZXZpbGx1
switch(config-ta-cert)# MQwwCgYDVQQKDANIUEUxEzARBgNVBASMCk5ldmcxFTATBgNVBAMMDFRl
...
switch(config-ta-cert)# rvadRXSAsUlevJRNN0yINrEJyOfUX2hAfLaiBYP+In6gKTAwVhlxLiXn
switch(config-ta-cert)# LlryAb2/go4BTYjil3eJyXxweUHheuBeesEslBawLv0cPCQPTTdbc970
switch(config-ta-cert)# iWbyAmfSpD/TS3AgCLnBFPKEKsms0f0LF3/C9dRUXjIHT/LDBr+lgzY3
switch(config-ta-cert)# m2NCvxY=
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Subject: C = US, ST = California, L = Roseville, O = HPE, OU = Networking,
        CN = Test CA root, emailAddress = generic@corp.com
Issuer:  C = US, ST = California, L = Roseville, O = HPE, OU = Networking,
```

```

CN =Test CA root, emailAddress = generic@corp.com
Serial Number: 0xBFC92197xxxxxxxx
TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
switch(config-ta-root)# exit

```

Create a CSR for a leaf certificate

```

switch(config)# crypto pki certificate leaf
switch(config-cert-leaf)# subject
Do you want to use the switch serial number as the common name (y/n)? y
Common Name: SG9Zxxxxxx
Org Unit:
Org Name:
Locality:
State:
Country:
switch(config-cert-leaf)# enroll terminal
You are enrolling a certificate with the following attributes:
Subject: C=<empty>, ST=<empty>, L=<empty>, OU=<empty>, O=<empty>,
        CN=SG9Zxxxxxx
Key Type: RSA (2048)

Continue (y/n)? y

```

```

-----BEGIN CERTIFICATE REQUEST-----
MIICWjCCAUICAQIwFTETMBEGA1UEAwKU0c5WktONDawSoZiHvcN
AQEBBQADggEPADCCAQoCggEBAMKdtoucDEMeuZjPGvCcWTm4D39A
WBA8K/bduJvM1E2B/uirU2TX7mF6lN30akClSxZOoofZAmBPCzI3
...
wZtb5c8fYCSR+TpLwZAdoXrvGJqJ1aGzV6/kVfb7rM6ulBksfBo/
JwO+7x8Vn5hldGCrsl9CPJienni/fq24+1CJzspMbY9BKu9EIL+P
5ND9BmN0IzEmDO26F+Ip74DqFCIYjXt13uPJk4cwJkXq121hlcrG
UlatpvjNEpZOtfoEryDJSs0pHXky7VjltYABIuDy
-----END CERTIFICATE REQUEST-----

```

Install the signed leaf certificate issued by an intermediate CA. The intermediate CA certificate is included after the signed leaf certificate.

```

switch(config-cert-leaf)# import terminal ta-profile root
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIKTCCAhGgAwIBAgIJA01LSoBmKxtbMA0GCSqGSIYxCzAJBgNV
switch(config-cert-import)# BAYTAkFVMRUwEwYDVQQIDAxJbnRlcm1lZG9BBAoMGEIudGVybmV0
switch(config-cert-import)# IFdpZGdpdHMgUHR5IEEx0ZDENMAsGA1UEAw0yMDA1MTQyMDI3MTla
...
switch(config-cert-import)# axnZcIaNp4eNi95in+TvckXA0eMLScNyR7IF+Wjn56H0fQKYsHp/
switch(config-cert-import)# jllbCkyB1xKnn6IpzIj/hvAx3NpA0jXx/qJA+V/cltaAL6+QPZmI
switch(config-cert-import)# vr5GZsoV72BHFOXxoteZlmWMUdVldYXXP2DzEUbttr9zojwz0MyK
switch(config-cert-import)# Qz5tc0BlGfJAAtghykw==
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIIFyzCCA7OgAwIBAgIJA01LSoBmKxtwMA0GCSqGCIGOMQswCQYD
switch(config-cert-import)# VQQGEwJVUzETMBEGA1UECAwKQ2FsaWZvc1UEBwwJUUm9zZXZpbGxl
switch(config-cert-import)# MQwwCgYDVQQKDANIUEUxEzARBgNVBAsMCmcxFTATBgNVBAMMDFRl
...
switch(config-cert-import)# LM9DV3YNWOM4UMMP2HXaDDfqxZPX9Zsj6G1/stRCh8SVfsF2duYR
switch(config-cert-import)# 5brLfEpiDhXrZVXxF91ljRABO2JPLSUufg7xr6M/K5aCuJxvYzK7

```

```
switch(config-cert-import)# DQaCEw5NlmC1vpY1Y2TG3dlUQPZDeQOAHwuBd4HewqDHWfp/T04=  
switch(config-cert-import)# -----END CERTIFICATE-----  
switch(config-cert-import)#  
Leaf certificate is validated with root and imported successfully.  
switch(config-cert-leaf)#
```

Installing a self-signed leaf certificate (created inside the switch)

This procedure describes how to create (wholly inside the switch) and install a self-signed X.509 leaf certificate. And associate it with one of the following switch features: syslog client, HTTPS server, or HSC (hardware switch controller).

Procedure

1. Create a leaf certificate context with the command `crypto pki certificate`. This switches to the leaf certificate configuration context.
2. Define leaf certificate properties with the command `subject`.
3. Set the encryption key type for the leaf certificate with the command `key-type`.
4. Generate and install the self-signed certificate with the command `enroll self-signed`.
5. Exit the leaf certificate context with the command `exit`.
6. Associate the leaf certificate with a switch feature (syslog client, HTTPS server, or HSC) with the command `crypto pki application`.

Example

This example:

- Creates the leaf certificate context.
- Defines the leaf certificate characteristics.
- Creates and installs the self-signed leaf certificate.
- Associates the leaf certificate with the syslog client (application) on the switch.

```
switch(config)# crypto pki cert SS_LC  
8400X(config-cert-SS_LC)# subject common-name SSLeaf country US  
state CA locality Rocklin org Company org-unit Site  
8400X(config-cert-SS_LC)# key-type rsa key-size 3072  
8400X(config-cert-SS_LC)# enroll self-signed  
You are enrolling a certificate with the following attributes:  
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Company,  
         CN=SSLeaf  
Key Type: RSA (3072)  
  
Continue (y/n)? y  
Self-signed certificate is created and enrolled successfully.  
8400X(config-cert-SS_LC)# exit  
switch(config)# crypto pki application syslog-client certificate SS_LC
```

Installing a self-signed leaf certificate (created outside the switch)

This procedure describes how to install a self-signed X.509 leaf certificate (that was created outside the switch). And then associate the certificate with one of the following switch features: syslog client, HTTPS server, or HSC (hardware switch controller).

Prerequisites

A self-signed leaf certificate (including private-key data) created outside the switch.

Procedure

1. Create the leaf certificate context with the command `crypto pki certificate` which then switches to the created leaf certificate context.
2. Import the leaf certificate data into the switch with the command `import (self-signed leaf certificate)`.
3. Exit the leaf certificate context with the command `exit`.
4. Associate the leaf certificate with a switch feature (syslog client, HTTPS server, or HSC) with the command `crypto pki application`.

Example

This example:

- Creates the leaf certificate context.
- Imports the self-signed leaf certificate.
- Associates the leaf certificate with the syslog client (application) on the switch.

```
switch(config)# switch(config)# crypto pki certificate SS_LC2
switch(config)# switch(config-cert-SS_LC)# import terminal self-signed
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIIFRDCCAYygAwIBAgIQP8nnS2Vp15u07xXMdktDJzANBgkqhkiG9
switch(config-cert-import)# MQswCQYDVQGEwJVUEOMAwGA1UECgwFXJlYmxDAOGNBAMMB1Jvb3gw
switch(config-cert-import)# HhcNMTkNDEwMjIwNT1WhcjIwMTA0MjIwNE1WjBzQswQYDVQGEwJV
...
switch(config-cert-import)# 1fIYZYQYla0AwFuPTTxBXHYwRxTPbUYU5tumJrfrwRPmE4OVY8S9D
switch(config-cert-import)# 1NGNm3NG03GqPScs/TF9bVyFA5BOrS5lmm7kNfRY1K8D/kMTfRreS
switch(config-cert-import)# YQ1ulNqShps=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIIFDjBABgkqhkiG9wBBQ0wMzAbBgkqhkiw0QwwDQImNpJMN7sVGwC
switch(config-cert-import)# MBQGCCqGSIB3DQMHAit+2qadNAASCMg5LYJ4AFm3EffhH5p51Ggr8
switch(config-cert-import)# IJ6L/UhEtH523nUkdV6gvoAWgoYaeD83PeswToAGv5VS8OMFTPttr
...
switch(config-cert-import)# OgSecqZsG6arbx0ESaYBirlc/6rPslpcjbDxw283DiD1MWOpes2a
switch(config-cert-import)# iKnXnUMpVPfLc74ty2S41DtH0X9Sgf6aa1LjiStg+N7cND9XfGtj/
switch(config-cert-import)# cb4=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
```

```
switch(config-cert-SS_LC2)# exit
switch(config)# crypto pki application syslog-client certificate SS_LC2
```

Installing a certificate of a root CA

Prerequisites

- A certificate of a root CA (that is used as the signer).
- Revocation checking URLs for the CA (optional).

Procedure

1. Create a TA profile with the command `crypto pki ta-profile` which then switches to the created TA profile context.

2.  **NOTE:** This step is optional and suggested only for advanced users.

Optionally enable certificate revocation checking with the command `revocation-check ocsp`. Most certificates contain revocation checking URLs for OCSP. If you want to override these URLs, configure custom revocation checking URLs with the command `ocsp url`.

3. Import the certificate of the root CA with the command `ta-certificate`.

Example

This example installs the certificate `root-cert` and defines custom revocation checking URLs:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# revocation-check ocsp
switch(config-ta-root-cert)# ocsp url primary http://ocsp-server.site.com
switch(config-ta-root-cert)# ocsp url secondary http://ocsp-server2.site.com
switch(config-ta-root-cert)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIDuTCCAqECCQCuoxeJ2ZNYcjANBgkqhkiG9w0BAQsFADCBqzELMAEBh
switch(config-ta-cert)# VVMxEzARBgNVBAGMCkNhbGlm3JuaWExEDAOBgNVBACMB1JvY2tsDAKBg
switch(config-ta-cert)# BAoMA0hQTjEVMBMGA1UECwwMSFBOUm9zZXZpbGx1MSowKAYDVQocG5zd
...
switch(config-ta-cert)# x3Wff3dFZ8o9sd5LVAHneH/ztb9MP34z+le1V346r12L2kpxmTOVJVyTO
switch(config-ta-cert)# BIzD/ST/HaWI+0S+S80rm93PSscEbb9GWk7vshh5EnW/moehBKcE401zy
switch(config-ta-cert)# 3LvMLZcssSe5J2Ca2XIhfDme8UaNZ7syGYMsAW0nG7yYHWkEOQu9s
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Issuer: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
        CN=8400/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert)#
```

Installing a CA-signed leaf certificate (initiated in the switch)

This procedure describes how to create and install an X.509 leaf certificate that is initiated inside the switch but signed outside the switch by a CA. And then associate the certificate with one of the following switch features: syslog client, HTTPS server, or HSC (hardware switch controller).

Prerequisites

Root CA certificate `root-cert` installed as described in [Installing a certificate of a root CA](#).

Procedure

1. Create a leaf certificate context with the command `crypto pki certificate` which then switches to the created leaf certificate configuration context.
2. Define leaf certificate properties with the command `subject`.
3. Set the encryption key type for the leaf certificate with the command `key-type`.
4. Generate the certificate signing request (CSR) with the command `enroll terminal`.
5. Use the CSR to obtain a leaf certificate from the root CA, using the root CA directly as the signer CA.
6. Import the leaf certificate into the switch with the command `import (CA-signed leaf certificate) .`
7. Exit the leaf certificate context with the command `exit`.
8. Associate the leaf certificate with a switch feature (syslog client, HTTPS server, or HSC) with the command `crypto pki application`.

Example

This example:

- Creates the leaf certificate context.
- Defines the leaf certificate characteristics.
- Generates the leaf certificate signing request in the switch for getting signed outside the switch by a CA.
- Imports the CA-signed leaf certificate into the switch.
- Associates the leaf certificate with the syslog client (application) on the switch.

```
switch(config)# crypto pki certificate lcert
switch(config-cert-lcert)# subject common-name Leaf country US state CA
locality Rocklin org Company org-unit Site
switch(config-cert-lcert)# key-type rsa key-size 3072
switch(config-cert-lcert)# enroll terminal
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site
        CN=Leaf
Key Type: RSA (2048)

Continue (y/n)? y

-----BEGIN CERTIFICATE REQUEST-----
```

```

MIIBozCCAQwCAQAwYzEVMBMGA1UEAxMMcG9kMDEtODQwMC0xMQ4wDAYDV
nViYTEMMAoGA1UEChMDSFBFMRIwEAYDVQQHEw1Sb3NldmlsbGUxCzAJBg
NBMQswCQYDVQQGEwJVUzCBnzANBgkqhkiG9w0BAQEFAAOBjQAwgYkCgYE
...
GBAJ4L3lFFfWBEL+KAKpOGjZcVmw1BMqSKftOFNF9nzmUmONmU3SKy6dz
7Au22mf3lWDxzrtCC/dj5RtWJeJekxp2LCIK/3eRXUwbYveQDKcxH7j9Z
ace+2tA68F2vlgRCQ/hcQH0YmNuaq4Ne3w0dhm7HlUrx
-----END CERTIFICATE REQUEST-----

```

```

switch(config-cert-lcert)# import terminal ta-profile root-cert
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIFRDCCAYygwIBAgIQPnnS2Vp5u07XMdktDJzANBgkqhkiG9w0Bv
switch(config-cert-import)# MQswCQYDVQGEwJVEOMAwG1UECgwFJ1YmxDAOGNBMMB1Jvb3QgQ0Ew
switch(config-cert-import)# HhcNMTkNDEwMjIwNTWcjIwMTA0MjwNE1WBzQswCQYDVQGEwJVUzEL
...
switch(config-cert-import)# 1fIYZYGQyla0AwFuTTxBXYwRXPbUYU5tumrfwRPmE4OVY8S9DQgcr
switch(config-cert-import)# 1NGNm3NG03GqPcs/T9bVyF5BOrS5lmm7kNfRYl8D/kMTfRreSdxis
switch(config-cert-import)# YQ1ulNqShps=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)#
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-lcert)# exit
switch(config)# crypto pki application syslog-client certificate lcert

```

Installing a CA-signed leaf certificate (created outside the switch)

This procedure describes how to install an X.509 leaf certificate that was created and signed (by a CA) outside the switch. And then associate the certificate with one of the following switch features: syslog client, HTTPS server, or HSC (hardware switch controller).

Prerequisites

- Root CA certificate `root-cert` installed as described in [Installing a certificate of a root CA](#).
- A CA-signed leaf certificate (including private-key data) created outside the switch.

Procedure

1. Create the leaf certificate context with the command `crypto pki certificate` which then switches to the created leaf certificate context.
2. Import the leaf certificate into the switch with the command `import (CA-signed leaf certificate)`.
3. Exit the leaf certificate context with the command `exit`.
4. Associate the leaf certificate with a switch feature (syslog client, HTTPS server, or HSC) with the command `crypto pki application`.

Example

This example:

- Creates the leaf certificate context.
- imports the CA-signed leaf certificate.
- Associates the leaf certificate with the syslog client (application) on the switch.

```
switch(config)# switch(config)# crypto pki certificate CA_LC
switch(config)# switch(config-cert-CA_LC)# import terminal ta-profile root-cert
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIFRDCCAYygAwIBAgIQP8nn2Vp15u07XMktDJANBgkqhkiG9w0Bv
switch(config-cert-import)# MQswCQYDVQGEwJVUEOMAw1UECgwFX1YmxDOgNBAMMB1Jvb3QgQ0Ew
switch(config-cert-import)# HhcNMTkNDEwMjIwNT1WhjIMTA0MjIwNE1jBzQswYDVQQGEwJVUzEL
...
switch(config-cert-import)# 1fIYZYGQyla0AwFuPTTxBXHYRxTPbUYUtmJrwrPmE4OVY8S9DQgcr
switch(config-cert-import)# 1NGNm3NG03GqPScs/TF9bVyFABOrlmm7kNfRlK8D/kMTfRreSdxis
switch(config-cert-import)# YQ1ulNqShps=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIFDjBAGkqhkiG9wBBQ0wMzAbBgkiwQwwQImNpJMN7sVGwCAgga
switch(config-cert-import)# MBQGCCqGS1b3DQMHAit+2qadNAASCMgLYJ4AFEfhH5p51Ggr86VqS
switch(config-cert-import)# IJ6L/UhEtH523nUkdV6gvoAWgoYaeD8eswAGv5VS8OMFTPttrn5/K
...
switch(config-cert-import)# OgSecqZsG6arbx0ESaYBirlc6rPs1pcbDx283DD1MW0peoS2aEmOX
switch(config-cert-import)# iKnXnUMpVPfLc74ty2S41tH0X9gfaaLiStg+N7cND9XfGtjaV2+/
switch(config-cert-import)# cb4=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-CA_LC)# exit
switch(config)# crypto pki application syslog-client certificate CA_LC
```

PKI commands



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

crypto pki application

Syntax

```
crypto pki application <APP-NAME> certificate <CERT-NAME>
```

```
no crypto pki application <APP-NAME> certificate
```

Description

Associates a leaf certificate with a feature (application) on the switch. By default, all features are associated with the default, self-signed certificate `local-cert`. This certificate is created by the switch the first time it starts.

The `no` form of this command associates the specified feature with the default certificate.

Command context

```
config
```

Parameters

<APP-NAME>

Specifies the name of a feature on the switch:

- `est-client`: EST client
- `hsc`: Hardware switch controller.
- `https-server`: HTTPS server.
- `syslog-client`: Syslog client.

<CERT-NAME>

Specifies the name of an installed leaf certificate.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the EST client with leaf certificate `leaf-cert1`:

```
switch(config)# crypto pki application est-client certificate leaf-cert1
```

Associating the syslog client with leaf certificate `leaf-cert`:

```
switch(config)# crypto pki application syslog-client certificate leaf-cert
```



NOTE: `syslog-client` communicates with syslog server over TLS.

You can associate a certificate with the `syslog-client` application by enrolling the certificate manually or through EST.

Setting the syslog client to use the default certificate:

```
switch(config)# no crypto pki application syslog-client certificate
```

Associating the HTTPS server with leaf certificate `leaf-cert2`:

```
switch(config)# crypto pki application https-server certificate leaf-cert2
```

crypto pki certificate

Syntax

```
crypto pki certificate <CERT-NAME>
```

```
no crypto pki certificate <CERT-NAME>
```

Description

Creates a leaf certificate and changes to its context `config-cert-<CERT-NAME>`. If the specified leaf certificate exists, this command changes to its context.

The first time the switch starts it creates a self-signed, default leaf certificate called `local-cert`. This certificate is used by any switch application that does not have an associated leaf certificate.

The `no` form of this command deletes the specified leaf certificate. The default leaf certificate `local-cert` cannot be deleted.

Command context

config

Parameters

<CERT-NAME>

Specifies the name of a leaf certificate. Range: 1 to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating leaf certificate **leaf-cert**:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)#
```

Deleting leaf certificate **leaf-cert**:

```
switch(config)# no crypto pki certificate leaf-cert
The leaf certificate has associated applications. Deleting the certificate
will make the applications use the default certificate local-cert.
Continue (y/n)? y
switch(config)#
```

crypto pki ta-profile

Syntax

crypto pki ta-profile <TA-NAME>

no crypto pki ta-profile <TA-NAME>

Description

Creates a trust anchor (TA) profile and changes to the `config-ta-<TA-NAME>` context for the profile. Each TA profile stores the certificate for a trusted CA. Up to 64 profiles can be defined.

If the specified TA profile exists, this command changes to the `config-ta-<TA-NAME>` context for the profile.

The `no` form of this command removes the specified TA profile.



NOTE: When creating a new profile, If you exit the `config-ta-<TA-NAME>` context without importing the TA certificate, the profile is discarded.

Command context

config

Parameters

<TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".



NOTE: The TA profile name cannot end with `est-ta<nn>` where `<nn>` is 00 to 99. For example, `company-trust-anchor-est-ta01` is not allowed. This TA profile name suffix is reserved for TA profiles that are created for CA certificates from EST servers.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)#
```

Removing TA profile **root-cert**:

```
switch(config)# no crypto pki ta-profile root-cert
```

enroll self-signed

Syntax

```
enroll self-signed
```

Description

Generates a key pair and generates a self-signed certificate with it.

The subject fields and key type of the current leaf certificate must be defined before running this command. If not, you are prompted to fill in the subject fields, and the key type is set to `RSA 2048`.

Command context

```
config-cert-<CERT-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# enroll self-signed  
You are enrolling a certificate with the following attributes:  
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,  
         CN=Leaf01  
Key Type: RSA (2048)  
  
Continue (y/n)? y  
Self-signed certificate is created and enrolled successfully.  
  
switch(config-cert-leaf-cert)#
```

enroll terminal

Syntax

```
enroll terminal
```

Description

Generates a key pair and certificate signing request (CSR) for the current leaf certificate. Use the CSR to obtain a signed certificate from a certificate authority (CA), and then import the certificate onto the switch with the command `import terminal`.

The key type, and the certificate common name in the subject fields of the current leaf certificate must be completed before running this command.

Command context

```
config-cert-<CERT-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# enroll terminal
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,
        CN=Leaf01
Key Type: RSA (2048)

Continue (y/n)? y

-----BEGIN CERTIFICATE REQUEST-----
MIIBozCCAQwCAQAwYzEVMBMGA1UEAxMMcG9kMDEtODQwMC0xMQ4wDAYDVQQLEwV
nViYTEMMAoGA1UEChMDSFBFMRIwEAYDVQQHEwlSb3Nldm1sbGUxCzAJBgNVBAGT
NBMQswCQYDVQQGEwJVUzCBnzANBgkqhkiG9w0BAQEFAAOBjQAwgYkCgYEAAtKcLS
...
GBAJ4L3lFFfWBEL+KAKpOGjZcVmw1BMqSKFtOFNF9nzmUmONmU3SKy6dzQ+6ynR
7Au22mf3lWDxzrtCC/dj5RtWJeJekxp2LCIK/3eRXUwbYveQDKcxH7j9ZB+BAp2
ace+2tA68F2vlgRCQ/hcQH0YmNuaq4Ne3w0dhm7HlUrx
-----END CERTIFICATE REQUEST-----
switch(config-cert-leaf-cert)#
```

import (CA-signed leaf certificate)

Syntax

```
import terminal ta-profile <TA-NAME> [password <PW>]
import <REMOTE-URL> ta-profile <TA-NAME> [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> ta-profile <TA-NAME> [password <PW>]
```

Description

Imports a CA-signed leaf certificate and then validates the certificate against the specified TA profile. If the imported data includes a private key, the private key must match the leaf certificate being imported. If the imported data does not include a private key, the certificate must match a CSR that was previously generated with the command `enroll terminal` and must be signed by the CA whose root certificate is installed in the specified TA profile. The TA profile must exist and have a TA certificate configured.

Parameters

terminal

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

ta-profile <TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

<REMOTE-URL>

Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

<STORAGE-URL>

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

password <PW>

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

vrf <VRF-NAME>

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

Command context

```
config-cert-<CERT-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The imported data must include all the intermediate CA certificates in the certificate chain leading to the certificate imported into the specified TA profile.
- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for `<REMOTE-URL>` and `<STORAGE-URL>`.
- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----  
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----  
-----END PRIVATE KEY-----  
  
-----BEGIN ENCRYPTED PRIVATE KEY-----  
-----END ENCRYPTED PRIVATE KEY-----
```

Examples

Importing a leaf certificate from the console:

```
switch(config)# crypto pki certificate leaf-cert  
switch(config-cert-leaf-cert1)# import terminal ta-profile root-cert  
Paste the certificate in PEM format below, then hit enter and ctrl-D:  
switch(config-cert-import)# -----BEGIN CERTIFICATE-----  
switch(config-cert-import)# MIIFRDCCAYygAwIBAgQP8nS2Vp15u0xXMdkDJzANBgkqhkiG9w0Bv  
switch(config-cert-import)# MQswCQYDVQGEwJVUEOMAwGA1UCgwFXJ1YmDAgNBAMM1Jvb3QgQ0Ew  
switch(config-cert-import)# HhcNMTkNDEwMjIwNT1lWhcjIwMT0MjwNE1WjzQswQDVQGEwJVUzEL  
...  
switch(config-cert-import)# 1fIYZYGQyla0AwFuPTTxBXHYWrxTPbUYU5umJfRPmE4VY8S9DQgcr
```

```

switch(config-cert-import)# 1NGNm3NG03GqPScs/TF9bVyFA5BOS5lmmkfRYK8D/kMTfRreSdxis
switch(config-cert-import)# YQ1u1NqShps=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIFDjBABgkqhkiG9wBBQ0wMzAbBgqkw0QwwDQIpJMN7sVGwCAgga
switch(config-cert-import)# MBQGCCqGSib3DQMHAit+2qadNAASCgLYJ4Am3EfhH5p51Ggr86VqS
switch(config-cert-import)# IJ6L/UhEtH523nUkdV6gvAgoYaD83PswToAGv5VS8OMFTPttrn5/K
...
switch(config-cert-import)# OgSecqZsG6arbx0ESaYBirlc/6rPspcjbx283iD1MWOpes2aEmOX
switch(config-cert-import)# iKnXnUMpVPfLc74ty2S41DtH0X9gf6aaljStg+7cND9XfGtjaV2+/
switch(config-cert-import)# cb4=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-leaf-cert)#

```

Importing a leaf certificate from a remote file:

```

switch(config)# crypto pki certificate leaf-cert2
switch(config-cert-leaf-cert2)# import tftp://1.1.1.2/c2.p12 ta-profile root-cert
% Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             %         Dload  Upload   Total     Spent    Left     Speed
100  3722  100  3722    0      0   391k      0  --:--:-- --:--:-- --:--:--   391k
100  3722  100  3722    0      0   376k      0  --:--:-- --:--:-- --:--:--   376k
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-leaf-cert2)#

```

import (self-signed leaf certificate)

Syntax

```

import terminal self-signed [password <PW>]
import <REMOTE-URL> self-signed [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> self-signed [password <PW>]

```

Description

Imports a self-signed leaf certificate including its matching private key.

Parameters

terminal

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

<REMOTE-URL>

Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

<STORAGE-URL>

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

password *<PW>*

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

vrf *<VRF-NAME>*

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

Command context

`config-cert-<CERT-NAME>`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for *<REMOTE-URL>* and *<STORAGE-URL>*.
- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----  
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----  
-----END PRIVATE KEY-----  
  
-----BEGIN ENCRYPTED PRIVATE KEY-----  
-----END ENCRYPTED PRIVATE KEY-----
```

Example

Importing a self-signed leaf certificate from the console:

```
switch(config)# crypto pki certificate ss-leaf-cert  
switch(config-cert-ss-leaf-cert)# import terminal self-signed  
Paste the certificate in PEM format below, then hit enter and ctrl-D:  
switch(config-cert-import)# -----BEGIN CERTIFICATE-----  
switch(config-cert-import)# MIID2TCCAsGgAwIBAgIJAKcrqokm6p9GMA0GCSqGSIb3DQEBCwUAM  
switch(config-cert-import)# tDCCA5ygAwIBAgICEAEwDQYJKoZIhvcNAQELBQAwYgxCzABAYTA1  
switch(config-cert-import)# VQQGEGwJVUzELMAkGA1UECAwCQ0ExDTALBgNVBACMBFJvc2UxDDAKB  
...  
switch(config-cert-import)# +fWQLxhp+jKJGZGOZz/FENt2uSfZHzlXiu8n3g+EgqExenY1pBRJr  
switch(config-cert-import)# VuEEoNb/YfkPXHHva4Zfx223q+f694wlVsHkENSzqr2goHpa2fOzq  
switch(config-cert-import)# alewwdmVqCES+x8bvhf3C/6IB6ePkEsnMlHNTeM=  
switch(config-cert-import)# -----END CERTIFICATE-----  
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----  
switch(config-cert-import)# MIIFDjBABgkqhkiG9w0BBQ0wMzAbBgkqhkiG9w0BBQwwDgQIT8Ni3  
switch(config-cert-import)# MBQGCCqGSIb3DQMHBAiBHrejkdPdASCBMjVxrrYYPNt3Vlabr9k8  
switch(config-cert-import)# 5GE0U99awh9ys4360WR95xOFGThvjKTyRWG511nGwVeLZs/7TPXWI  
...  
switch(config-cert-import)# hzc5ZT/w2F08icRI5mFbGoTAAw9IIWMOXGweaWQJDyKGrhg89GrnV  
switch(config-cert-import)# M2UuP/tYuuO328QcenKZEJmZKCbx78oFRR+pgma4oeMaFTIyXE6Pr  
switch(config-cert-import)# GAdCK8tkDiJ9DKbqdM5W0/nTJfqwUQ1fl127dNrBAodsHdwr3UR99H  
switch(config-cert-import)# SPo=  
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
```



```
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert)#
```

Importing a leaf certificate from a remote file:

```
switch(config)# crypto pki certificate ss-leaf-cert2
switch(config-cert-ss-leaf-cert2)# import tftp://1.1.1.2/ss2.p12 self-signed
% Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
             %                   Dload  Upload  Total   Spent    Left   Speed
100 3230 100 3230    0     0  875k      0  --:--:-- --:--:-- --:--:--  875k
100 3230 100 3230    0     0  831k      0  --:--:-- --:--:-- --:--:--  831k
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert2)#
```

key-type

Syntax

```
key-type {rsa [key-size <K-SIZE>] | ecdsa [curve-size <C-SIZE>]}
```

Description

Sets the key type and key size for the current leaf certificate. The key type of the default certificate `local-cert` cannot be changed.

Command context

```
config-cert-<CERT-NAME>
```

Parameters

rsa

Specifies the key type as RSA.

key-size <K-SIZE>

Specifies the RSA key size in bits. Supported values: 2048, 3072, 4096. Default: 2048

ecdsa

Specifies the key type as ECDSA.

curve-size <C-SIZE>

Specifies the ECDSA elliptic curve size in bits. Supported values: 256, 348, 521. Default: 256

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting RSA encryption on the leaf certificate `leaf-cert`:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type rsa key-size 3072
```

Setting ECDSA encryption on the leaf certificate `leaf-cert`:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type ecdsa curve-size 521
```

ocsp disable-nonce

Syntax

```
ocsp disable-nonce
```

```
no ocsp disable-nonce
```

Description

Configures exclusion of the nonce from OCSF requests. A nonce is a unique identifier that an OCSF client inserts in an OCSF request and expects the OCSF responder to include it in the corresponding OCSF response. The nonce mechanism helps prevent replay attacks in which a malicious player attempts to masquerade as the OCSF responder. Although the nonce is included by default, it can be excluded. Some OCSF responders choose to not support the use of the nonce due to performance considerations.

The `no` form of this command re-enables nonce inclusion in OCSF requests.

Command context

```
config-ta-<TA-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# ocsp disable-nonce
```

Enable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# no ocsp disable-nonce
```

ocsp enforcement-level

Syntax

```
ocsp enforcement-level {strict | optional}
```

```
no enforcement-level
```

Description

Sets either strict or reduced enforcement of the OCSF check of certificates. Strict enforcement is enabled by default.

The `no` form of this command resets enforcement to its default of `strict`.

Command context

```
config-ta-<TA-NAME>
```

Parameters

strict

Sets strict OCSP checking of certificates. The certificate is accepted only if all possible checking (including validation failures, software system errors, configuration errors, transactional errors) is successful.

optional

Sets reduced OCSP checking of certificates. The certificate is accepted unless one or more of these validation errors occur:

- Response signature invalid.
- Nonce in response mismatch.
- Certificate revoked, but only when revocation checking is possible. If revocation check is not possible, the certificate is still accepted if there are no other validation errors.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting reduced OCSP checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# ocsp enforcement-level optional
```

Setting strict OCSP checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# ocsp enforcement-level strict
```

ocsp url

Syntax

```
ocsp url {primary | secondary} <URL>
```

```
no ocsp url {primary | secondary}
```

Description

Configures the OCSP responder URLs that the current TA profile uses to verify the revocation status of an X.509 digital certificate. These URLs override the OCSP responder URL contained within the peer certificate being verified (as well as URLs defined in any intermediate CAs in the chain of trust).

If no OCSP responder URLs are defined for a TA profile (default setting), then the OCSP responder URL in the peer certificate is used for revocation status checking. (The OCSP responder URL is contained in a certificate's Authority Information Access field, which is an X.509 v3 certificate extension.)

The `no` form of this command deletes the specified OCSP responder URL (primary or secondary) from the current TA profile.

Command context

```
config-ta-<TA-NAME>
```

Parameters

{primary | secondary} <URL>

Specify the HTTP URL of the primary or secondary OCSP responder using either a fully qualified domain name or IPv4 address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining the primary OCSP URL for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# revocation-check oosp  
switch(config-ta-root-cert)# oosp url primary http://oosp-server.site.com
```

Removing the primary OCSP URL from the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile oot-cert  
switch(config-ta-root-cert)# revocation-check oosp  
switch(config-ta-root-cert)# no oosp url primary
```

oosp vrf

Syntax

oosp vrf <VRF-NAME>

no oosp vrf

Description

Sets the VRF that the switch uses to communicate with OCSP responders for OCSP checking. VRF **mgmt** is used by default.

The **no** form of this command resets the VRF to its default **mgmt**.

Command context

config-ta-<TA-NAME>

Parameters

<VRF-NAME>

Specifies the name of the VRF the switch uses to communicate with OCSP responders. Default: **mgmt**.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Reverting the OCSP responder VRF to its default:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# no oosp vrf
```

revocation-check oosp

Syntax

```
revocation-check oosp
```

```
no revocation-check
```

Description

Enables certificate revocation checking for the current profile using the online certificate status protocol (OCSP).

The `no` form of this command disables certificate revocation checking for the current profile.

Command context

```
config-ta-<TA-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling revocation checking for the TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# revocation-check oosp
```

Disabling revocation checking for the TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# no revocation-check
```

show crypto pki application

Syntax

```
show crypto pki application
```

Description

Shows certificate information for all features (applications) using leaf certificates that are managed by PKI.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing certificate information for all features (applications) using leaf certificates:

```
switch# show crypto pki application1
```

Associated Applications	Certificate Name	Cert Status
https-server		not configured, using local-cert

syslog-client	local-cert	valid
hsc	xhsccert	invalid, using local-cert

show crypto pki certificate

Syntax

```
show crypto pki certificate [<CERT-NAME> [plaintext | pem]]
```

Description

Shows a list of all configured leaf certificates, or detailed information for a specific leaf certificate.

Possible values for Cert Status are: CSR pending, expired, expires soon, installed, malformed, not yet known.

Possible values for EST Status are: enroll failed, enroll pending, enroll retrying, enroll success, n/a (certificate is not EST-enrolled), reenroll failed, reenroll pending, reenroll retrying.

Command context

Manager (#)

Parameters

<CERT-NAME>

Specifies the leaf certificate name. Range: 1 to 32 alphanumeric characters excluding ".".

plaintext

Shows certificate information in plain text.

pem

Shows certificate information in PEM format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured leaf certificates:

```
switch# show crypto pki certificate
```

Certificate Name	Cert Status	EST Status	Associated Applications
device-identity	installed	n/a	none
pod01-99-1	installed	n/a	https-server, est-client
syslog-1	CSR pending	enroll retrying	syslog-client
leaf-cert1	installed	enroll success	none
leaf-cert2	CSR pending	enroll failed	none

Showing detailed information (in plaintext format) for leaf certificate **pod01-99-1**:

```
switch# show crypto pki certificate pod01-99-1 plaintext
```

```
Certificate Name: pod01-99-1
Associated Applications:
  https-server, est-client
Certificate Status: installed
EST Status: n/a
Certificate Type: regular
```

```

Intermediates:
  Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = DeviceCA
  Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
  Serial Number: 0x02
  Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
  Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-Root
  Serial Number: 0x01
Certificate:
  Data:
    Version: 1 (0x0)
    Serial Number: 14529416756121781768 (0xc9a2db8f3e3f4608)
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C=US, ST=CA, OU=Lab-IT, O=Company, CN=DeviceCA
  Validity
    Not Before: Jan 12 23:36:57 2018 GMT
    Not After : Nov 1 23:36:57 2020 GMT
  Subject: C=US, ST=CA, OU=Lab-IT, O=Company, CN=pod01-99-1
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:a0:cd:ef:1b:f9:b8:bd:39:fc:7a:0e:00:17:ff:
      2b:72:d8:4e:d4:df:49:36:ca:3a:f9:05:05:d7:e3:
      d1:97:29:71:e6:33:b8:bb:8e:f0:ee:a6:e4:4a:f8:
      ...
      fe:dd:d9:a0:af:59:47:25:b4:34:06:af:03:1d:33:
      30:c3:85:fe:5c:e7:19:7f:ff:3a:b2:21:b8:e8:ed:
      83:09
    Exponent: 65537 (0x10001)
  Signature Algorithm: sha256WithRSAEncryption
    39:f6:03:86:03:d9:05:61:39:25:5f:0d:75:cc:05:ae:04:7e:
    4c:a3:13:0b:f0:1e:af:68:0e:40:9f:ed:48:b6:5e:56:8c:53:
    46:5b:c9:a4:e0:b0:bc:31:4b:a7:5d:0a:ed:7c:9c:f6:bf:1e:
    ...
    39:f5:26:58:68:e2:13:ec:94:ac:60:8e:4b:b0:ba:45:cf:d6:
    6a:4b:9f:7d:ae:3f:e5:2e:81:fe:ac:b3:65:44:35:47:a5:2f:
    89:e7:58:a0

```

Showing detailed information (in PEM format) for leaf certificate **leaf-cert1** with a status of CSR pending:

```

switch# show crypto pki certificate leaf-cert1 pem

Certificate Name: leaf-cert1

Associated Applications:
  syslog-client
Certificate Status: CSR pending
EST Status: enroll retrying
Certificate Type: regular
-----BEGIN CERTIFICATE REQUEST-----

MIICtTCAZ0CAQAwcDEWMBQGA1UEAxMNc3lzbG9nLTg0MBYGA1UECxMPQ
XJlYmEtUm9zZXZpbGx1MQ4wDAYDVQQKEYTESMBAGA1EBxMJUm9zZXZpbG
xlMQswCQYDVQQIEWJDQTELMAGA1UEBhMCVVMwggEiMSIb3DQEBAQUAA4I
...
cw2ytN6Idgh81k59x6DH7V/eORaKd5lq+oO7nkr6+QBf5L3f5Kb+TOFio
lei+EdCHMxxc07MK0n3dkziSW25HFUGsyEXVMK+BiD3zbKDoUe6XVhvqI
mamXyghigLYDcbsn6WVw==
-----END CERTIFICATE REQUEST-----

```

show crypto pki ta-profile

Syntax

```
show crypto pki ta-profile [<TA-NAME>]
```

Description

Shows a list of all configured TA profiles, or detailed information for a specific profile.



NOTE: This command shows information for both directly-configured TA profiles and TA profiles that were dynamically downloaded from EST servers.

Command context

Manager (#)

Parameters

<TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured TA profiles:

```
switch# show crypto pki ta-profile
```

Profile Name	TA Certificate	Revocation Check
BASE_CA	Installed,valid	disabled
BASE02_CA	Installed,expired	disabled
root-cert	Installed,valid	OCSP
ROOT-A_CA	Not Installed	OCSP
EST-Service1	Installed,valid	None
EST-Service2	Installed,valid	None

Showing detailed information for TA profile **root-cert**:

```
switch# show crypto pki ta-profile root-cert
```

```
TA Profile Name      : root-cert
Revocation Check     : OCSP
  OSCP Primary URL   : http://ocsp1.domain.com
  OSCP Secondary URL : Not Configured
  OSCP Disable-nonce : false
  OSCP Enforcement Level: strict
  OSCP VRF           : mgmt
TA Certificate: Installed and valid
Version: 3 (0x2)
Serial Number:
  74:e6:6d:22:3f:52:cc:94:43:41:ab:66:a8:8d:47:b1
Signature Algorithm: sha1withRSAEncryption
Issuer: OU=DeviceTrust, OU=Operations, O=Site, C=US,
       CN=Site Trusted Computing Root CA 1.0
Validity
  Not Before: Sep 14 03:12:06 2007 GMT
  Not After : Sep 14 03:21:14 2032 GMT
Subject: OU=DeviceTrust, OU=Operations, O=Site, C=US,
```



```

        CN=Site Trusted Computing Root CA 1.0
Subject Public Key Info:
  Public Key Algorithm: rsaEncryption
    RSA Public Key: (2048 bit)
      Modulus (2048 bit):
        30:0d:06:09:2a:86:48:86:f7:0d:01:01:01:05:33:
        03:82:01:0f:00:30:82:01:3a:02:82:01:01:00:ac:
        3d:60:3a:2e:ca:a4:34:db:5c:3b:6b:07:df:73:62:
        ...
        20:c8:df:63:14:5a:e8:d3:ea:83:d8:47:a3:b5:2e:
        bb:64:51:f0:be:13:b6:91:e4:32:45:58:5e:1f:0d:
        02:03:01:00:01
      Exponent: 65537 (0x10001)
X509v3 extensions:
  X509v3 Key Usage:
    Digital Signature, Certificate Signing, CRL Signing
  X509v3 Basic Constraints:
    CA:TRUE, pathlen:4
  X509v3 Subject Key Identifier:
    eb:d7:ec:db:8a:cb:f2:51:d5:06:e1:42:7b:39:a7:d0:1e:31:6e:bf

Signature Algorithm: sha1withRSAEncryption
  1c:90:f3:a4:f0:0d:e2:e3:e9:ae:01:e1:7d:a7:13:e2:cc:0b:
  17:31:26:92:a2:5d:1d:19:60:54:03:13:9b:e1:73:6c:e4:b3:
  01:4f:4e:ae:61:bd:ae:b6:12:d3:ab:08:ae:8c:47:92:d7:0d:
  ...
  ca:cf:11:78:55:6d:06:49:fa:d4:8d:f3:ef:7f:79:38:35:5d:
  16:5a:57:7f:a8:dc:b0:f8:a2:04:0d:17:0b:bb:58:32:30:e0:
  2d:a8:37:a2

```

ta-certificate

Syntax

```
ta-certificate { [import [terminal]] | import {<REMOTE-URL> | <STORAGE-URL>} }
```

Description

Imports a CA certificate for use in the current TA profile. The certificate must be in PEM format. The PEM data must be delimited with these lines:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```



NOTE: Only the first certificate in the PEM data is imported. Any additional certificates are ignored.

Command context

```
config-ta-<TA-NAME>
```

Parameters

[import [terminal]]

Import the certificate by pasting at the console (the default). This form of importing is selected whether `ta-certificate` is entered without parameters or if only `import` is entered or if `import terminal` is entered. Upon execution, the `config-ta-cert` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary..

import <REMOTE-URL>

Import the certificate from a file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

import <STORAGE-URL>

Available on switch families that provide USB device file import capability, import the certificate from a file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Importing a certificate into the TA profile **root-cert** by pasting PEM-format certificate data at the console:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIDuTCCAqECCQCuoxeJ2ZNYcjANBgkqhkiG9w0BAQsFADCBQzELMAEBh
switch(config-ta-cert)# VVMxEzARBgNVBAgMCKNhbgGlm3JuaWEExEDAOBgNVBAcMB1JvY2tsDAKBg
switch(config-ta-cert)# BAoMA0hQTjEVMBMGA1UECwwMSFBOUm9zZXZpbGxlMSowKAYDVQocG5zd
...
switch(config-ta-cert)# x3Wff3dFZ8o9sd5LVAHneH/ztb9MP34z+le1V346r12L2kpxmTOVJVyTO
switch(config-ta-cert)# BIzD/ST/HaWI+0S+S80rm93PSscEbb9GWk7vshh5EnW/moehBKcE40lzy
switch(config-ta-cert)# 3LvMLZcssSe5J2Ca2XIhfdMe8UaNZ7syGYMsAW0nG7yYHWkEOQu9s
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Issuer: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
        CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert)#
```

Importing a certificate into the TA profile **root-cert2** from file **rcert2-data** on the USB device:

```
switch(config)# crypto pki ta-profile root-cert2
switch(config-ta-root-cert2)# ta-certificate import usb:/rcert2-data
The certificate you are importing has the following attributes:
Issuer: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
        CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert2)#
```

subject

Syntax

```
subject [common-name <COMMON-NAME>] [country <COUNTRY>] [locality <LOCALITY>]  
        [org <ORG-NAME>] [org-unit <ORG-UNIT>] [state <STATE>]
```

Description

Sets the subject fields for the current leaf certificate. If the `common-name` parameter is not specified, then you are prompted to define a value for each field. If a configured value exists for any field, it is presented as the default.

The subject fields of the default certificate `local-cert` cannot be changed.

Command context

```
config-cert-<CERT-NAME>
```

Parameters

common-name <COMMON-NAME>

Specifies the common name.

country <COUNTRY>

Specifies the region.

locality <LOCALITY>

Specifies the locality.

org <ORG-NAME>

Specifies the organization.

org-unit <ORG-UNIT>

Specifies the organizational unit.

state <STATE>

Specifies the state.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting subject fields for the leaf certificate `leaf-cert`:

```
switch(config-cert-leaf-cert)# subject common-name Leaf01 country US  
locality CA org Company org-unit Site state CA
```

Setting subject fields for the leaf certificate `leaf-cert` interactively:

```
switch(config-cert-leaf-cert)# subject  
Do you want to use the switch serial number as the common name (y/n)? n  
Enter Common Name : Leaf01  
Enter Org Unit : Site  
Enter Org Name : Company  
Enter Locality : Rocklin  
Enter State : CA  
Enter Country : US  
switch(config-cert-leaf-cert)#
```

PKI EST commands



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

arbitrary-label

Syntax

```
arbitrary-label <LABEL>
```

```
no arbitrary-label
```

Description

Within the EST profile context, configures the generic optional label (also known as arbitrary label) to be concatenated to the EST server URL that is configured with the `url` command. There is no arbitrary label configured by default. Any existing arbitrary label is replaced by this command. The use of arbitrary labels is optional.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with this `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the generic arbitrary label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the generic arbitrary label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the URL and generic arbitrary label. Note that with the URL and arbitrary label configured in this example, the final URL the switch uses to request CA certificates from the EST server is `https://est-service999.com/.well-known/est/rsa2048/cacerts`.

```
switch(config)# crypto pki est-profile EST-service1
switch(config)# url https://est-service999.com/.well-known/est
switch(config-est-EST-service1)# arbitrary-label rsa2048
```

Removing the generic arbitrary label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label
```

arbitrary-label-enrollment

Syntax

```
arbitrary-label-enrollment <LABEL>
```

```
no arbitrary-label-enrollment
```

Description

Within the EST profile context, configures the arbitrary enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the enrollment operation. There is no arbitrary enrollment label configured by default. Any existing arbitrary enrollment label is replaced by this command. The use of arbitrary enrollment labels is optional.

When the enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with this `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the arbitrary enrollment label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the arbitrary enrollment label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the arbitrary enrollment label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# arbitrary-label-enrollment ipsec-v7
```

Removing the arbitrary enrollment label :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label-enrollment
```

arbitrary-label-reenrollment

Syntax

```
arbitrary-label-reenrollment <LABEL>
```

```
no arbitrary-label-reenrollment
```

Description

Within the EST profile context, configures the arbitrary re-enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the re-enrollment operation. There is no arbitrary re-enrollment label configured by default. Any existing arbitrary re-enrollment label is replaced by this command. The use of arbitrary re-enrollment labels is optional.

When the re-enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for re-enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with this `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the arbitrary re-enrollment label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the arbitrary re-enrollment label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the arbitrary re-enrollment label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# arbitrary-label-reenrollment ipsec-v7
```

Removing the arbitrary re-enrollment label :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label-reenrollment
```

crypto pki est-profile

Syntax

```
crypto pki est-profile <EST-NAME>
```

```
no crypto pki est-profile <EST-NAME>
```

Description

Creates a certificate Enrollment over Secure Transport (EST) profile and changes to the `config-est-<EST-NAME>` context for the profile. Each EST profile stores information about the EST service, including EST server URL. Up to 16 profiles can be created.

If the specified EST profile exists, this command changes to the `config-est-<EST-NAME>` context for the profile.

The `no` form of this command deletes the specified EST profile. It also deletes the TA profiles whose CA certificates were downloaded from the corresponding EST server, and the leaf certificates that were enrolled using this EST profile.



NOTE: The deletion of the related TA profiles and enrolled certificates is permanent. If the EST profile is in the startup configuration and the EST profile is deleted but this deletion is not updated in the startup configuration before a switch reboot, the EST profile will still exist after the reboot but the related TA profiles and enrolled certificates will not exist.

Command context

```
config
```

Parameters

<EST-NAME>

Specifies the EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating EST profile **EST-Service1**:

```
switch(config)# crypto pki est-profile EST-Service1  
switch(config-est-service1)#
```

Removing EST profile **service1**:

```
switch(config)# no crypto pki est-profile EST-Service1
```

enroll est-profile

Syntax

```
enroll est-profile <EST-NAME>
```

Description

Enrolls a leaf certificate through a remote EST (Enrollment over Secure Transport) server.

Per RFC 7030, EST enables clients to request certificate signing services over secure TLS connections. The switch generates a key pair and the corresponding CSR. The CSR is sent to the EST server to request signing, and the signed certificate is returned to the switch where it is validated. If the whole process succeeds, the certificate can be used as a leaf certificate on the switch. When the leaf certificate approaches its expiry date, it will be renewed automatically through the same EST server.

Each enrollment or re-enrollment attempt starts with a `/cacerts` request sent to the EST server to get the latest chain of CA certificates. After the enrollment or re-enrollment succeeds, this chain of CA certificates will be compared with those downloaded previously from the same EST server. Updates will be made as appropriate.

The subject fields of the current leaf certificate must be defined before running this command. If the common name subject field is not configured, this command is rejected.

This command cannot be used to enroll or renew the default certificate "local-cert."

Command context

```
config-cert-<CERT-NAME>
```

Parameters

<EST-NAME>

Specifies an existing EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling leaf certificate **leaf-cert1** through the EST server identified in EST profile **EST-service1**:

```
switch(config-cert-leaf-cert1)# enroll est-profile EST-service1
You are enrolling a certificate with the following attributes:
  Subject: C=US, ST=CA, L=Roseville, OU=Aruba-Roseville, O=Aruba,
           CN=leaf-cert1
  Key Type: RSA (2048 bits)

Continue (y/n)? y
Certificate enrollment via EST-service1 has been initiated.
Please use `show crypto pki certificate leaf-cert1` to check its status.

switch(config-cert-leaf-cert1)#
```

reenrollment-lead-time

Syntax

```
reenrollment-lead-time <LEAD-TIME>
```

```
no reenrollment-lead-time
```

Description

Within the EST profile context, sets the certificate re-enrollment lead time which is the number of days before certificate expiry date that certificate re-enrollment will be initiated.

The no form of this command resets the EST server re-enrollment lead time to its default of 2 days.

Command context

```
config-est-<EST-NAME>
```


Parameters

<LEAD-TIME>

Specifies the certificate re-enrollment lead time in days. Range: 0 to 30 days. Default: 2 days.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the certificate re-enrollment lead time to 15 days:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# reenrollment-lead-time 15
```

Resetting the certificate re-enrollment lead time to its default of 2 days :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no reenrollment-lead-time
```

retry-count

Syntax

```
retry-count <RETRIES>
```

```
no retry-count
```

Description

Within the EST profile context, sets the maximum number of retries to be attempted after the initial certificate enrollment request fails.

The no form of this command resets the maximum number of certificate enrollment request retries to its default of 3.

Command context

```
config-est-<EST-NAME>
```

Parameters

<RETRIES>

Specifies the maximum number of certificate enrollment request retries. Range: 0 to 32 retries. Default: 3 retries.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the retry count to 5 retries:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# retry-count 5
```

Resetting the retry count to its default of 3 retries:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no retry-count
```

retry-interval

Syntax

```
retry-interval <INTERVAL>
```

```
no retry-interval
```

Description

Within the EST profile context, sets the interval at which a failed certificate enrollment request is retried. The no form of this command resets the enrollment request retry interval to its default of 30 seconds.

Command context

```
config-est-<EST-NAME>
```

Parameters

<INTERVAL>

Specifies the enrollment request retry interval in seconds. Range: 30 to 600 seconds. Default: 30 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the certificate enrollment request retry interval to 45 seconds:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# retry-interval 45
```

Resetting the retry interval to its default of 30 seconds:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no retry-interval
```

show crypto pki est-profile

Syntax

```
show crypto pki est-profile [<EST-NAME>]
```

Description

Shows a list of all configured EST profiles, or detailed information for a specific profile.

Command context

Manager (#)

Parameters

<EST-NAME>

Specifies the EST profile name. Range: Up to 32 alphanumeric characters excluding ".

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured EST profiles:

```
switch# show crypto pki est-profile
```

Profile Name	Downloaded TA Profiles	Enrolled Certificates
EST-service1	2	3
EST-service2	1	2
EST-service3	2	0

Showing detailed information for EST profile **EST-service1**:

```
switch# show crypto pki est-profile EST-service1
Profile Name           : EST-service1
Service VRF            : mgmt
Service URL             : https://est-service999.com
  Arbitrary Label       : not configured
  Arbitrary Label Enrollment : /ipsec-VP7
  Arbitrary Label Reenrollment : not configured
Authentication Username : est1
Authentication Password :
  AQBapREALpWYm2z7L1LanOtR3vGkqhBN1hBUU2CuvQXUF/ggYgAAnAnGTnKq49P4c
  dNQ6UqPbjHL4XzCO0T04djkhsUXPKGfnsWuFEONveh+JbEobqKImfwJjc3eWHiaUb
  eNpPx2zN2Q1DdyxAAQi4rmKr8LITMTTmd7qr
Retry Interval          : 45 seconds
Retry Count             : 5 times
Reenrollment Lead Time : 2 days
Downloaded TA Profiles  : 2
Enrolled Certificates   :
  leaf-cert1
  leaf-cert2
  leaf-cert3
```

url

Syntax

```
url <URL>
```

```
no url
```

Description

Within the EST profile context, configures the URL of the certificate enrollment EST server. This is not configured by default. Any existing URL is replaced by this command.

The no form of this command removes the EST server URL within the selected EST profile. The removal of the URL does not affect the TA profiles and enrolled certificates from the EST server.

Command context

```
config-est-<EST-NAME>
```

Parameters

<URL>

Specifies the EST server URL. Range: Up to 192 characters.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The configuration and update of the EST profile URL triggers the sending of a `/cacerts` request to the EST server. A successful request will result in a chain of trusted CA certificates being downloaded from the EST server. Each CA certificate, either root CA certificates or intermediate CA certificates, will be saved as a TA profile, with TA profile name `<est-name>-est-taNN` with NN representing two numerical digits. This TA profile naming scheme with the `-est-taNN` suffix is reserved for TA profiles downloaded from EST servers.
- Upon connection with an EST server, the switch authenticates the server by validating the server certificate. For this validation to succeed, a TA profile needs to pre-exist in the switch with a CA certificate from the issuer chain of the server certificate. Once the server is authenticated, all CA certificates in its `/cacerts` response will be trusted, with no further validation occurring for them.
- The TA profiles with CA certificates downloaded from an EST server will have their revocation check set to OCSP, enforcement set to optional, and the OCSP VRF set to the same as that of the EST profile.

Examples

Configuring the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# url https://est-service999.com/.well-known/est
```

Removing the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no url
```

username

Syntax

```
username <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> |  
                               plaintext <PLAINTEXT-PASSWORD>]
```

```
no username
```

Description

Within the EST profile context, configures the user account information for the EST server that is used to authenticate the switch before accepting requests from the switch. This is not configured by default. Any existing username and password is replaced by this command.

When entered without either optional `ciphertext` or `plaintext` parameters, the plaintext password is prompted for twice, with the characters entered masked with "*" symbols.

The `no` form of this command removes the user account information within the selected EST profile.

There are two ways the EST client on a CX switch can prove itself to an EST server: a certificate, and/or username and password. At least one of the two must be configured for the EST request to succeed. If both are configured, certificate authentication will be used. If a certificate is not configured or certificate

authentication fails, and username and password is configured, the username and password will be sent to the EST server for authentication.

Command context

`config-est-<EST-NAME>`

Parameters

<USERNAME>

Specifies the EST server account user name. The exact user name requirements are set by the chosen EST service. Range: Up to 32 alphanumeric characters.

ciphertext <CIPHERTEXT-PASSWORD>

Specifies the EST server account password as Base64 ciphertext. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user.



NOTE: The ciphertext password must be gotten from the EST service.

plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. The exact password requirements are set by the chosen EST service. Range: Up to 64 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an EST user with prompted cleartext password entry :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# username est1 password
Enter password: *****
Confirm password: *****
switch(config-est-EST-service1)#
```

Configuring an EST user with direct cleartext password entry:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# username est1 password plaintext concept_leap739
```

Configuring an EST user with ciphertext password entry :

```
switch(config)# crypto pki est-profile EST-service3
switch(config-est-EST-service3)# username est1 password ciphertext
AQBpRALpWYm2z7L1LanOtr3vGkqhN1hBU2CuvQXUF/ggYgAAAHWaPqxU6nAnGTnKq49P4cdNQ6U
qPbjHL4Xz00T04djkUPKGfnsWuFEONveh+JbEobq63+1k80qBKImfwJjc3eWHiaUbeNpPx2zN2Q
1DdyxAAQi4rmKr8LITMTTmd7qr
```

Removing the EST user account information for EST profile EST-service2:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# no username
```

vrf

Syntax

```
vrf <VRF-NAME>
```

```
no vrf
```

Description

Within the EST profile context, selects the VRF through which the EST server can be reached. Any existing VRF selection is replaced by this command. When this command is not used, VRF `mgmt` is used by default on switch families supporting the `mgmt` VRF, otherwise the default VRF named `default` is used.

The no form of this command selects the default VRF either `mgmt` or `default`.

Command context

```
config-est-<EST-NAME>
```

Parameters

<VRF-NAME>

Specifies the name of the VRF to use for EST server communication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Selecting VRF `it-services` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# vrf it-services
```

Resetting the VRF to its default of `mgmt` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no vrf
```



NOTE: On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

About enhanced security

Several measures can be taken to enhance switch security, including setting secure mode to enhanced in the Service OS. For maximum security, perform all the configuration described in this chapter.

Configuring enhanced security

Prerequisites

If you have switch configuration that you want to retain, create a backup. This procedure erases all configuration, including the current running configuration, the startup configuration, and all historical configuration checkpoints.

Procedure

1. Set enhanced security mode:
 - a. Reboot the switch into the Service OS with command `boot system serviceos`.
If on an 8400 Switch with both Management Modules:
 - I. Issue the `boot` command only on the active Management Module. This command ensures that both Management Modules are booted into the Service OS.
 - II. Perform steps b to e on both modules starting with the active module.
 - b. Log in to the Service OS as `admin`.
 - c. Enter command `secure-mode enhanced`.
 - d. When prompted about the mode change, respond with `y` for "yes."
 - e. Wait for the reboot and zeroization to complete. The switch firmware boots automatically.
2. Ensure adequate password requirements:
 - a. Before adding users, enable and configure password complexity as described in **password complexity**. To maintain enhanced security, configure the `password complexity` subcommand settings no smaller than their defaults.
 - b. Configure passwords for all users, including `admin`. To make your password complexity settings applicable to the default admin user, change the admin password after enabling password complexity. The new admin password must respect your password complexity settings.
3. Ensure proper login management as follows:

- a. Configure local user session management as described in **CLI user session management commands** using `cli-session` and its subcommands `max-per-user`, `timeout`, and `tracking-range` to achieve the wanted configuration. To maintain enhanced security, configure `cli-session` subcommand settings no smaller than their defaults.
 - b. Restrict remote SSH connections to only use certified crypto algorithms using `ssh certified-algorithms-only`.
 - c. Configure pre- and post-login banners using respectively, `banner motd`, and `banner exec`.
4. Ensure that the switch date and time is accurately set using `clock datetime <DATE> <TIME>`.
 5. When logging to a remote syslog server is required, ensure that the connection to the server is cryptographically secure. See **Configuring remote logging using SSH reverse tunnel**.

To ensure that enhanced security is maintained, also respect these requirements:

- Do not configure remote logging with a remote server directly without setting up an SSH tunnel.
- Do not configure passwords and secret keys using the plaintext option.



NOTE: When in enhanced security mode, the switch (Product OS) `start-shell` command is disabled for security purpose. If you attempt to use this command while in enhanced security mode, it is rejected and the following error message is displayed:

The `start-shell` command is not available in enhanced secure mode.



NOTE: When in enhanced security mode, the following Service OS commands are disabled for security purposes: `config-clear`, `password`, `sh`, and `update`. If you attempt to use any of these Service OS commands while in enhanced security mode, the command is rejected and an error message is displayed:

password complexity

Syntax

```
password complexity
```

```
no password complexity
```

Description

Enters the password-complexity context (shown in the switch prompt as `config-pwd-cplx`) for the purpose of enabling and configuring password complexity. Password complexity enhances security by enforcing specific password complexity requirements. Password complexity is disabled by default and must be enabled by execution of the `enable` command.

The no form of this command reverts all settings to their default values and disables password complexity enforcement.



NOTE: To ensure that enhanced security is maintained, it is recommended that you do not set any values to less than their defaults.



NOTE: Password complexity applies only to local authentication. For remote authentication, you may choose to set up an equivalent of password complexity according to whatever is supported on your particular TACACS+ or RADIUS server.

Command context

config

Subcommands

These subcommands are available within the password complexity context (shown in the switch prompt as `config-pwd-cplx`).

enable

Enables password complexity enforcement. The enforcement only applies to passwords created after this enabling. Existing passwords are not checked against password complexity.

disable

Disables password complexity enforcement.

[no] history-count <COUNT>

Specifies the number of previous passwords checked to prevent excessive reuse. Not applicable when adding new users. The no form of this subcommand resets the value to its default. Default: 5. Range: 1 to 5.

[no] minimum-length <LENGTH>

Specifies the minimum password length. The no form of this subcommand resets the value to its default. Default: 8. Range: 1 to 32.

[no] position-changes <POSITIONS>

Specifies the minimum number of characters that must change in the new password compared to the previous password. Not applicable if no previous password exists, including when adding new users. The no form of this subcommand resets the value to its default. Default: 8. Range: 1 to 32.

The number of password position changes is based on the number of simple character insertions, deletions, or replacements. For example:

```
Old password: abCD4$ New password: abCD$      Position changes=1 ("4" deleted)
Old password: abCD4$ New password: abCDEF4$    Position changes=2 ("EF" inserted)
Old password: abCD4$ New password: ebCD4$1     Position changes=2 ("a" replaced with "e," "1" added)
Old password: abCD4$ New password: abC$#       Position changes=3 ("D4" deleted, "#" added)
```

[no] lowercase-count <COUNT>

Specifies the minimum lowercase character count for new passwords. The no form of this subcommand resets the value to its default. Default: 1. Range: 0 to 32.

[no] uppercase-count <COUNT>

Specifies the minimum uppercase character count for new passwords. The no form of this subcommand resets the value to its default. Default: 1. Range: 0 to 32.

[no] numeric-count <COUNT>

Specifies the minimum numeric digit count for new passwords. The no form of this subcommand resets the value to its default. Default: 1. Range: 0 to 32.

[no] special-char-count <COUNT>

Specifies the minimum special character count for new passwords. The no form of this subcommand resets the value to its default. Default: 1. Range: 0 to 32.

list

List the subcommands available within the password complexity context.

exit

Exits the password complexity context.

end

Exits the password complexity context and then the config context.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Password complexity is only for use with plaintext passwords. With password complexity enabled, existing ciphertext passwords will continue working until a password is changed. All new passwords must be entered in plaintext form and be compliant with your password complexity configuration.
- The effective minimum password length may be larger than the configured `minimum-length` value. The effective minimum password length is calculated as follows:

```
LARGEST-of:(minimum-length, position-changes,  
            (SUM-of:lowercase-count+uppercase-count+numeric-count+special-char-count))
```

For example, with `minimum-length=8`, and `position-changes=10` (and the sum of the other four count settings ≤ 9), the **effective minimum-length is 10** (because `position-changes` is largest). Similarly, with a `minimum-length=12`, `position-changes=8`, `lowercase-count=8`, `uppercase-count=4`, `numeric-count=1`, `special-char-count=1`, the **effective minimum-length is 14** ($8+4+1+1=14$) (because sum off the four counts is largest).

Examples

Configuring password complexity settings with an effective minimum length of 10 (because `position-changes` is 10):

```
switch(config)# password complexity  
switch(config-pwd-cplx)# history-count 3  
switch(config-pwd-cplx)# minimum-length 8  
switch(config-pwd-cplx)# position-changes 10  
switch(config-pwd-cplx)# lowercase-count 2  
switch(config-pwd-cplx)# uppercase-count 2  
switch(config-pwd-cplx)# numeric-count 2  
switch(config-pwd-cplx)# special-char-count 2  
switch(config-pwd-cplx)# enable  
switch# exit
```

Configuring password complexity settings with an effective minimum length of 14 (because the sum of the four count items is 14):

```
switch(config)# password complexity  
switch(config-pwd-cplx)# history-count 4  
switch(config-pwd-cplx)# minimum-length 12  
switch(config-pwd-cplx)# position-changes 8  
switch(config-pwd-cplx)# lowercase-count 8  
switch(config-pwd-cplx)# uppercase-count 4  
switch(config-pwd-cplx)# numeric-count 1  
switch(config-pwd-cplx)# special-char-count 1  
switch(config-pwd-cplx)# enable  
switch# exit
```

Enabling password complexity (with default settings) and changing a user (admin1) password successfully but failing to change another user (admin2) password due to not meeting complexity requirements:

```
switch(config)# password complexity  
switch(config-pwd-cplx)# enable  
switch(config-pwd-cplx)# exit
```

```

switch(config)#
switch(config)# user admin1 password
Changing password for user admin1
Enter old password:*****
Enter new password:*****
Confirm new password:*****
switch(config)#
switch(config)# user admin2 password
Changing password for user admin2
Enter old password:*****
Enter new password:*****
Confirm new password:*****
User password not changed.
The new password does not meet one or more of the following complexity requirements:
Minimum length           : 8
Position changes         : 8
Numeric count            : 1
Lowercase count          : 1
Uppercase count          : 1
Special character count  : 1
switch(config)#

```

With password complexity already enabled, attempting to change an existing user password but failing because the new password is identical to a recently used one (`history-count`).

```

switch(config)# user admin1 password
Changing password for user admin1
Enter old password:*****
Enter new password:*****
Confirm new password:*****
User password not changed.
The new password is the same as a recently used password.
switch(config)#

```

With password complexity already enabled, creating a new admin user (admin3) with a plaintext password that meets complexity requirements.

```

switch(config)# user admin3 group administrators password
Adding user admin3
Enter password:*****
Confirm password:*****
switch(config)#

```

With password complexity already enabled, attempting to create a new admin user (admin4) with a ciphertext password but failing because ciphertext passwords are not supported with password complexity enabled.

```

switch(config)# user admin4 group administrators password ciphertext AQBapPd...==
Ciphertext passwords cannot be used when password complexity is enabled.
switch(config)#

```

Configuring remote logging using SSH reverse tunnel

Logging to a remote syslog server can be made cryptographically secure by using SSH reverse tunnel. The `syslog` daemon on the switch forwards log messages to the SSH tunnel, and the SSH tunnel endpoint on the remote server host forwards messages to the listening `syslog` server.



NOTE: This procedure includes sample configuration commands for a user-supplied syslog server based on Ubuntu 14.04.5 LTS with `rsyslog`. It is up to the user to check their server documentation and adjust the sample commands as required. Optionally see your server documentation for information on how to use the `systemd` and `autossh` services to automatically restore the SSH tunnel after system reboot.

Prerequisites

The user-supplied remote syslog server must be on a network that can reach the switch management interface.

Procedure

1. Configure SSH server on the switch.

- a. Enter these commands (although this example uses the mgmt VRF, other VRFs can be used):

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip address <switch_mgmt_IP>
switch(config-if-mgmt)# exit
switch(config)# ssh server vrf mgmt
```

- b. If public key authentication is desired for remote SSH users, configure it on the switch:

```
switch(config)# user admin authorized-key <PUBKEY>
```

2. Configure logging on the switch to forward to localhost:

```
switch(config)# logging localhost tcp <switch_tcp_port> vrf mgmt
include-auditable-events
```

3. Configure the rsyslog server on the remote host:

- a. Make `rsyslog` accept TCP connections and specify the log file, by adding the following to `/etc/rsyslog.conf`:

```
$ModLoad imtcp
$InputTCPServerRun <server_tcp_port>
$template RemoteLogs, "/var/log/remote.log"
*. * ?RemoteLogs
```

- b. To activate the added configuration, restart the `rsyslog` server:

```
root@Ubuntu4479:~#sudo service rsyslog restart
```

4. Establish an SSH reverse tunnel from the remote host to the switch:

```
root@Ubuntu4479:~#ssh -nNTx -R
<switch_tcp_port>:127.0.0.1:<server_tcp_port>
admin@<switch_mgmt_IP>
```

CLI user session management commands

cli-session

Syntax

```
cli-session
```

```
no cli-session
```

Description

Enters the CLI session context (shown in the switch prompt as `config-cli-session`) for the purpose of configuring CLI user session management. Session management enhances security by enforcing specific CLI user session requirements. The following information is provided at time of successful login:

- When applicable, the number of failed login attempts since the most recent successful login.
- The date, time, and location (console or IP address or hostname) of the most recent previous successful login.
- The count of successful logins within the past (configurable) time period.

For example:

```
switch login: admin
Password:
```

```
There were 3 failed login attempts since the last successful login
Last login: 2019-04-20 08:51:33 from the console
User "admin" has logged in 73 times in the past 30 days
```

The no form of this command disables concurrent CLI user session restrictions and reverts `timeout` and `tracking-range` to their default values.



NOTE: To ensure that enhanced security is maintained, it is recommended that you keep CLI user session management fully enabled by setting `max-per-user` to a nondefault value.



NOTE: The `cli-session` command applies only to SSH/console login connection types. It does not apply to other connection types such as REST.

Command context

```
config
```

Subcommands

These subcommands are available within the CLI session context.

[no] max-per-user <SESSIONS>

Specifies the maximum number of concurrent CLI sessions per user. The no form of this subcommand disables concurrent CLI user session restrictions. Default: Disabled (no value). Range: 1 to 5.



NOTE: When the same user name is configured for both local and remote authentication, both users, regardless of privilege level, are considered to be the same user for the purpose of counting concurrent CLI sessions. For example, with `max-per-user` set to 1 and user `admin1` configured for local and remote authentication, only the local user `admin1` or the remote user `admin1` can be logged in at any given moment. Both `admin1` users cannot be logged in simultaneously unless `max-per-user` is increased to at least 2.

[no] timeout <MINUTES>

Specifies the number of minutes a CLI session can be idle before the session is automatically terminated and the user is logged out. A value of 0 minutes disables the session timeout. The `no` form of this subcommand sets the timeout value to the default. Default 30: Range 0 to 4320.



NOTE: This subcommand is the recommended replacement for the `session-timeout` command.

[no] tracking-range <DAYS>

Specifies the maximum number of days to track CLI user session logins. The `no` form of this subcommand resets the value to its default. Default 30: Range 1 to 30.

exit

Exits the CLI session context.

end

Exits the CLI session context and then the config context.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring CLI user session settings for a maximum of one concurrent session, a 20-minute timeout, and tracking for a maximum of 25 days.

```
switch(config)# cli-session
switch(config-cli-session)# max-per-user 1
switch(config-cli-session)# timeout 20
switch(config-cli-session)# tracking-range 25
switch# exit
```

After successful earlier logins, logging in from the console without any intervening unsuccessful logins.

```
switch login: admin1
Password:
```

```
Last login: 2019-04-15 14:10:21 from the console
User 'admin1' has logged in 65 times in the past 25 days
```

Attempting to log in as `admin1` when already logged in as `admin1` from elsewhere.

```
switch login: admin1
Password:
Too many logins for 'admin1'
```

After successful earlier logins, attempting to log in twice with an invalid password, followed by a successful login.

```
switch login: admin1
Password:
```

```
Login incorrect
```

```
switch login: admin1  
Password:
```

```
Login incorrect  
switch login: admin1  
Password:
```

```
There were 2 failed login attempts since the last successful login  
Last login: 2019-04-15 17:22:45 from 192.168.1.1  
User 'admin1' has logged in 72 times in the past 25 days
```

Port access general commands

aaa authentication port-access auth-mode

Syntax

```
aaa authentication port-access auth-mode {client-mode | device-mode}
```

Description

Configures the authentication mode for the port. By default, client mode is enabled.

Command context

```
config-if
```

Parameters

client-mode

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

The maximum number of clients allowed to connect to the port is limited by the client limit value configured with the `aaa authentication port-access client-limit` command.

device-mode

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring device mode authentication for a port:

```
switch(config-if)# aaa authentication port-access auth-mode device-mode
```

aaa authentication port-access auth-precedence

Syntax

```
aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]  
no aaa authentication port-access auth-precedence
```

Description

Configures the per port authentication precedence using the space separator.

By default, 802.1X authentication (`dot1x`) takes a higher precedence than MAC authentication (`mac-auth`).

The `no` form of the command resets the port access authentication precedence to the default, 802.1X authentication followed by MAC authentication.

Command context

`config-if`

Parameters

`dot1x mac-auth`

Specifies that the port access authentication precedence is 802.1X authentication followed by MAC authentication.

`mac-auth dot1x`

Specifies that the port access authentication precedence is MAC authentication followed by 802.1X authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring MAC authentication precedence on a port:

```
switch(config-if)# aaa authentication port-access auth-precedence mac-auth dot1x
```

Resetting the authentication precedence to the default value:

```
switch(config-if)# no aaa authentication port-access auth-precedence mac-auth dot1x
```

aaa authentication port-access auth-priority

Syntax

```
aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-priority
```

Description

Configures the authentication priority using the space separator to specific interface.

Default `auth-priority` with concurrent onboarding is 802.1x followed by MAC authentication. With authentication precedence, the default `auth-priority` follows the `auth-precedence` order.

The `no` form of the command resets the port access authentication priority to the default, is same as the configured `auth-precedence` order.

The authentication priority is useful in deployments where clients such as wireless access points (APs), IT-compliant-laptops or phones, or laptops without pre-loaded supplicant software must download the supplicant software or firmware patches before attempting 802.1X authentication. In such cases, configure the MAC authentication as the primary authentication method followed by 802.1X for the authentication order. Meanwhile, configure 802.1x as the primary authentication priority and MAC authentication as secondary to enforce access based on 802.1X. Thus the client (or end access device) will initially be authenticated by MAC authentication with the access required to onboard and install the software or patches, and subsequently attempt the 802.1X authentication.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

Command context

config-if

Parameters

dot1x mac-auth

Specifies that the port access authentication priority is 802.1X authentication followed by MAC authentication.

mac-auth dot1x

Specifies that the port access authentication priority is MAC authentication followed by 802.1X authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring MAC authentication priority on a port:

```
switch(config-if) # aaa authentication port-access auth-priority mac-auth dot1x
```

Resetting the authentication priority to the default value:

```
switch(config-if) # no aaa authentication port-access auth-priority mac-auth dot1x  
switch(config-if) # no aaa authentication port-access auth-priority
```

Sample configuration:

```
interface 1/1/1  
  no shutdown  
  no routing  
  vlan access 1  
  aaa authentication port-access auth-precedence mac-auth dot1x  
  aaa authentication port-access auth-priority dot1x mac-auth
```

aaa authentication port-access client-limit

Syntax

```
aaa authentication port-access client-limit <CLIENTS>  
no aaa authentication port-access client-limit
```

Description

Configures the maximum number of clients that can simultaneously connect to a port.

The no form of this command resets the number of clients to the default.

Command context

config-if

Parameters

<CLIENTS>

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the client limit for a port:

```
switch(config-if)# aaa authentication port-access client-limit 25
```

aaa authentication port-access (role)

Syntax

```
aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role] <ROLE-NAME>
```

```
no aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role]
```

Description

Configures the role to assign to the clients depending on the client authentication state.

The `no` form of the command disassociates the roles that you assign to clients based on the authentication state.

Command context

config-if

Parameters

critical-role

Specifies the role that is applied when the RADIUS server is unreachable for authentication or when there is a request timeout.

preauth-role

Specifies the role that is applied when authentication is still in progress.

reject-role

Specifies the role that is applied when authentication has failed.

auth-role

Specifies the role that is applied to authenticated clients when a specific role is not assigned in the RADIUS server.

<ROLE-NAME>

Specifies the role name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring critical role for clients:

```
switch(config-if)# aaa authentication port-access critical-role role1
```

port-access allow-flood-traffic

Syntax

```
port-access allow-flood-traffic {enable | disable}
```

Description

Enables or disables transmission of flood traffic, such as broadcast, multicast, and unknown unicast messages through a security enabled port on which no client has been authenticated.

By default, transmission of flood traffic is disabled.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command can be used to allow Wake-on-LAN packets on security enabled ports, before a client is authenticated.

Examples

Enabling flood traffic on a port:

```
switch(config-if) # port-access allow-flood-traffic enable
```

port-access fallback-role

Syntax

```
port-access fallback-role <ROLE-NAME>
```

```
no port-access fallback-role <ROLE-NAME>
```

Description

Configures the fallback role to assign to the clients onboarding on a port. This role is applied only when no derived role is applied to the clients.

The `no` form of the command resets the fallback role.

Command context

```
config-if
```

Parameters

<ROLE-NAME>

Specifies the fallback role name. The maximum number of characters supported is 64.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Following are the conditions for the fallback role to be applied on onboarding devices:

- The device profile local MAC match feature with block-until-profile-applied mode is configured.
- Device profile along with AAA is configured but no match was found for the device profile client.
- AAA method with no reject or critical role is configured, and the connection to RADIUS server failed.
- 802.1X authentication is enabled on the port, but the supplicant of the device timed out to respond to the authentication request.

Example

Configuring fallback role for a port:

```
switch(config)# interface 1/1/3
switch(config-if)# port-access fallback-role fallback01
```

port-access log-off client

Syntax

```
port-access log-off client mac <MAC-ADDRESS>
port-access log-off client interface <INTERFACE-NAME>
port-access log-off client role <ROLE-NAME>
```

Description

Logs off the client connected to a port access-enabled interface.

Command context

Manager (#)

Parameters

mac <MAC-ADDRESS>

Specifies the client MAC address.

interface <INTERFACE-NAME>

Specifies the client interface.

role <ROLE-NAME>

Specifies the client role.

Authority

Administrators or local user group members with execution rights for this command.

Example

Logging a client off from the switch, specifying the MAC address:

```
switch# port-access log-off client mac 00:50:56:bd:04:2d
```

Logging a client off from the switch, specifying the interface:

```
switch# port-access log-off client interface 1/1/1
```

Logging a client off from the switch, specifying the role:

```
switch# port-access log-off client role r1
```

port-access onboarding-method precedence

Syntax

```
port-access onboarding-method precedence [aaa device-profile | device-profile aaa]

no port-access onboarding-method precedence [aaa device-profile | device-profile aaa]
```

Description

Configures the precedence for the method to be used to authenticate onboarding devices for each interface.

The `no` form of the command resets the authentication method precedence to the default precedence of AAA followed by device profile.

AAA includes the 802.1X and MAC authentication methods whose precedence can be configured using the `aaa authentication port-access auth-precedence` command. Here, the default precedence is 802.1X authentication.

For example, if you configure AAA (both 802.1X and MAC) authentication methods and device profile on a port, by default, the authentication precedence would be 802.1X, then MAC, and lastly device profile.

Command context

config-if

Parameters



NOTE: aaa here refers to the authentication precedence configured using the `aaa authentication port-access auth-precedence` command.

aaa device-profile

Specifies that the precedence for per port onboarding authentication method is AAA followed by device profile.

device-profile aaa

Specifies that the precedence for per port onboarding authentication method is device profile followed by AAA.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring AAA method precedence on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method precedence device-profile aaa
```

Resetting the authentication method precedence:

```
switch(config)# interface 1/1/1
switch(config-if)# no port-access onboarding-method precedence device-profile aaa
```

port-access onboarding-method concurrent

Syntax

```
port-access onboarding-method concurrent <enable | disable>
```

Description

Configures all methods to start concurrently for faster onboarding process. If authentication priority is not configured when enabling concurrent onboarding, the priority will be 802.1x followed by `mac-auth` and `device-profile`.

Default priority for concurrent onboarding is 802.1x followed by `mac-auth` and `device-profile`.

When enabling concurrent onboarding on the port, existing clients will be de-authenticated and freshly onboarded concurrently.

When concurrent onboarding is enabled, then auth-precedence will be ignored.

If concurrent onboarding is configured, the client will stay in pre-auth role till it gets succeeded by one authentication method or gets failed by all the authentication methods.

When the authentication method with the highest priority fails, the profile of the next successful authentication method is applied.

If all methods fail, the reject or critical role is applied based on the 802.1X authentication failure reason and continues to reauthenticate with the 802.1X method.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

Some RADIUS server may block the client when it receives two requests, `mac-auth` and 802.1X, from the same client at the same time. This is because the RADIUS server allows only one authentication request. In such cases, concurrent onboarding is not feasible. To prevent such scenarios, configure `auth-precedence` with `auth-priority`.

Command context

```
config-if
```

Parameters

enable

Enable clients to be onboarded concurrently.

disable

Disable clients to be onboarded concurrently.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1  
switch(config-if)# port-access onboarding-method concurrent enable
```

Disabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1  
switch(config-if)# port-access onboarding-method concurrent disable
```

Sample configuration:

```
interface 1/1/1
  no shutdown
  no routing
  vlan access 999
  !aaa authentication port-access auth-precedence mac-auth dot1x
  port-access onboarding-method concurrent enable
```

port-access reauthenticate interface

Syntax

```
port-access reauthenticate interface <INTERFACE-NAME>
```

Description

Forcefully reauthenticates all clients connected to an interface.



NOTE: Clients that are in the `HELD` state are ignored.

Command context

Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication of all clients on a port (6200 and 6300 Switch Series):

```
switch# port-access reauthenticate interface 1/1/1
```

Configuring reauthentication of all clients on a port (6400 Switch Series):

```
switch# port-access reauthenticate interface 1/3/1
```

port-access security violation action

Syntax

```
port-access security violation action [notify | shutdown]
```

```
no port-access security violation action
```

Description

Configures the action that the switch must take whenever a security violation occurs at a port, such as the number of clients exceeds the configured client limit.

The `no` form of the command resets the action to the default action, `notify`.

Command context

config-if

Parameters

notify

Specifies that the switch notifies any security violation as an event or log in the syslog server. This action is the default.

shutdown

Specifies that the switch shuts down the port where the client limit has exceeded.

A port that is shut down can be configured to auto-recover after a recovery period that can be configured with the `port-access security violation action shutdown auto-recovery` and `port-access security violation action shutdown recovery-timer` commands.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the shutdown security violation action for a port:

```
sswitch(config-if)# port-access security violation action shutdown
```

Resetting the security violation action to the default value:

```
switch(config-if)# no port-access security violation action
```

port-access security violation action shutdown auto-recovery

Syntax

```
port-access security violation action shutdown auto-recovery [enable | disable]
```

```
no port-access security violation action shutdown auto-recovery
```

Description

Configures auto-recovery of the port when the security violation action is configured as shutdown.

This configuration allows the port that is shut down when a security violation occurs to automatically enable after the recovery timer expires.

The `no` form of the command resets auto-recovery to the default, disable.

Command context

config-if

Parameters

enable

Enables auto-recovery of port when the security violation action is configured as shutdown.

disable

Disables auto-recovery of port when the security violation action is configured as shutdown.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling auto-recovery of port:

```
switch(config-if)# port-access security violation action shutdown auto-recovery enable
```

Disabling auto-recovery of port:

```
switch(config-if)# no port-access security violation action shutdown auto-recovery
```

port-access security violation action shutdown recovery-timer

Syntax

```
port-access security violation action shutdown recovery-timer <RECOVERY-TIME>
```

```
no port-access security violation action shutdown recovery-timer
```

Description

Configures security violation recovery timer for the port when the security violation action is configured as shutdown.

The `no` form of the command resets the shutdown recovery timer to the default, 10.

Command context

config-if

Parameters

<RECOVERY-TIME>

Specifies the recovery timer (in seconds) after which the port, which is shut down because of security violation, is automatically enabled. Default: 10. Range: 10 to 600.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the shutdown recovery-timer on a port:

```
switch(config-if)# port-access security violation action shutdown recovery-timer 60
```

Resetting the shutdown recovery-timer to the default value:

```
switch(config-if)# no port-access security violation action shutdown recovery-timer
```

show aaa authentication port-access interface client-status

Syntax

```
show aaa authentication port-access interface {all|<INTERFACE-NAME>}  
      client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about active port access sessions. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show aaa authentication port-access interface all client-status
```

```
Port Access Client Status Details
```

```
Client 00:50:56:96:93:d6, John Doe
```

```
=====
```

```
Session Details
```

```
-----
```

```
Port          : 1/1/13
```

```
Session Time  : 30s
```

```
Authentication Details
```

```
-----
```

```
Status          : dot1x Authenticated
```

```
Auth Precedence : dot1x - Authenticated, mac-auth - Not attempted
```

```
Authorization Details
```

```
-----
```

```
Role   : Employee
```

```
Status : Applied
```

```
Client 00:50:56:96:50:28
```

```
=====
```

```
Session Details
```

```
-----
```

```
Port          : 1/1/14
```

```
Session Time  : 10s
```

```
Authentication Details
```

```
-----
```

```
Status          : mac-auth Authenticated
```

```
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated
```

Authorization Details

Role : RADIUS_773420618
Status : Applied

show port-access clients

Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>]
```

Description

Shows summarized active port access client information. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all clients.

```
switch# show port-access clients
```

Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/4	00:50:56:bd:04:c8	port-security	Success	
1/1/5	00:50:56:bd:32:07		Success	reject-role, Reject
1/1/5	00:50:56:bd:32:08		Fail	critical-..., Critical
1/1/6	00:50:56:bd:50:43	mac-auth	Success	auth-role, Auth
1/1/6	00:50:56:bd:50:45	dot1x	Success	RADIUS_773420618
1/1/19	08:97:34:ad:e4:00	device-profile	Success	ap-role
1/1/20	00:50:56:bd:32:08		In-Progress	preauth-role, Preauth
1/1/20	00:50:56:bd:32:06		In-Progress	
1/1/20	00:50:56:bd:32:09		Fail	
1/1/212	00:60:56:bd:50:43	mac-auth	Success	fallback-role, Fallback

Showing information for clients on a particular interface:

```
switch# show port-access clients interface 1/1/5
```

Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/5	00:50:56:bd:32:07		Success	reject-role, Reject
1/1/5	00:50:56:bd:32:08		Fail	critical-..., Critical

Showing information for a particular client MAC address:

```
switch# show port-access clients mac 00:50:56:bd:50:43
```

Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/6	00:50:56:bd:50:43	mac-auth	Success	auth-role, Auth

show port-access clients detail

Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>] detail
```

Description

Shows detailed active port access clients information. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing detailed information for clients on a particular interface (one client):

```
switch# show port-access clients interface 1/1/7 detail
```

Port Access Client Status Details:

```
-----
Client 2c:41:38:7f:35:b9, John Doe
=====
Session Details
-----
Port       : 1/1/7
Session Time : 203s
IPv4 Address : 10.10.10.10
```

IPv6 Address :

Authentication Details

Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated

Authorization Details

Role : RADIUS_21963402
Status : Applied

Role Information:

Name : RADIUS_21963402
Type : radius

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Description :

Access VLAN : 10
Native VLAN :
Allowed Trunk VLANs :
Access VLAN Name :
Native VLAN Name :
Allowed Trunk VLAN Names :
MTU :
QOS Trust Mode :
STP Administrative Edge Port :
PoE Priority : low

Policy : PERMIT-ALL_87364653

Access Policy Details:

Policy Name : PERMIT-ALL_87364653
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
10	dns	ipv4	permit
20	dhcp	ipv4	permit
30	clearpass-web	ipv4	cir kbps 1024 exceed drop

Class Details:

class ip dns
10 match tcp any any
class ip dhcp
20 match any any any
class ip clearpass-web
30 match any any any

Showing information for a particular client MAC address:

```
switch# show port-access clients mac 2c:41:38:7f:35:c8 detail
```

Port Access Client Status Detail

Client 2c:41:38:7f:35:c8, Jane Q Public

Session Details

Port : 1/1/8
Session Time : 33s
IPv4 Address :
IPv6 Address :

Authentication Details

Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated

Authorization Details

Role : RADIUS_21963402
Status : Applied

Role Information:

Name : RADIUS_21963402
Type : radius

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Description :

Access VLAN : 10
Native VLAN :
Allowed Trunk VLANs :
Access VLAN Name :
Native VLAN Name :
Allowed Trunk VLAN Names :
MTU :
QOS Trust Mode :
STP Administrative Edge Port :
PoE Priority : low

Policy : PERMIT-ALL_87364653

Access Policy Details:

Policy Name : PERMIT-ALL_87364653
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
10	dns	ipv4	permit
20	dhcp	ipv4	permit

Class Details:

```

class ip dns
  10 match tcp any any
class ip dhcp
  20 match any any any
class ip clearpass-web
  30 match any any any
class ip web-traffic
  40 match any any any

```

show port-access clients onboarding-method

Syntax

```
show port-access clients onboarding-method <METHOD>
```

Description

Shows active port access client information for the specified onboarding method.

Command context

Operator (>) or Manager (#)

Parameters

<METHOD>

Selects the onboarding method. Available methods: device-profile, dot1x, mac-auth, port-security.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for clients onboarded using MAC authentication.

```
switch# show port-access clients onboarding-method mac-auth
```

Port Access Clients

Status codes: device-mode

Port	MAC-Address	Onboarding Method	Status	Role
1/1/6	00:50:56:bd:50:43	mac-auth	Success	auth-role, auth
1/1/212	00:60:56:bd:50:43	mac-auth	Success	fallback-role, fallback

show port-access security violation client-limit-exceeded interface

Syntax

```
show port-access security violation client-limit-exceeded interface {all|<INTERFACE-NAME>}
```


Description

Shows information related to information on the number of client-limit-exceeded security violations that have occurred. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<INTERFACE-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show port-access security violation client-limit-exceeded interface all
```

Client Limit Exceeded Violation Status Details

```
Port          : 1/1/1
Violation Count : 10
Violation State : Violation Occurred
```

```
Port          : 1/1/2
Violation Count : 10
Violation State : None
```

Showing information for a particular port.

```
switch# show port-access security violation client-limit-exceeded interface 1/1/1
```

Client Limit Exceeded Violation Status Details

```
Port          : 1/1/1
Violation Count : 10
Violation State : None
```

Port access 802.1X authentication

Overview

IEEE 802.1X is a standard for port-based authentication. This standard provides administrators with an authentication mechanism for devices trying to access a LAN or WLAN. 802.1X defines the encapsulation of the Extensible Authentication Protocol (EAP) over IEEE 802, which is known as EAP over LAN (EAPOL).

802.1X authentication involves the following entities:

- **Supplicant:** Device that tries to access the LAN.
- **Authenticator:** A network device, such as an Ethernet switch that authenticates the supplicant.
- **Authentication Server:** Typically a host running software supporting the RADIUS and EAP protocols that provides an authentication service to the authenticator.

Until the supplicant is authenticated, the authenticator allows only EAPOL traffic through the port to which the supplicant is connected. Only after the authentication is successful, the authenticator allows normal traffic from the supplicant.

802.1X port-based authentication provides port-level security. It allows LAN access only on ports where a single 802.1X-capable client (supplicant) has entered authorized RADIUS user credentials. 802.1X authentication is recommended for applications where only one client can connect to the port at a time. Using this option, the port processes all IP traffic as if it comes from the same client.

Port access 802.1X authentication commands

`aaa authentication port-access dot1x authenticator`

Syntax

```
aaa authentication port-access dot1x authenticator {enable | disable}
```

```
no aaa authentication port-access dot1x authenticator
```

Description

Enables or disables 802.1X authentication globally or at the port-level.

The `no` form of the command deletes global 802.1X configuration details and disables 802.1x authentication.

Command context

```
config
```

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling global 802.1X authentication:

```
switch(config)# no aaa authentication port-access dot1x authenticator
```

Enabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling 802.1X authentication configuration on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator
```

```
aaa authentication port-access dot1x authenticator auth-method
```

Syntax

```
aaa authentication port-access dot1x authenticator auth-method eap-radius
```

```
no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

Description

Configures the authentication mechanism used to control access to the network. The configured authentication method will be used to authenticate 802.1X clients.

The `no` form of the command resets the authentication mechanism to the default, `eap-radius`.

Command context

```
config
```

Parameters

eap-radius

Specifies the EAP RADIUS as the 802.1X authentication method.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# aaa authentication port-access dot1x authenticator auth-method eap-radius
```

Resetting the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

```
aaa authentication port-access dot1x authenticator cached-reauth
```

Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth
```

```
no aaa authentication port-access dot1x authenticator cached-reauth
```

Description

Enables cached reauthentication on a port. Cached reauthentication allows 802.1X reauthentications to succeed when the RADIUS server is unavailable. Users already authenticated retain their currently assigned RADIUS attributes.

The `no` form of the command disables the cached reauthentication on a port.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth
```

```
aaa authentication port-access dot1x authenticator cached-reauth-  
period
```

Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator cached-reauth-period
```

Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the cached reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth-period
```

```
aaa authentication port-access dot1x authenticator discovery-period
```

Syntax

```
aaa authentication port-access dot1x authenticator discovery-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator discovery-period
```

Description

Configures the period the port waits to retransmit the next EAPOL request identity frame on an 802.1X enabled port that has no authenticated clients.

The `no` form of the command resets the discovery period to the default, 30 seconds.

Command context

`config-if`

Parameters

<PERIOD>

Specifies the discovery period (in seconds). Default: 30. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the discovery period on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator discovery-period 120
```

Resetting the discovery period to the default value:

```
switch(config-if) # no aaa authentication port-access dot1x authenticator discovery-period  
aaa authentication port-access dot1x authenticator eapol-timeout
```

Syntax

```
aaa authentication port-access dot1x authenticator eapol-timeout <EAPOL-TIMEOUT>
```

```
no aaa authentication port-access dot1x authenticator eapol-timeout
```

Description

Configure the period the switch waits for a response from a client before retransmitting an EAPOL PDU.

If the value is 0, the time period is calculated as per RFC 2988.



NOTE: As per RFC 2988 2.1: Before Round-Trip Time (RTT) measurement, set Retransmission Timeout (RTO) to 3 seconds for initial retransmission and then double the RTO to provide back off as per section 5.5. Limit the maximum RTO (RTOmax) to 20 seconds as per section 4.3 of RFC 3748.

The `no` form of the command resets the timeout period to the default.

Command context

`config-if`

Parameters

<EAPOL-TIMEOUT>

Specifies the EAPOL timeout period (in seconds). Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring EAPOL timeout on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator eapol-timeout 120
```

Resetting the EAPOL timeout to the default value:

```
switch(config-if) # no aaa authentication port-access dot1x authenticator eapol-timeout  
aaa authentication port-access dot1x authenticator max-eapol-requests
```

Syntax

```
aaa authentication port-access dot1x authenticator max-eapol-requests <MAX-EAPOL-REQUESTS>
```

```
no aaa authentication port-access dot1x authenticator max-eapol-requests
```

Description

Configures the number of EAPOL requests to send to a supplicant that must time out before authentication fails and the authentication session ends.

The `no` form of the command resets the maximum number of EAPOL requests to the default, 5.

Command context

config-if

Parameters

<MAX-EAPOL-REQUESTS>

Specifies the maximum number of EAPOL requests. Default: 5. Range: 1 to 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring maximum EAPOL requests on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator max-eapol-requests 3
```

Resetting the maximum EAPOL requests on a port to default:

```
switch(config-if) # no aaa authentication port-access dot1x authenticator max-eapol-requests  
port-access dot1x authenticator max-retries
```

Syntax

```
aaa authentication port-access dot1x authenticator max-retries <max-retries>
```

```
no aaa authentication port-access dot1x authenticator max-retries
```

Description

Configures the maximum number of retries that the switch attempts to authenticate a client on a port before marking the client as unauthenticated.

The `no` form of the command resets the maximum number of retries to the default, 2.

Command context

`config-if`

Parameters

<max-retries>

Indicates the number of authentication attempts. Default: 2. Range: 1 to 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring maximum authentication attempts on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator max-retries 5
```

Resetting the maximum authentication attempts on a port to default:

```
switch(config-if) # no aaa authentication port-access dot1x authenticator max-retries
```

```
aaa authentication port-access dot1x authenticator quiet-period
```

Syntax

```
aaa authentication port-access dot1x authenticator quiet-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator quiet-period
```

Description

Configures the period during which the port does not try to acquire a supplicant. This period begins after the last authentication attempt, authorized by the maximum retries parameter, fails.

You can configure the number of maximum retries with the `aaa authentication port-access dot1x authenticator max-retries` command.

The `no` form of the command resets the quiet period to the default, 60 seconds.

Command context

`config-if`

Parameters

<PERIOD>

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring quiet period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator quiet-period 100
```

Resetting the quiet period on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator quiet-period
```

```
aaa authentication port-access dot1x authenticator radius server-group
```

Syntax

```
aaa authentication port-access dot1x authenticator radius server-group <GROUP-NAME>
```

```
no aaa authentication port-access dot1x authenticator radius server-group
```

Description

Configures the switch to use an existing RADIUS server group for 802.1X authentication.

The **no** form of the command resets the server group to the default, **radius**.

Command context

config

Parameters

<GROUP-NAME>

Specifies the name of the RADIUS server group.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the switch to use RADIUS server group **employee**:

```
switch(config)# aaa authentication port-access dot1x authenticator radius server-group employee
```

Resetting RADIUS server group configuration to default:

```
switch(config)# no aaa authentication port-access dot1x authenticator radius server-group
```

```
aaa authentication port-access dot1x authenticator reauth
```

Syntax

```
aaa authentication port-access dot1x authenticator reauth
```

```
no aaa authentication port-access dot1x authenticator reauth
```

Description

Enables periodic reauthentication of authenticated clients on the port.

The **no** form of the command disables periodic reauthentication.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling periodic reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth
```

Disabling periodic reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth
```

```
aaa authentication port-access dot1x authenticator reauth-period
```

Syntax

```
aaa authentication port-access dot1x authenticator reauth-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator reauth-period
```

Description

Configures the period after which the authenticated clients are reauthenticated on the port. You must enable reauthentication on the port before configuring the reauthentication period.

The `no` form of the command resets the reauthentication period to the default, 3600 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth-period 100
```

Resetting the reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth-period
```

```
clear dot1x authenticator statistics interface
```

Syntax

```
clear dot1x authenticator statistics [interface <IF-NAME>]
```

Description

Clears the 802.1X authentication statistics associated with the port and all the authenticator clients attached to this port.

If no interface is specified, the statistics is cleared for all 802.1X enabled ports.

Command context

Manager (#)

Parameters

<IF-NAME>

Specifies the interface name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/1/1
```

Clearing authentication statistics on a port (6400 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/3/1
```

Clearing authentication statistics on all ports:

```
switch# clear dot1x authenticator statistics
```

```
show aaa authentication port-access dot1x authenticator interface  
client-status
```

Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about active 802.1X authentication sessions. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all client-status
```

```
Client  FE:04:D7:50:89:37, johndoe, 1/1/1
=====
```

Authentication Details

```
-----
Status           : Authenticated
Type             : Pass-Through
EAP-Method       : MD5
Time Since Last State Change : 10s
```

Authentication Statistics

```
-----
Authentication           : 0
Authentication Timeout   : 0
EAP-Start While Authenticating : 0
EAP-Logoff While Authenticating : 0
Successful Authentication : 0
Failed Authentication     : 0
Re-Authentication        : 0
Successful Re-Authentication : 0
Failed Re-Authentication  : 0
EAP-Start When Authenticated : 0
EAP-Logoff When Authenticated : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication  : 0
```

```
Client  9A:B4:59:97:D0:7E, janedoe, 1/1/1
=====
```

Authentication Details

```
-----
Status           : Authenticated
Type             : Pass-Through
EAP-Method       : TLS
Time Since Last State Change : 5s
```

Authentication Statistics

```
-----
Authentication           : 0
Authentication Timeout   : 0
EAP-Start While Authenticating : 0
EAP-Logoff While Authenticating : 0
Successful Authentication : 0
Failed Authentication     : 0
Re-Authentication        : 0
Successful Re-Authentication : 0
Failed Re-Authentication  : 0
EAP-Start When Authenticated : 0
EAP-Logoff When Authenticated : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication  : 0
```

show aaa authentication port-access dot1x authenticator interface port-statistics

Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>} port-statistics
```

Description

Shows information about 802.1X ports. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all port-statistics
```

```
Port 1/1/1
```

```
=====
```

```
Client Details
```

```
-----
```

```
Number of Clients           : 1
Number of Authenticated Clients : 1
Number of Unauthenticated Clients : 0
```

```
Statistics
```

```
-----
```

```
EAPOL Frames Received           : 4
EAPOL Frames Transmitted         : 3
EAPOL Start Frames Received      : 1
EAPOL Logoff Frames Received     : 0
EAPOL Response ID Frames Received : 2
EAPOL Response Frames Received   : 1
EAPOL Request ID Frames Transmitted : 2
EAPOL Request Frames Transmitted : 1
EAPOL Invalid Frames Received    : 0
EAPOL EAP Length Error Frames Received : 0
EAPOL Last Received Frame Version : 0
EAPOL Last Received Frame Client MAC : 0
```

```
Port 1/1/2
```

```
=====
```

```
Client Details
```

```
-----
```

```
Number of Clients           : 1
Number of Authenticated Clients : 1
```

```
Number of Unauthenticated Clients : 0
```

Statistics

EAPOL Frames Received	: 4
EAPOL Frames Transmitted	: 3
EAPOL Start Frames Received	: 1
EAPOL Logoff Frames Received	: 0
EAPOL Response ID Frames Received	: 2
EAPOL Response Frames Received	: 1
EAPOL Request ID Frames Transmitted	: 2
EAPOL Request Frames Transmitted	: 1
EAPOL Invalid Frames Received	: 0
EAPOL EAP Length Error Frames Received	: 0
EAPOL Last Received Frame Version	: 0
EAPOL Last Received Frame Client MAC	: 0

Port access MAC authentication

Overview

MAC authentication is designed to be used at the edge of a network. It provides port-based security measures for protecting private networks and switches from unauthorized access. Because this method does not require clients to run special supplicant software (unlike 802.1X authentication), it is suitable to be used in legacy systems, IoT devices, and temporary access situations where introducing supplicant software is not an attractive option. Only a MAC address is required for authentication.

MAC authentication relies on a RADIUS server to authenticate clients. This technique simplifies access security management by using a master database on a single server to control client access. Up to three RADIUS servers can be used for backup in case access to the primary server fails. It also means that the same credentials can be used for authentication, regardless of which switch or switch port is the current access point into the LAN.

On a port configured for MAC authentication, the switch operates as a port-access authenticator using a RADIUS server, and the Challenge-Handshake Authentication Protocol (CHAP) and Password Authentication Protocol (PAP) protocols. Inbound traffic is processed by the switch alone, until authentication occurs. Some traffic from the switch to an unauthorized client is supported (for example, broadcast or unknown destination packets) before authentication occurs.

MAC authentication allows wireless clients to move between switch ports (for example, moving from one access point to another) without having to reauthenticate.

How MAC authentication works

MAC authentication grants access to a secure network by authenticating devices. When a device connects to the switch, either by direct link or through the network, the switch forwards the device MAC address to the RADIUS server for authentication. The RADIUS server uses the device MAC address as the user name and password, and grants or denies network access in the same way that it does for clients capable of interactive logons. The process does not use a client device configuration or a logon session. MAC authentication is well suited for clients not capable of providing interactive logons, such as telephones, printers, and wireless access points. Also, because most RADIUS servers allow for authentication to depend on the source switch and port through which the client connects to the network, you can use MAC authentication to lock a particular device to a specific switch and port.



NOTE: 802.1X port access and MAC authentication can be configured at the same time on a port. A total of 32 clients can be configured per port and 1024 clients on the entire switch, irrespective of the authentication method. After the limit of 1024 clients is reached, no additional authentication clients are allowed on any port for any method. The default is one client.

MAC authentication, MAC lockout, and port security are mutually exclusive on a given port. If you configure any of these authentication methods on a port, you must disable LACP on the port.

How RADIUS server is used in MAC authentication

MAC authentication uses a RADIUS server to temporarily assign a port to a static VLAN to support an authenticated client. During client authentication, the switch port membership is determined according to the following hierarchy:

- A RADIUS-assigned VLAN.
- A static, port-based, untagged VLAN to which the port is configured. A RADIUS-assigned VLAN has priority over switch-port membership in any VLAN.

Port access MAC authentication commands

`aaa authentication port-access mac-auth`

Syntax

```
aaa authentication port-access mac-auth {enable | disable}
```

Description

Enables or disables MAC authentication globally or at the port-level.

Command context

```
config
```

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# enable
```

Disabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# disable
```

Enabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# enable
```

Disabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# disable
```

aaa authentication port-access mac-auth addr-format

Syntax

```
aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |  
    multi-dash | multi-colon | no-delimiter-uppercase | single-dash-uppercase |  
    multi-dash-uppercase | multi-colon-uppercase}  
  
no aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |  
    multi-dash | multi-colon | no-delimiter-uppercase | single-dash-uppercase |  
    multi-dash-uppercase | multi-colon-uppercase}
```

Description

Configures the MAC address format that the switch must use in the RADIUS request message. The `no` form of the command resets the MAC address format to the default, `no-delimiter`.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MAC address format on the switch:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# addr-format single-dash
```

Resetting the MAC address format on the switch to its default:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# no addr-format
```

aaa authentication port-access mac-auth auth-method

Syntax

```
aaa authentication port-access mac-auth auth-method {chap | pap}  
  
no aaa authentication port-access mac-auth auth-method
```

Description

Configures the RADIUS authentication method for MAC authentication. Following are the MAC authentication methods supported:

- CHAP
- PAP



NOTE: The PEAP-MSCHAPv2 method of authentication is not supported.

The `no` form of the command resets the authentication method to the default, `chap`.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the RADIUS authentication method on the switch:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# auth-method pap
```

Resetting the RADIUS authentication method on the switch:

```
switch(config)# no aaa authentication port-access mac-auth auth-method
```

```
aaa authentication port-access mac-auth cached-reauth
```

Syntax

```
aaa authentication port-access mac-auth cached-reauth
```

```
no aaa authentication port-access mac-auth cached-reauth
```

Description

Enables cached reauthentication on a port. Cached reauthentication allows MAC reauthentications to succeed when the RADIUS server is unavailable. Users who are already authenticated, retain their currently assigned RADIUS attributes.

The `no` form of the command disables cached reauthentication.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no cached-reauth
```


aaa authentication port-access mac-auth cached-reauth-period

Syntax

```
aaa authentication port-access mac-auth cached-reauth-period <PERIOD>
```

```
no aaa authentication port-access mac-auth cached-reauth-period
```

Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

Command context

```
config-if
```

Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring cached reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no cached-reauth-period
```

aaa authentication port-access mac-auth password

Syntax

```
aaa authentication port-access mac-auth password {plaintext|ciphertext}<PASSWORD>
```

```
no aaa authentication port-access mac-auth password
```

Description

Enables and configures the global password that the switch must use for MAC authentication. The password can be either in ciphertext or plaintext format.

The `no` form of the command disables the password for MAC authentication.

Command context

```
config
```

Parameters

<PASSWORD>

Specifies the global password to be used by all MAC authenticating devices.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# password {ciphertext|plaintext}
```

Disabling the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# no password
```

aaa authentication port-access mac-auth quiet-period

Syntax

```
aaa authentication port-access mac-auth quiet-period <PERIOD>
```

```
no aaa authentication port-access mac-auth quiet-period
```

Description

Configures the period during which the switch does not try to authenticate a rejected client.

The **no** form of the command resets the quiet period to the default, 60 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the quiet period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# quiet-period 65
```

Resetting the quiet period on a port to default:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no quiet-period
```

aaa authentication port-access mac-auth radius server-group

Syntax

```
aaa authentication port-access mac-auth radius server-group <GROUP-NAME>

no aaa authentication port-access mac-auth radius server-group
```

Description

Configures the MAC authentication server group.

The `no` form of the command resets the authentication server group to the default value, `radius`.

Command context

`config`

Parameters

<GROUP-NAME>

Specifies the MAC authentication server group name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the MAC authentication server group:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# radius server-group group1
```

aaa authentication port-access mac-auth reauth

Syntax

```
aaa authentication port-access mac-auth reauth

no aaa authentication port-access mac-auth reauth
```

Description

Enables periodic MAC reauthentication of authenticated clients on the port.

The `no` form of the command disables periodic MAC reauthentication on the port.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# reauth
```

Disabling reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no reauth
```

aaa authentication port-access mac-auth reauth-period

Syntax

```
aaa authentication port-access mac-auth reauth-period <PERIOD>
```

```
no aaa authentication port-access mac-auth reauth-period
```

Description

Configures the period after which MAC authenticated clients must be reauthenticated on the port. You must first enable MAC reauthentication on the port before configuring the MAC reauthentication period.

The **no** form of the command resets the MAC reauthentication period to the default, 3600 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the MAC reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the MAC reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# reauth-period 60
```

Resetting the MAC reauthentication period to its default:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no reauth-period
```

clear mac-auth statistics

Syntax

```
clear mac-auth statistics [interface <IF-NAME>]
```

Description

Clears the MAC authentication statistics associated with the port and all the authenticator state machines associated to this port.

If no interface is specified, the statistics is cleared for all MAC authentication enabled ports.

Command context

Operator (>) or Manager (#)

Parameters

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing MAC authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear mac-auth statistics interface 1/1/1
```

Clearing MAC authentication statistics on a port (6400 Switch Series):

```
switch# clear mac-auth statistics interface 1/3/1
```

Clearing MAC authentication statistics on all ports:

```
switch# clear mac-auth statistics
```

```
show aaa authentication port-access mac-auth interface client-status
```

Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about MAC authentication clients status. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports:

```
switch# show aaa authentication port-access mac-auth interface all client-status
```

Port Access Client Status Details

Client AB:CD:DE:FF:AA:BB, 1/1/1

=====

Authentication Details

Status	: Authenticated
Type	: Pass-Through
Auth-Method	: CHAP
Time Since Last State Change	: 10 secs

Authentication Statistics

Authentication	: 1
Authentication Timeout	: 0
Successful Authentication	: 1
Failed Authentication	: 0
Re-Authentication	: 0
Successful Re-Authentication	: 0
Failed Re-Authentication	: 0
Re-Auths When Authenticated	: 0
Cached Re-Authentication	: 0

Client DD:CD:AB:CS:EE:OI, 1/1/2

=====

Authentication Details

Status	: Unauthenticated
Type	: Pass-Through
Auth-Method	: CHAP
Auth Failure reason	: Server reject/ Server timeout
Time Since Last State Change	: 15 secs

Authentication Statistics

Authentication	: 1
Authentication Timeout	: 0
Successful Authentication	: 0
Failed Authentication	: 1
Re-Authentication	: 0
Successful Re-Authentication	: 0
Failed Re-Authentication	: 0
Re-Auths When Authenticated	: 0
Cached Re-Authentication	: 0

Showing status information for a client:

```
switch# show aaa authentication port-access mac-auth interface 1/1/1 client-status mac ab:cd:de:ff:aa:bb
```

Port Access Client Status Details

Client AB:CD:DE:FF:AA:BB, 1/1/1

=====

Authentication Details

Status	: Authenticated
Type	: Pass-Through
Auth-Method	: CHAP
Time Since Last State Change	: 10 secs

Authentication Statistics

Authentication	: 1
Authentication Timeout	: 0
Successful Authentication	: 1
Failed Authentication	: 0

```
Re-Authentication          : 0
Successful Re-Authentication : 0
Failed Re-Authentication   : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication   : 0
```

show aaa authentication port-access mac-auth interface port-statistics

Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>} port-statistics
```

Description

Shows information about MAC authentication ports. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show aaa authentication port-access mac-auth interface all port-statistics
```

```
Port 1/1/1
```

```
=====
```

Client Details

```
-----
```

```
Number of Clients          : 3
Number of authenticated clients : 2
Number of unauthenticated clients : 1
Number of authenticating clients : 0
```

```
Port 1/1/2
```

```
=====
```

Client Details

```
-----
```

```
Number of Clients          : 4
Number of authenticated clients : 2
Number of unauthenticated clients : 2
Number of authenticating clients : 0
```

Port access policy

Overview

Port access policy allows network administrators to define a set of rules. These rules are used to restrict or alter the passage of traffic for clients onboarding to a switch that has port security (802.1X, MAC authentication) enabled.

Unlike classifier policies, which are associated with individual front plane port, Link Aggregation Group (LAG), and VLAN or tunnel interface, port-access policies are associated with roles. Based on the role associated with a user after authentication, the policy is applied to the user.

The switch can obtain policies from any of the following sources:

- Local: Policies configured locally on the switch.
- RADIUS: Policies configured using the NAS-Filter-Rule or Aruba-NAS-Filter-Rule RADIUS attributes.



NOTE: Both local and downloaded type of policies do not have any standards associated with them. Policies that are obtained from the RADIUS server must support all criteria that can be defined using the `NAS-Filter-Rule` attribute.

Classes and actions supported by port access policies

Port access policies support IPv4- and IPv6-based classes, and the following actions.

- `cir`: Set the bandwidth limit for guaranteed traffic.
- `drop`: Drop the packet.
- `dscp`: Remark the 6-bit field in the IP header for packet classification.
- `ip-precedence`: Remark the 3-bit field in the IP header which denotes the priority of the datagram.
- `local-priority`: Change the internal priority that is used to queue the packets for transmission.

Port access policy commands

port-access policy

Syntax

```
port-access policy <POLICY-NAME>
```

```
[<SEQUENCE-NUMBER>]
class {ip|ipv6} <CLASS-NAME>
    action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
```

```
[<SEQUENCE-NUMBER>]
comment ...
```

Description

Creates or modifies policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6 class and one or more policy actions associated with it.

A policy must be applied to a role using the `associate policy` command.

The `no` form of the command can be used to delete either a policy (use `no` with the `policy` command) or an individual policy entry (use `no` with the sequence number).

Command context

`config`

The `policy` command takes you into the `config-pa-policy` context where you enter the policy entries.

Parameters

<POLICY-NAME>

Specifies the name of the policy.

<SEQUENCE-NUMBER>

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

comment

Specifies the description of an entry in the policy. Optional.

class {ip|ipv6} <CLASS-NAME>

Specifies a type of class, `ip` for IPv4 and `ipv6` for IPv6 policy. And specifies a class name.

<REMARK-ACTIONS>

Remark actions can be any of the following options: {`ip-precedence <IP-PRECEDENCE-VALUE>` | `dscp <DSCP-VALUE>` | `local-priority <LOCAL-PRIORITY-VALUE>`} where:

ip-precedence <IP-PRECEDENCE-VALUE>

Specifies the numeric IP precedence value. Range: 0 to 7.

dscp <DSCP-VALUE>

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

Keyword	Status	Syntax	Description
AF11	Optional	Keyword	DSCP 10 (Assured Forwarding Class 1, low drop probability)
AF12	Optional	Keyword	DSCP 12 (Assured Forwarding Class 1, medium drop probability)
AF13	Optional	Keyword	DSCP 14 (Assured Forwarding Class 1, high drop probability)
AF21	Optional	Keyword	DSCP 18 (Assured Forwarding Class 2, low drop probability)
AF22	Optional	Keyword	DSCP 20 (Assured Forwarding Class 2, medium drop probability)

Table Continued

Keyword	Status	Syntax	Description
AF23	Optional	Keyword	DSCP 22 (Assured Forwarding Class 2, high drop probability)
AF31	Optional	Keyword	DSCP 26 (Assured Forwarding Class 3, low drop probability)
AF32	Optional	Keyword	DSCP 28 (Assured Forwarding Class 3, medium drop probability)
AF33	Optional	Keyword	DSCP 30 (Assured Forwarding Class 3, high drop probability)
AF41	Optional	Keyword	DSCP 34 (Assured Forwarding Class 4, low drop probability)
AF42	Optional	Keyword	DSCP 36 (Assured Forwarding Class 4, medium drop probability)
AF43	Optional	Keyword	DSCP 38 (Assured Forwarding Class 4, high drop probability)
CS0	Optional	Keyword	DSCP 0 (Class Selector 0: Default)
CS1	Optional	Keyword	DSCP 8 (Class Selector 1: Scavenger)
CS2	Optional	Keyword	DSCP 16 (Class Selector 2: OAM)
CS3	Optional	Keyword	DSCP 24 (Class Selector 3: Signaling)
CS4	Optional	Keyword	DSCP 32 (Class Selector 4: Real time)
CS5	Optional	Keyword	DSCP 40 (Class Selector 5: Broadcast video)
CS6	Optional	Keyword	DSCP 48 (Class Selector 6: Network control)
CS7	Optional	Keyword	DSCP 56 (Class Selector 7)
EF	Optional	Keyword	DSCP 46 (Expedited Forwarding)

local-priority *<LOCAL-PRIORITY-VALUE>*

Specifies a local priority value. Range: 0 to 7.

<POLICE-ACTIONS>

Police actions can be the following {cir kbps *<RATE-BPS>* exceed} where:

cir kbps *<RATE-BPS>*

Specifies a Committed Information Rate (CIR) value in Kilobits per second. Range: 1 to 4294967295.

exceed

Specifies action to take on packets that exceed the rate limit.

<OTHER-ACTIONS>

Other actions can be the following:

drop

Specifies drop traffic.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- An applied policy processes the packet sequentially against policy and class entries in the list, until either the last policy entry in the list has been evaluated or the packet matches an entry. If there is no match, the packet will be dropped by one of the implicit `deny all` IPv4 and IPv6 entries.
- Entering an existing `<POLICY-NAME>` value will cause the existing policy to be modified, with any new `<SEQUENCE-NUMBER>` value creating an additional policy entry, and any existing `<SEQUENCE-NUMBER>` value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10. The sequence numbers may be reordered with the `port-access policy <POLICY-NAME> resequence <STARTING-SEQU-NUM> <INCREMENT>` command.
- If a policy is configured without any action, the default action, `permit`, is applied for that policy.

Examples

Creating a policy with several class entries:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip dns
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 30 class ip test action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
10	dns	ipv4	permit
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Adding a comment to an existing class entry:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 20 comment DHCP-PERMIT
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1
```

```
port-access policy POL1
  10 class ip dns
  20 class ip dhcp
  20 comment DHCP-PERMIT
  30 class ip test action cir kbps 1024 exceed drop
```

Removing a comment from an existing class entry:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# no 20 comment
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1

port-access policy POL1
  10 class ip dns
  20 class ip dhcp
  30 class ip test action cir kbps 1024 exceed drop
```

Modifying a policy by replacing one class with another at the same sequence number:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip mds action dscp af21
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:
=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
10	mds	ipv4	dscp AF21
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Removing a class:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# no 10
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:
=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	clearpass-web	ipv4	cir kbps 1024 exceed drop

port-access policy copy

Syntax

```
port-access policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

Description

Copies details of an existing policy to a new policy. Copying a policy copies all its entries as well.

Command context

config

Parameters

<POLICY-NAME>

Specifies the existing policy name.

<DESTINATION-POLICY>

Specifies the destination policy name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a policy:

```
switch(config)# port-access policy POL1 copy POL1_copy
switch(config)# show port-access policy
```

Access Policy Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Policy Name : POL1_copy
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

port-access policy resequence

Syntax

port-access policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>

Description

Resequences numbering in a policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy where you want to resequence policy entries.

<STARTING-SEQ-NUM>

Specifies the sequence number to start resequencing from. Range: 1-4294967295.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing a policy starting at 5 with an increment of 10:

```
switch(config)# port-access policy POL1 resequence 5 10
switch(config)# show port-access policy POL1
```

Access Policy Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
5	dhcp	ipv4	permit
15	test	ipv4	cir kbps 1024 exceed drop

port-access policy reset

Syntax

```
port-access policy <POLICY-NAME> reset
```

Description

Resets the policy configuration to match the current hardware configuration of the policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the name of the policy to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy:

```

switch(config)# port-access policy POL2
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 40 class test2 action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1-V2

```

Access Policy Details:
=====

Policy Name : POL2
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION
20	dhcp	ipv4 permit
40	test2	ipv4 cir kbps 1024 exceed drop

```

switch(config)# port-access policy POLV2
switch(config-pa-policy)# 50 class ip test3 action cir kbps 1024 exceed drop
switch(config-pa-policy)# no 20
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL2

```

Access Policy Details:
=====

Policy Name : POL2
Policy Type : Local
Policy Status : Rejected

SEQUENCE	CLASS	TYPE ACTION
40	test2	ipv4 cir kbps 1024 exceed drop
50	test3	ipv4 cir kbps 1024 exceed drop

```

switch(config)# port-access policy POL2 reset
Following policy entries will be removed:
class ip test3 action cir kbps 1024 exceed drop

```

Following policy entries will be added:
20 class ip dhcp

Do you want to continue (y/n)? **y**
switch(config)# show port-access policy POL2

Access Policy Details:
=====

Policy Name : POL1-V2
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION
20	dhcp	ipv4 permit
40	test2	ipv4 cir kbps 1024 exceed drop

clear port-access policy hitcounts

Syntax

```
clear port-access policy <POLICY-NAME> hitcounts {port | client}
```

Description

Clears statistics and conform rate of policy applied on a port or client.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

port

Clears the policy statistics matching policy name in the port mode.

client

Clears the policy statistics matching policy name in the client mode.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Clearing policy hit counts:

```
switch# show port-access policy POL6 hitcounts port
```

Port Access Policy Hit-Counts Details:

=====

Policy Name : POL4
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION	CUR-RATE (kbps)
3	test8	ipv4 cir kbps 1024 exceed drop	512

Class Name : dhcp
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match icmp any any count	0

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
15	match udp any any count	15101830

Class Name : web-traffic
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	241
20	match any 10.1.1.1 10.1.1.2 dscp AF11 count	50

Class Name : class6
Class Type : ipv6

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	173
20	match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count	32

switch#

switch# **clear port-access policy POL6 hitcounts port**

switch#

switch# **show port-access policy POL6 hitcounts port**

Port Access Policy Hit-Counts Details:

=====

Policy Name : POL4
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION	CUR-RATE (kbps)
3	test8	ipv4 cir kbps 1024 exceed drop	512

Class Name : dhcp
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match icmp any any count	0

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
15	match udp any any count	0

Class Name : web-traffic
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	0
20	match any 10.1.1.1 10.1.1.2 dscp AF11 count	0

Class Name : class6
Class Type : ipv6

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	0
20	match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count	0

show port-access policy

Syntax

show port-access policy [<POLICY-NAME>]

Description

Shows various aspects of policies and their current usage. Details of a policy including the content of a specific policy is shown.

Following are the possible values for policy types:

- **Local**—User configured policy
- **Downloaded**—Downloaded user policy
- **RADIUS**—Policy obtained from the RADIUS server

Following are the possible values for policy status:

- **Applied**—Policy is successfully applied in the hardware.
- **Rejected**—Policy is not supported in the hardware.
- **In-Progress**—Policy is being processed in the hardware.
- **Failed**—Displayed when the switch fails to apply the policy configuration because the TCAM resources are unavailable or full.



NOTE: If a policy is configured without any action, the `show` command will represent such an entry with the action as `permit`.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all policies:

```
switch(config)# show port-access policy
```

```
Access Policy Details:
```

```
=====
```

```
Policy Name   : POL1
Policy Type    : Local
Policy Status  : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

```
Policy Name   : POL1_copy
Policy Type    : Local
Policy Status  : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Showing information for a particular policy:

```
switch(config)# show port-access policy POL1
```

Access Policy Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

show port-access policy hitcounts

Syntax

Syntax that shows statistical information in the form of hit counts:

```
show port-access policy <POLICY-NAME> hitcounts {port | client}
```

Description

Shows the statistics and the current rate of policy applied on the port or the client.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

port

Shows policy statistics matching the policy name in the port mode.

client

Shows policy statistics matching the policy name in the client mode.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing policy hit counts (statistics) with current rate:

```
switch# show port-access policy POL6 hitcounts port
```

Port Access Policy Hit-Counts Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION	CUR-RATE (kbps)
30	test8	ipv4 cir kbps 1024 exceed drop	512

Class Name : dhcp
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match icmp any any count	982150

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
70	match udp any any count	15101830

Class Name : web-traffic
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
4	match any any any count	3194
5	match any 10.1.1.1 10.1.1.2 dscp AF11 count	1716

Class Name : class6
Class Type : ipv6

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	0
20	match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count	0

Port access role

Overview

Every device that connects to a port is associated with a role. Roles are associated with all clients, both authenticated and unauthenticated, and applied to each user session. By default, roles are enabled on a switch.

Following are a few examples of user role names and the access privileges that can be configured:

- Employee—Provide complete access to network resources.
- Contractor—Provide limited access to network resources.
- Guest—Provide only Internet browsing access.

Each user role determines the client network privileges, frequency of reauthentication, applicable bandwidth contracts, and other permissions.



NOTE: Active user roles applied on clients are created only on Ternary Content-Addressable Memory (TCAM) resource availability of the switch.

A user role consists of the following optional parameters:

- ***Ingress user policy***
L3 (IPv4 and/or IPv6) ordered list of classes with actions.
- ***inactivity-timeout***
The inactivity timeout period in seconds with a range of 300 to 4294967295 for the authenticated client for an implicit logoff.
- ***reauth-period***
Sets the reauthentication period in seconds or 0 to disable.
- ***vlan access***
Sets the untagged VLAN ID.
- ***vlan trunk***
Sets the tagged VLAN ID.
- ***auth-mode***
Sets the configuration in user role to either device-mode or port-mode. The following are the attributes:
 - ***poe-priority***
Specifies the PoE priority for the interface.
 - ***mtu***
Configures the MTU support for the client.
 - ***vlan trunk allowed***
Specifies the list of tagged VLANs configured for the interface.
 - ***trust-mode***
Configures the QoS trust mode for the client.

Operational notes

Following are some of the operational notes to be considered for port access roles:

- On the 6100 Switch Series, a maximum of 1024 authenticated clients are supported.
- When roles are enabled, they are applied to all devices connected to ports where authentication is configured.
- Special roles, such as, critical, reject, pre-auth, and auth are applied depending on the authentication state of the device.
- Roles can be applied in one of the following two ways:
 - Vendor-Specific Attribute (VSA)-Derived Role
Type: RADIUS: Aruba
Name: Aruba-User-Role

ID: 25

Value: `<myUserRole>`

See **Vendor-Specific Attributes supported in session authorization**.

The RADIUS server (ClearPass Policy Manager server) determines how the VSA-Derived Role is applied to the user. The role is sent to the switch through a RADIUS VSA. The VSA derived role will have the same precedence order as the authentication type (802.1x, MAC authentication).

- User Derived Role (UDR)

The UDR is applied when the roles are enabled.

UDR will have the same precedence order as the authentication type (802.1x, MAC authentication).

Special roles

Special roles are always local roles that are created to support instances when the deployments are not yet complete. They are also helpful when any network issues occur during authentication of devices. Special roles are always purpose-based, that is, they are applied only in instances such as the RADIUS server is not reachable for authentication or when a single role has to be applied across all switches.

The following special roles are available:

- Critical role
- Reject role
- Pre-authentication role
- Auth-role
- Fallback role

Critical role

The critical role is applied to devices when the RADIUS server is unreachable during the first authentication process or during reauthentication. This role helps ensure that the devices have limited access to the network even though the authentication is not completed. Once the RADIUS server is available for authentication, the devices are authenticated and the ultimate role is applied.

Reject role

The reject role is applied when the RADIUS server rejects a device during authentication. The reject role gives restricted access to the device compared to a full access role.

Pre-authentication role

The pre-authentication (pre-auth) role allows a device, such as an IP phone, to have network access before the device is authenticated. The pre-auth role is triggered when a MAC-based client is connected to a switch before being authenticated by the RADIUS server. Devices must be assigned a VLAN to provide network connectivity. Two new VLANs are created for pre-auth role functionality, one for voice traffic and one for data traffic. Pre-auth role VLANs can be configured on the switch individually or within a user-role. Devices that can be connected to the switch without authentication are divided into two categories:

- Devices that send voice traffic.
- Devices that send data traffic.



NOTE: Either one of pre-auth role VLANs (voice and/or data) or a pre-auth role can be configured for a port. However, both a VLAN and role cannot coexist for an interface. Initial traffic on the port is restricted only by Access Control Lists (ACLs) configured for the port or for VLANs or ACLs in the role.

Impact of pre-auth role on existing features

Unauthenticated devices

Configuring pre-auth role VLAN will change the behavior of unauthenticated devices. Normally, authentication-enabled ports will not provide unauthenticated client any network access until the device is authenticated by the RADIUS server. With pre-auth role VLAN configured, the client will be assigned to the pre-auth role VLAN until the RADIUS server authenticates the device.

Unauthenticated clients will be placed into the VLAN specified in the pre-auth role. After authenticated by the RADIUS server, the client will be placed into the VLAN specified in the RADIUS authentication command string or as specified in the RADIUS authentication accept string.

LLDP-bypass

When LLDP-bypass is enabled on the switch, Aruba APs are not authenticated. Therefore pre-auth role VLAN is not applicable.

Bypass using device-identity

Pre-auth role VLAN is not applicable to VoIP devices because they do not need authentication. It is applicable to PCs that need authentication.

ACLs applied on an interface

If an ACL rule is applied on an interface, which is part of a pre-auth role VLAN, traffic coming through that interface will be affected. Traffic will be affected based on the rule in the ACL.

ACLs applied on a VLAN

If an ACL rule is applied on a pre-auth role VLAN, traffic entering that VLAN will be affected. Traffic will be affected based on the rule in the ACL.

Rate-limiting on an interface

If the traffic is rate-limited on an interface as part of a pre-auth role VLAN, the traffic will be impacted. The traffic will be affected based on the rule in the rate-limiting configuration command.

Authenticated or rejected clients

Clients that are authenticated or rejected by the RADIUS server are given different VLANs. These clients are moved from pre-auth role to new VLANs based on the authentication by the RADIUS server.

MAC pinning

Clients whose MAC addresses are pinned and have undergone authentication will always be treated as authenticated. Pre-auth role VLAN is not applicable in this scenario.

Effect of RADIUS tracking on pre-auth role

If RADIUS tracking is enabled and no RADIUS server is available for authentication, the port will be changed from a pre-auth role VLAN to a critical VLAN. The time taken to move from pre-auth role VLAN to critical VLAN depends on the time it takes for RADIUS tracker to inform the subsystem.

Restrictions

The pre-auth role restrictions are as follows:

- It will not support more than one tagged or untagged VLAN membership either through direct VLAN configuration or through user-roles.
- It is not applicable for authentication methods other than MAC-based.
- It is not available to be configured from WebUI, Menu, or REST.

Auth-role

After the RADIUS server authenticates a device, if no role is configured on the server, it sends an empty access accept packet to the switch. The switch translates this empty packet to assign an auth-role to the device. The switch then checks if an auth-role is configured on that port and assigns this role to the device.

Fallback role

The fallback role is applied to onboarding devices when there is no derived role available for the devices. Following are the conditions for the fallback role to be applied on onboarding devices:

- The device profile local MAC match feature with block-until-profile-applied mode is configured.
- Device profile along with AAA is configured but no match was found for the device profile client.
- AAA method with no reject or critical role is configured, and the connection to RADIUS server failed.
- 802.1X authentication is enabled on the port, but the supplicant of the device timed out to respond to the authentication request.

Port access role commands

associate policy

Syntax

```
associate policy <POLICY-NAME>
```

```
no associate policy <POLICY-NAME>
```

Description

Associates the policy with the current role.

The `no` form of this command dissociates the policy from the role.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<POLICY-NAME>

Specifies the policy name to associate with the current role. The maximum number of characters allowed is 64.



NOTE: Only those policies created by using the `port-access policy` command are allowed to be associated with a role.

Policies created using the `policy` command are not allowed to be associated with a role.

Policies that are of the downloaded type are not allowed to be associated with a role.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating a policy with a role:

```
switch(config)# port-access role role01  
switch(config-pa-role)# associate policy policy01
```

Dissociating a policy from the role:

```
switch(config-pa-role)# no associate policy policy01
```

auth-mode

Syntax

```
auth-mode {client-mode | device-mode}
```

Description

Configures the authentication mode for the clients that are associated with the current role.

Command context

```
config
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

client-mode

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

device-mode

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the client authentication mode:

```
switch(config-pa-role)# auth-mode client-mode
```

cached-reauth-period

Syntax

```
cached-reauth-period [<PERIOD>]
```

```
no cached-reauth-period
```

Description

Enables cached reauthentication, setting the period after which clients that associated with the current role must be reauthenticated.

The `no` form of this command disables cached authentication.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 30 to 4294967295 .

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication and setting its period to 200 seconds:

```
switch(config-pa-role)# cached-reauth-period 200
```

Disabling cached reauthentication:

```
switch(config-pa-role)# no cached-reauth-period
```

client-inactivity timeout

Syntax

```
client-inactivity timeout {<CLIENT-INACTIVITY-PERIOD> | none}
```

```
no client-inactivity timeout
```

Description

Configures the period that the switch waits for a response from a client after which it removes the client from the role.

The `no` form of the command resets the timeout period to the default.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<CLIENT-INACTIVITY-PERIOD>

Specifies the client inactivity time (in seconds). Default: 300. Range: 300 to 4294967295.

none

Specifies that the client must not be deleted because of inactivity.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client inactivity timer for a role:

```
switch(config-pa-role)# client-inactivity timeout 3600
```

description

Syntax

```
description <ROLE-DESCRIPTION>
```

Description

Configures the role description.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<ROLE-DESCRIPTION>

Specifies the role description. Up to 255 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the role description:

```
switch(config-pa-role)# description student role
```

mtu

Syntax

```
mtu <MTU-SIZE>
```

Description

Configures the MTU size of a client for a role.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<MTU-SIZE>

Specifies the MTU size in bytes of a client for a role. Range: 68 to 9198.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client MTU size:

```
switch(config-pa-role) # mtu 9198
```

poe-priority

Syntax

```
poe-priority {critical | high | low}
```

```
no poe-priority
```

Description

Configures the power distribution priority for the port access roles. High power consumption can be prevented using `poe-priority` control mechanism.

The `no` form of this command restores the power distribution to its default priority.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

critical

Specifies critical priority.

high

Specifies high priority.

low

Specifies low priority.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring PoE priority for a new role:

```
switch(config) # port-access role role01  
switch(config-pa-role) # poe-priority critical
```

Resetting PoE priority for the role to its default:

```
switch(config) # port-access role role01  
switch(config-pa-role) # no poe-priority
```

port-access role

Syntax

```
port-access role <ROLE-NAME>
```

```
no port-access role <ROLE-NAME>
```

Description

Creates a new port access role or modifies an existing role. This command takes you into the `config-pa-role` context. A maximum of 32 port access roles can be created.

The `no` form of this command deletes a role.

Command context

```
config
```

Parameters

<ROLE-NAME>

Specifies the role name. The maximum number of characters allowed is 64.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a new role:

```
switch(config)# port-access role basic01
```

reauth-period

Syntax

```
reauth-period <PERIOD>
```

```
no reauth-period
```

Description

Configures the period after which clients that associated with the current role must be reauthenticated.



NOTE: The reauthentication period configured here takes precedence over the reauthentication period configured at the port level.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<PERIOD>

Specifies the reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 1 to 86400.



NOTE: Reauthentication period of less than 60 seconds is not recommended.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication period:

```
switch(config-pa-role) # reauth-period 3000
```

session timeout

Syntax

```
session-timeout <SESSION-TIMEOUT>
```

```
no session-timeout
```

Description

Configures the session timeout for the role. After the timeout period, the session will be disconnected.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<SESSION-TIMEOUT>

Specifies the session timeout (in seconds). Range: 1 to 4294967295.



NOTE: A session timeout period of less than 60 seconds is not recommended.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring session timeout for a role:

```
switch(config-pa-role) # session timeout 3600
```

show aaa authentication port-access interface client-status

Syntax

```
show aaa authentication port-access interface {all | <IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about the status of the role applied on ports.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the instance name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information about a client:

```
switch# show aaa authentication port-access interface all client-status mac 00:00:00:00:00:01

Port Access Client Status Details

Client 00:00:00:00:00:01
=====
Session Details
-----
Port           : 1/7/24
Session Time   : 151s

Authentication Details
-----
Status          : mac-auth Authenticated
Auth Precedence : mac-auth - Authenticated, dot1x - Not attempted

Authorization Details
-----
Role            : UserRole_1
Status          : Applied
```

show port-access role

Syntax

```
show port-access role {local | radius | name <ROLE-NAME>}
```

Description

Shows information about roles configured locally, or downloaded from RADIUS server.

Command context

Operator (>) or Manager (#)

Parameters

local

Shows information about locally configured roles.

radius

Shows information about roles downloaded from the RADIUS server.

<ROLE-NAME>

Specifies the role name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing locally configured role information:

```
switch# show port-access role local
```

Role Information

```
Name   : local_role_01
Type    : local
-----
Reauthentication Period      : 333 secs
Authentication Mode          :
Session Timeout              :
Client Inactivity Timeout    :
Tunneled Node Server Zone    :
Tunneled Node Server Secondary Role :
Access VLAN                  :
Native VLAN                  :
Allowed Trunk VLANs          :
MTU                           :
QoS Trust Mode                :
PoE Priority                   : low
Policy                        :
```

Showing information for roles downloaded from a RADIUS server:

```
switch# show port-access role radius
```

Role Information

```
Name   : RADIUS_21963402
Type    : radius
-----
Reauthentication Period      : 333 secs
Authentication Mode          :
Session Timeout              :
Client Inactivity Timeout    :
Tunneled Node Server Zone    :
Tunneled Node Server Secondary Role :
Access VLAN                  : 10
Native VLAN                  :
Allowed Trunk VLANs          :
MTU                           :
QoS Trust Mode                :
PoE Priority                   : low
Policy                        : PERMIT-ALL_87364653
```


stp-admin-edge-port

Syntax

```
stp-admin-edge-port
```

```
no stp-admin-edge-port
```

Description

Configures the port as a spanning tree administrative edge port for the role. This configuration removes the port participation from STP interactions when onboarding devices. This in turn helps in faster onboarding of devices.

The `no` form of the command disables STP edge port functionality.



NOTE: If the port receives STP BPDU on the STP administrative edge configured port, the port will move to the STP state. You must configure the port as an STP administrative edge port only if you are sure that the connected device will not participate in STP interactions.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring STP edge port for a role:

```
switch(config)# port-access role role01  
switch(config-pa-role)# stp-admin-edge-port
```

trust-mode

Syntax

```
trust-mode [dscp | cos | none]
```

```
no trust-mode
```

Description

Configures QoS trust mode for the role.

The `no` form of this command configures the default trust mode for the role.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

dscp

Specifies to trust DSCP and preserve the 802.1p priority.

cos

Specifies to trust 802.1p priority and retain DSCP or IP-ToS.

none

Specifies to not trust any priority fields.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring DSCP trust mode for a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# trust-mode dscp
```

vlan

Syntax

```
vlan {access | trunk native | trunk allowed} <VLAN-ID>
```

```
no vlan {access | trunk native | trunk allowed} <VLAN-ID>
```

Or

```
vlan {access name | trunk native name | trunk allowed name} <VLAN-NAME>
```

```
no vlan {access name | trunk native name | trunk allowed name} [<VLAN-NAME>]
```

Description

Configures VLAN IDs or VLAN names, and VLAN modes for a port access role. You can configure either VLAN IDs or VLAN names, or a combination of both for a role.

The `no` form of the command deletes the VLAN configuration from the role. For trunk allowed VLAN names, you can delete the VLAN names individually or all names at once.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

access <VLAN-ID>

Specifies the VLAN ID for the access VLAN. Supports a single VLAN ID in the range 1 to 4094.

trunk native <VLAN-ID>

Specifies the native VLAN ID on the trunk interface. Supports a single VLAN ID in the range 1 to 4094.

trunk allowed <VLAN-ID>

Specifies the list of tagged or allowed VLANs on the trunk interface. Supports a list of VLAN IDs in the range 1 to 4094. The 6300, 6400 Switch Series support a maximum of 1024 trunk allowed VLAN IDs. The 6200 Switch Series supports a maximum of 256 trunk allowed VLAN IDs.

access name <VLAN-NAME>

Specifies the VLAN name for the access VLAN. Supports a single VLAN name with a maximum length of 32 characters.

trunk native name <VLAN-NAME>

Specifies the native VLAN name on the trunk interface. Supports a single VLAN name with a maximum length of 32 characters.

trunk allowed name <VLAN-NAME>

Specifies the tagged or allowed VLAN name on the trunk interface. Supports a single VLAN name in each CLI with a maximum length of 32 characters. The switch supports a maximum of 50 trunk allowed VLAN names.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Note the following points when configuring the VLAN IDs and names for a role:

- For VLAN access and VLAN trunk native respectively, it is recommended to configure only one of either VLAN ID or name for a role. In case both VLAN ID and name are configured, then VLAN ID takes precedence and is applied with the role.
- For VLAN trunk allowed, you can collectively configure a maximum of 50 names and 1024 VLAN IDs on the 6300, 6400 Switch Series, or 256 VLAN IDs on the 6200 Switch Series. In case this limit is exceeded in the role, then that role is rejected when applying it to an onboarding device.

Examples

Configuring VLAN modes and VLAN IDs for a new role:

```
switch(config)# port-access role role01
switch(config-pa-role)# vlan trunk native 10
switch(config-pa-role)# vlan trunk allowed 11-15
switch(config-pa-role)# vlan access 50
```

Configuring VLAN modes and VLAN names for a new role:

```
switch(config)# port-access role role10
switch(config-pa-role)# vlan trunk native name hpe01
switch(config-pa-role)# vlan trunk allowed name data
switch(config-pa-role)# vlan trunk allowed name voice
switch(config-pa-role)# vlan trunk allowed name video
```

Deleting VLAN configuration from a role:

```
switch(config-pa-role)# no vlan trunk native 10
switch(config-pa-role)# no vlan trunk allowed 10-15
switch(config-pa-role)# no vlan access 50
```

Deleting trunk allowed VLAN names from a role individually:

```
switch(config-pa-role)# no vlan trunk native name hpe01
switch(config-pa-role)# no vlan trunk allowed name data
switch(config-pa-role)# no vlan trunk allowed name voice
switch(config-pa-role)# no vlan trunk allowed name video
```

Deleting trunk allowed VLAN names from a role all at once:

```
switch(config-pa-role)# no vlan trunk native name hpe01  
switch(config-pa-role)# no vlan trunk allowed name
```

ArubaOS-CX supports various RADIUS server attributes to be applied during authentication of clients. This section lists the attributes supported in the following features:

- 802.1X authentication
- MAC authentication
- Dynamic authorization
- Session authorization in 802.1X and MAC authentication, and CoA
- RADIUS server tracking
- RADIUS network accounting



NOTE: The following terms are used in the list of attributes:

- **Tx:** Attribute added in the request packets that are sent to the RADIUS server.
- **Rx:** Attribute processed in the response packets received from the RADIUS server.

Attributes supported in 802.1X authentication

Following are the RADIUS attributes supported in 802.1X authentication:

Attribute name	Tx	Rx	Notes
User-name	Y	N	RFC2865
Calling-station-id	Y	N	RFC2865/3580
Called-station-id	Y	N	RFC2865/3580
NAS-port-id	Y	N	RFC2869
NAS-port	Y	N	RFC2865
Service-Type	Y	N	RFC2865
EAP-Message	Y	N	RFC2869
State	Y	Y	RFC2865
Session-Timeout	N	Y	RFC2865 - Used as EAP timeout in Challenge packet
NAS-IP-Address	Y	N	RFC2865
NAS-Identifier	Y	N	RFC2865
NAS-Ipv6-Address	Y	N	RFC3162
Message-Authenticator	Y	Y	RFC2869

Attributes supported in MAC authentication

Following are the RADIUS attributes supported in MAC authentication:

Attribute name	Tx	Rx	Notes
User-name	Y	N	RFC2865
Calling-station-id	Y	N	RFC2865/3580
Called-station-id	Y	N	RFC2865/3580
NAS-port-id	Y	N	RFC2869

NAS-port	Y	N	RFC2865	
Service-Type	Y	N	RFC2865	
State	Y	Y	RFC2865	
NAS-IP-Address	Y	N	RFC2865	
NAS-Identifier	Y	N	RFC2865	
NAS-IPv6-Address	Y	N	RFC3162	
Message-Authenticator	Y	Y	RFC2869	
Chap-Challenge	Y	N	RFC2865	
Chap-Password	Y	N	RFC2865	
User-Password	Y	N	RFC2865	

Attributes supported in dynamic authorization

Following are the RADIUS attributes supported in dynamic authorization:

Attribute name	Tx	Rx	Notes	
-----	----	----	-----	
User-name	N	Y	RFC2865	
Calling-station-id	N	Y	RFC2865/3580	
Called-station-id	N	Y	RFC2865/3580	
NAS-port-id	N	Y	RFC2869	
NAS-port	N	Y	RFC2865	
NAS-IP-Address	N	Y	RFC2865	
NAS-Identifier	N	Y	RFC2865	
NAS-IPv6-Address	N	Y	RFC3162	
Error-cause	Y	N	RFC5176	
Acct-Terminate-Cause	Y	Y	RFC2866	
Acct-Session-Id	N	Y	RFC2866	
Message-Authenticator	N	Y	RFC2869	
Event-Timestamp	N	Y	RFC2869	



NOTE:

- One or more of the following attributes, `NAS-IP-Address`, `NAS-Identifier`, `NAS-IPv6-Address`, is mandatory for processing Change of Authorization (CoA) requests from dynamic authorization clients.
- Either `Acct-Session-Id` attribute, or one or both of `NAS-port` and `NAS-port-id` attributes and the `Calling-station-id` attribute is mandatory to identify the user session for processing CoA requests from dynamic authorization clients.
- The `User-name` attribute is mandatory for processing all CoA requests.
- The `Acct-Terminate-Cause` attribute is used in the CoA disconnect request and response messages.
- All session authorization attributes (both VSA and standard) are supported in the CoA message only, including `Aruba-Port-Bounce`.

Session authorization attributes supported in 802.1X and MAC authentication, and CoA

Standard session attributes supported

Following are the standard session attributes supported in 802.1X and MAC authentication, and CoA:

Attribute name	ID	Type	Notes
Filter-Id	11	String	RFC3580
Egress-VLANID	56	Octet	RFC4675
Egress-VLAN-Name	58	String	RFC4675
Tunnel-Type	64	Octet	RFC2868
Tunnel-Medium-Type	65	Octet	RFC2868
Tunnel-Private-Group-ID	81	String	RFC2868
NAS-Filter-Rule	92	String	RFC4849
Framed-MTU	12	Octet	RFC2865
Session-Timeout	27	Octet	RFC2865
Terminate-Action	29	Octet	RFC2865
Idle-Timeout	28	Octet	RFC2865



NOTE: When multiple clients request a different MTU value using the Framed-MTU attribute, the highest MTU value requested among the clients will be programmed on the port.

Vendor-Specific Attributes supported in session authorization

Following are the Vendor-Specific Attributes (VSAs) supported in session authorization:

Attribute Name	Length	Type	Aruba Vendor ID	Aruba Attribute Type
Aruba-PoE-Priority	4	integer	14823	49
Aruba-Port-Auth-Mode	4	integer	14823	50
Aruba-NAS-Filter-Rule	<=247	string	14823	51
Aruba-QoS-Trust-Mode	4	integer	14823	52
Aruba-User-Role	<=63	string	14823	1
Aruba-Port-Bounce	4	Integer	14823	40
Aruba-STP-Admin-Edge-Port	4	Integer	14823	55



NOTE:

- Change of Authorization of specific attributes in the user role is not supported. Only entire role can be changed.
- Similarly if the session is using RADIUS attributes, CoA can change only the RADIUS session attributes.
- Change of Authorization to user role for a session using RADIUS attributes is not supported either. If this action is attempted, a NAK message is sent.

Description of VSAs

Aruba-PoE-Priority

Specifies the PoE priority of onboarding devices post authentication. Following are the supported values:

- 0: Critical
- 1: Medium
- 2: Low

This attribute overrides the PoE priority configured on the port where the device onboards. This attribute is typically used for infrastructure devices. When multiple clients request different priorities, the critical priority takes precedence over medium priority and the medium priority takes precedence over low priority setting.

Aruba-Port-Auth-Mode

Specifies the authentication mode of the port post authentication. Following are the supported values:

- 1: Device mode—In this mode, an infrastructure device, for example, switch or access point, is authenticated first, and all devices connecting to this authenticated device are allowed access. Here, the policy and VLAN attributes are applied at the port-level. In device mode, it is expected that only one device is active and authenticated at any instant. Untagged VLAN will override the port VLAN ID and the tagged VLANs will override the tagged VLANs that are configured on the port using the CLI.
- 2: Client mode—In this mode, all devices trying to onboard on that port are authenticated. Here, the policy and VLAN attributes are applied per client. Untagged VLAN is configured using MAC-based VLAN. Tagged VLANs are arbitrated among all clients and the result is applied to the port.

Aruba-NAS-Filter-Rule

This attribute is similar to the NAS-Filter-Rule RFC attribute but with additional functionality to support vendor-specific actions in the rule. This attribute can be used to perform actions such as count and rate-limiting. Multiple instances of this attribute are supported, however, the maximum number of filter rules (including this VSA and NAS-Filter-Rule) supported per client is 128. A single NAS-Filter-Rule attribute split across multiple VSAs is not supported as servers such as FreeRadius and ClearPass Policy Manager do not terminate filter rule with '\0' value. Most policy servers use one filter rule per VSA.

Aruba-QoS-Trust-Mode

Specifies how the switch assigns local priority values to ingress packets. Following are the supported values:

- 0: Trust mode DSCP
- 1: Trust mode QoS
- 2: No trust mode configuration

This attribute overrides the trust mode configured on the port where the device onboards. This attribute is typically used for infrastructure devices. When multiple clients request different trust modes, the DSCP trust mode takes precedence over QoS trust mode and the QoS trust mode takes precedence over no trust mode configuration setting.

Aruba-User-Role

Specifies the role that must be applied for the devices post authentication. The role must be defined on the switch. All the session attributes can be defined in the role. Session authorization attributes (both standard and VSAs) sent with this attribute are ignored.

Aruba-Port-Bounce

Used in the CoA message to signal the switch to shut down the port for the duration specified.

Aruba-STP-Admin-Edge-Port

When enabled, the port will be treated as STP edge port. Even if STP is configured on the port, it will not be executed. Traffic forwarding will occur immediately when a device connects to the port and completes authentication. Following are the supported values:

- 0: Disable STP admin edge port.
- 1: Enable STP admin edge port.

Attributes supported in RADIUS network accounting

Following are the attributes supported in RADIUS network accounting:

Attribute name	Tx	Rx	Notes
User-name	N	N	RFC2865
Calling-station-id	Y	N	RFC2865
Calling-station-id	Y	N	RFC2865
NAS-port-id	Y	N	RFC2869
NAS-port	Y	N	RFC2865
NAS-port-Type	Y	N	RFC2865
Service-Type	Y	N	RFC2865
NAS-IP-Address	Y	N	RFC2865
NAS-Identifier	Y	N	RFC2865
NAS-Ipv6-Address	Y	N	RFC3162
Acct-Authentic	Y	N	RFC2866
Acct-Session-Id	Y	N	RFC2866
Acct-Status-Type	Y	N	RFC2866
Acct-Input-Octets	Y	N	RFC2866 - In Interim and Stop packets
Acct-Output-Octets	Y	N	RFC2866 - In Interim and Stop packets
Acct-Input-Packets	Y	N	RFC2866 - In Interim and Stop packets
Acct-Output-Packets	Y	N	RFC2866 - In Interim and Stop packets
Acct-Input-Gigawords	Y	N	RFC2869 - In Interim and Stop packets
Acct-Output-Gigawords	Y	N	RFC2869 - In Interim and Stop packets
Acct-Session-Time	Y	N	RFC2866 - In Interim and Stop packets
Acct-Terminate-Cause	Y	N	RFC2866 - in Stop packet
Class	Y	N	RFC2865

Attributes supported in RADIUS server tracking

Following are the attributes supported in RADIUS server tracking:

Attribute name	Tx	Rx	Notes
User-name	Y	N	RFC2865
NAS-IP-Address	Y	N	RFC2865
NAS-Identifier	Y	N	RFC2865
NAS-Ipv6-Address	Y	N	RFC3162
Chap-Challenge	Y	N	RFC2865
Chap-Password	Y	N	RFC2865
User-Password	Y	N	RFC2865

Overview

Port security enables you to configure each switch port with a unique list of the MAC addresses of devices that are authorized to access the network through that port. This security enables individual ports to detect, prevent, and log attempts by unauthorized devices to communicate through the switch.

MAC Lockdown, also known as Static Addressing, is used to prevent station movement and MAC address hijacking, by allowing a given MAC address to use only an assigned port on the switch. MAC Lockdown also restricts the client device to a specific VLAN.

MAC Lockout enables blocking a specific MAC address so that the switch drops all traffic to or from the specified address.



NOTE: Port security does not prevent intruders from receiving broadcast and multicast traffic. MAC Lockdown has a higher priority over port security.

Basic operation

Default port security operation

The default port security setting for each port is dynamic mode in which the switch learns addresses from inbound traffic from any connected device.

Intruder protection

A port that detects an intruder blocks the intruding device from transmitting to the network through that port.

General operation for port security

On a per-port basis, you can configure security measures to block unauthorized devices, and to send notice of security violations. Once port security is configured, you can then monitor the network for security violations through one or more of the following:

- Alert flags captured by network management tools.
- Alert Log entries in the WebAgent
- Event Log entries in the console interface

For any port, you can configure the following:

- Action—Used when a port detects an intruder. Specifies whether to send an SNMP trap to a network management station and whether to disable the port.
- Address limit—Sets the number of authorized MAC addresses allowed on the port.
- Address learn mode—Specify how the port acquires authorized MAC addresses. After configuring the maximum secure MAC address limit, the MAC address can be learned by one of the following ways:

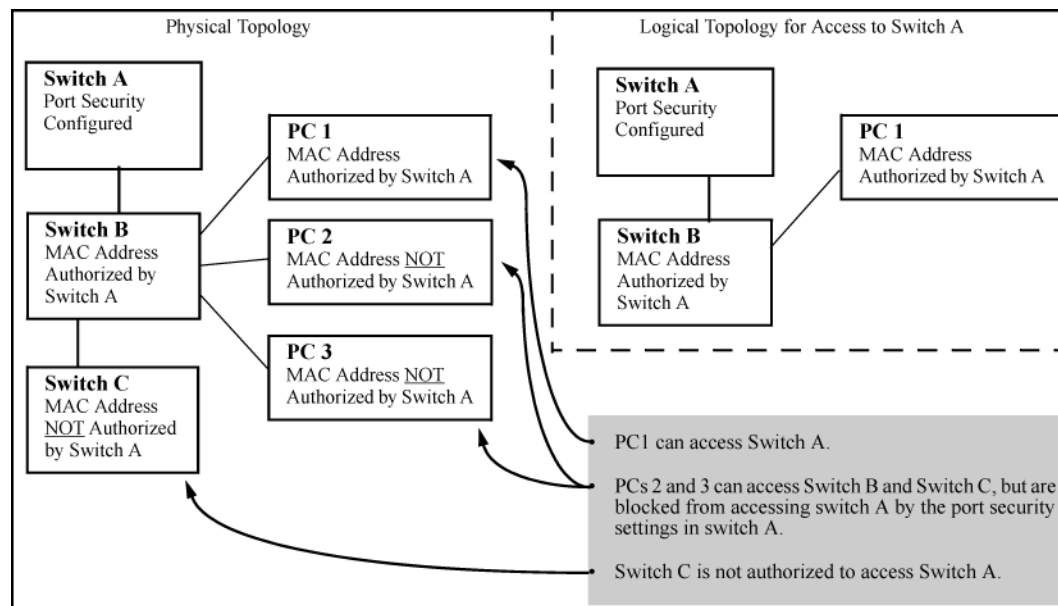
- Configure all MAC addresses by using the `mac-address` command.
- Allow the port to dynamically learn all MAC addresses.
- Configure some MAC addresses and allow the rest to be learned dynamically.
- Static—Enables you to set a fixed limit on the number of MAC addresses authorized for the port and to specify some or all the authorized addresses. (If you specify only some of the authorized addresses, the port learns the remaining authorized addresses from the traffic it receives from connected devices.)
- Configured—Requires that you specify all MAC addresses authorized for the port. The port is not allowed to learn addresses from inbound traffic.
- Authorized (MAC) Addresses—Specify up to eight devices (MAC addresses) that are allowed to send inbound traffic through the port. This feature:
 - Closes the port to inbound traffic from any unauthorized devices that are connected to the port.
 - Provides the option for sending an SNMP trap notifying of an attempted security violation to a network management station and, optionally, disables the port.
- Port Access—Allows only the MAC address of a device authenticated through the switch 802.1X Port-Based access control.

Blocking unauthorized traffic

Unless you configure the switch to disable a port on which a security violation is detected, the switch security measures block unauthorized traffic without disabling the port. This implementation enables you to apply the security configuration to ports on which hubs, switches, or other devices are connected, and to maintain security while also maintaining network access to authorized users.

Example

Figure 1: How port security controls access



NOTE: Broadcast and Multicast traffic is always allowed, and can be read by intruders connected to a port on which you have configured port security.

Trunk group exclusion

Port security does not operate on either a static or dynamic trunk group. If you configure port security on one or more ports that are later added to a trunk group, the switch resets the port security parameters for those ports to the factory default configuration. Ports configured for either Active or Passive LACP, and which are not members of a trunk, can be configured for port security.

Port security commands

port-access port-security

Syntax

```
port-access port-security {enable | disable}
```

Description

Enables or disables port security globally or at the port level.

Command context

```
config
```

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling port security globally:

```
switch(config)# port-access port-security enable
```

Disabling port security globally:

```
switch(config)# port-access port-security disable
```

Enabling port security on a port:

```
switch(config-if)# port-access port-security enable
```

Disabling port security on a port:

```
switch(config-if)# port-access port-security disable
```

port-access port-security client-limit

Syntax

```
port-access port-security client-limit <CLIENTS>
```

```
no port-access port-security client-limit
```

Description

Configures the maximum number of clients that are allowed on a port. After configuring the maximum clients limit, the MAC addresses of the clients can be learned by one of the following methods:

- User can manually configure all MAC addresses by using the `mac-address` command.
- User can allow the port to dynamically learn all MAC addresses.
- User can configure a fixed number of MAC addresses and allow the switch to learn the remaining addresses dynamically.

The `no` form of the command resets the number of clients to the default, 1.

Command context

`config-if`

Parameters

<CLIENTS>

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client limit on a port:

```
switch(config-if)# port-access port-security enable
switch(config-if-port-security)# client-limit 64
```

port-access port-security mac-address

Syntax

```
port-access port-security mac-address <MAC-ADDRESS>
no port-access port-security mac-address <MAC-ADDRESS>
```

Description

Configures the MAC addresses of static clients.

The `no` form of this command removes an authorized static client from the port.

Command context

`config-if`

Parameters

<MAC-ADDRESS>

Specifies the MAC addresses of the static clients.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# mac-address aa:bb:cc:dd:ee:ff
```

Deleting a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# no mac-address aa:bb:cc:dd:ee:ff
```

show port-access port-security interface client-status

Syntax

```
show port-access port-security interface {all|<IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

Description

Shows port security clients status information for the ports. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports:

```
switch# show port-access port-security interface all client-status
```

Port Security Client Status Details

Authorized-Clients	Port
AB:CD:DE:FF:AA:BB	1/1/1
DD:CD:AB:CD:EE:01	1/1/2

Showing client status information for a client:

```
switch# show port-access port-security interface 1/1/1 client-status mac ab:cd:de:ff:aa:bb
```

Port Security Client Status Details

Authorized-Clients	Port
AB:CD:DE:FF:AA:BB	1/1/1

Showing client status information for a port:

```
switch# show port-access port-security interface 1/1/1 client-status
```

Port Security Client Status Details

Authorized-Clients	Port
70:00:02:b8:50:11	1/1/1
70:00:02:b8:50:3d	1/1/1
70:00:02:b8:50:46	1/1/1
70:00:02:b8:50:3b	1/1/1
70:00:02:b8:50:36	1/1/1
70:00:02:b8:50:48	1/1/1
70:00:02:b8:50:1e	1/1/1
70:00:02:b8:50:1a	1/1/1
70:00:02:b8:50:13	1/1/1
70:00:02:b8:50:26	1/1/1
70:00:02:b8:50:2d	1/1/1

show port-access port-security interface port-statistics

Syntax

```
show port-access port-security interface {all|<IF-NAME>} port-statistics
```

Description

Shows port security statistics for the ports in a switch. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show port-access port-security interface all port-statistics
```

```
Port 1/1/1
=====
```

```
Client Details
-----
```

```
Number of authorized clients    : 2
```

Overview

ArubaOS-CX switches consist of automatic detection and control for certain link errors and excessive traffic conditions. Fault monitor can be used to log an event or send SNMP traps for these conditions and temporarily disable the port to protect the network. Monitoring can be enabled for all recognized faults or for individual faults after the threshold action and auto-enable parameters for each can be configured.

Fault monitor applies only to physical ports and not to LAGs, tunnels, VSF links, or other types of interfaces. Faults can be applied to the individual members of a LAG.

Fault monitoring conditions

The following fault conditions are monitored:

Excessive oversize packets

An excessive oversized packet fault is reported when the amount of ingress oversized frames per 10,000 received frames exceeds the configured threshold value in a twenty second interval. In oversize packet fault, the packets size is more than the configured MTU on the interface with good cyclic redundancy check (CRC).

Excessive jabbers

An excessive jabbers fault is reported when the amount of ingress jabber frames per 10,000 received frames exceeds the configured threshold value in a twenty second interval. In jabbers fault, the packets size is more than the configured MTU on the interface with bad CRC.

Excessive fragments

An excessive fragment fault is reported when the amount of ingress fragment frames per 10,000 received frames exceeds the configured threshold value in a twenty second interval. In fragments fault, the packet size is lesser than the configured MTU on the interface with bad CRC.

Excessive CRC errors

An excessive CRC error fault is reported when the amount of ingress `crc-error` frames per 10,000 received frames exceeds the configured threshold value in a twenty second interval.

Excessive TX drops

Excessive TX drops fault is an example of over bandwidth. It is reported when egress dropped packets per 10,000 transmitted frames exceeds the configured threshold value in a twenty second interval.

Excessive link flaps

A link flap fault is reported when the count of transitions between link-up and link-down state exceeds the configured threshold in a ten second interval.

Excessive broadcasts

A broadcast storm fault is reported when the average ingress traffic rate of broadcast packets exceeds the configured threshold in a twenty second interval.

The default threshold level is configured as a percentage of the bandwidth of the port. Larger the frame size, smaller the converted threshold value in PPS. Hence larger frames require lower threshold percent configurations to hit the fault.

Excessive multicasts

A multicast storm fault is reported when the average ingress traffic rate of multicast packets exceeds the configured threshold in a twenty second interval.

Excessive collisions

An excessive collision fault is an example of over bandwidth, it gets reported when egress `collision` frames per 10,000 transmitted frames exceeds the configured threshold value in a 20 second interval.

Excessive Late Collisions

An excessive late collision fault is reported when ingress `late-collision` frames per 10,000 received frames exceeds the configured threshold value in a 300 second interval.

Excessive alignment errors

A full duplex mismatch fault is reported when ingress `alignment-error` frames per 10,000 received frames exceeds the configured threshold value in a 20 second interval.

Fault monitor commands

Command for creating a fault monitoring profile

Syntax

```
fault-monitor profile <profile-name>
no fault-monitor profile <profile-name>
```

Description

Creates a fault monitoring profile. You can create maximum of 16 fault monitoring profiles to monitor different types of interfaces.

The `no` form of this command deletes the fault monitoring profile.

Command context

`config`

Parameters

<profile-name>

Specifies the fault monitor profile name. Profile name accepts the special characters and the name must not exceed 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a fault monitor profile:

```
switch(config)# fault-monitor profile noisy-ports  
switch(config-fault-monitor-profile)#
```

Deleting a fault monitor profile:

```
switch(config)# no fault-monitor profile noisy-ports  
switch(config)#
```

Commands for enabling and disabling the faults

Syntax

```
all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |  
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |  
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors
```

```
no all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |  
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |  
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors
```

Description

Enables the faults for a profile.



NOTE: If user has not configured threshold, action or auto-enable parameters, then this command enables fault with the default values.

By default, the faults are disabled in a profile. Configuring threshold, action, or auto-enable will not enable the fault.

The `no` form of this command disables the faults for a profile.

Command context

config-fault-monitor-profile

Parameters

all

Enables all the faults for a profile.

excessive-oversize-packets

Enables excessive oversize packets fault for a profile.

excessive-jabbers

Enables excessive jabbers fault for a profile.

excessive-fragments

Enables excessive fragments fault for a profile.

excessive-crc-errors

Enables excessive CRC errors fault for a profile.

excessive-tx-drops

Enables excessive transmit fault for a profile.

excessive-link-flaps

Enables excessive link flaps fault for a profile.

excessive-broadcasts

Enables excessive broadcasts fault for a profile.

excessive-multicasts

Enables excessive multicasts fault for a profile.

excessive-collisions

Enables excessive collisions fault for a profile.

excessive-late-collisions

Enables excessive late collisions fault for a profile.

excessive-alignment-errors

Enables excessive alignment errors fault for a profile.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the faults for a profile:

```
switch(config-fault-monitor-profile) # all
switch(config-fault-monitor-profile) # excessive-oversize-packets
switch(config-fault-monitor-profile) # excessive-jabbers
switch(config-fault-monitor-profile) # excessive-fragments
switch(config-fault-monitor-profile) # excessive-crc-errors
switch(config-fault-monitor-profile) # excessive-tx-drops
switch(config-fault-monitor-profile) # excessive-link-flaps
switch(config-fault-monitor-profile) # excessive-broadcasts
switch(config-fault-monitor-profile) # excessive-multicasts
switch(config-fault-monitor-profile) # excessive-collisions
switch(config-fault-monitor-profile) # excessive-late-collisions
switch(config-fault-monitor-profile) # excessive-alignment-errors
```

Disabling the faults for a profile:

```
switch(config-fault-monitor-profile) # no all
switch(config-fault-monitor-profile) # no excessive-oversize-packets
switch(config-fault-monitor-profile) # no excessive-jabbers
switch(config-fault-monitor-profile) # no excessive-fragments
switch(config-fault-monitor-profile) # no excessive-crc-errors
switch(config-fault-monitor-profile) # no excessive-tx-drops
switch(config-fault-monitor-profile) # no excessive-link-flaps
switch(config-fault-monitor-profile) # no excessive-broadcasts
switch(config-fault-monitor-profile) # no excessive-multicasts
switch(config-fault-monitor-profile) # no excessive-collisions
switch(config-fault-monitor-profile) # no excessive-late-collisions
switch(config-fault-monitor-profile) # no excessive-alignment-errors
```

Commands for applying or changing the action and configuring auto-enable for a fault

Syntax

```
[all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors]
[action {notify | notify-and-disable [auto-enable <timeout>]}]
```

```
no [all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors]
[action {notify | notify-and-disable [auto-enable]}]
```

Description

Applies an action and configures the auto-enable timer for the fault. Default action: `notify` with auto-enable timer disabled.

The `no` form of this command removes the action and auto-enable timer for the fault.

Command context

`config-fault-monitor-profile`

Parameters

notify

Specifies the action as `notify` for the faults. Notifies through events, DLOGs, and SNMP trap. It is enabled by default.

notify-and-disable

Specifies the action as `notify-and-disable` for the faults. Notifies through events, DLOGs, and SNMP trap, and then disables the port.

auto-enable <timeout>

Re-enables the port automatically after the specified time. Specifies the time value in seconds. Time range: 1-604800 seconds.



NOTE: The fault parameter values are saved in the database even after the fault is disabled from the profile. These values are used when you enable the fault in the profile again.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring `notify-and-disable` action for the faults:

```
switch(config-fault-monitor-profile)# all action notify-and-disable
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable
switch(config-fault-monitor-profile)# excessive-late-collisions action notify-and-disable
switch(config-fault-monitor-profile)# excessive-alignment-errors action notify-and-disable
```

Configuring `notify` action for the faults:

```
switch(config-fault-monitor-profile)# all action notify
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify
switch(config-fault-monitor-profile)# excessive-jabbers action notify
switch(config-fault-monitor-profile)# excessive-collisions action notify
```

Configuring `notify-and-disable` action with `auto-enable` timer for the faults:

```
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable auto-enable 80
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable auto-enable 100
switch(config-fault-monitor-profile)# excessive-collisions action notify-and-disable auto-enable 70
```

Changing the fault action to default:

```
switch(config-fault-monitor-profile)# no excessive-oversize-packets action
switch(config-fault-monitor-profile)# no excessive-jabbers action
switch(config-fault-monitor-profile)# no excessive-alignment-errors action
```

Removing the auto-enable timer configuration for the fault:

```
switch(config-fault-monitor-profile)# no all action notify-and-disable auto-enable
switch(config-fault-monitor-profile)# no excessive-jabbers action notify-and-disable auto-enable
switch(config-fault-monitor-profile)# no excessive-alignment-errors action notify-and-disable auto-enable
```

Commands for configuring the threshold for the fault

Syntax

```
excessive-jabbers | excessive-crc-errors | excessive-oversize-packets |
excessive-fragments | excessive-tx-drops | excessive-collisions |
excessive-late-collisions | excessive-alignment-errors {threshold value <value>}

no excessive-jabbers | excessive-crc-errors | excessive-oversize-packets |
excessive-fragments | excessive-tx-drops | excessive-collisions |
excessive-late-collisions | excessive-alignment-errors {threshold}

excessive-link-flaps {threshold count <count>}

no excessive-link-flaps {threshold}

excessive-broadcasts | excessive-multicasts {threshold [percent <value> | pps <value>]}

no excessive-broadcasts | excessive-multicasts {threshold}
```

Description

Configures and changes the fault threshold value for the profile.

The **no** form of this command sets the threshold to its default value.

Command context

config-fault-monitor-profile

Parameters

threshold value <value>

Specifies the fault threshold value. Default threshold value: 25.

threshold count <value>

Specifies the fault threshold count. Default threshold count: 7.

threshold percent <value>

Specifies the fault threshold bandwidth percentage. Range: 1 to 100. Default threshold percent: 5.

threshold pps <value>

Specifies the fault threshold PPS. Range: 1-195312500.



NOTE: If excessive-broadcast or excessive-multicast faults are configured with the threshold higher than the `rate-limit` threshold, the following occurs:

- Fault reporting still happens as the port has actually received packets at a rate that violated its threshold.
- Traffic gets shaped as per `rate-limit` configuration and any packet exceeding the `rate-limit` threshold gets dropped.

These two fault-monitoring and `rate-limiting` features are orthogonal to one another.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the excessive jabbers, oversize packets, CRC, fragments, and TX faults with threshold value:

```
switch(config-fault-monitor-profile) # excessive-jabbers threshold value 30
switch(config-fault-monitor-profile) # excessive-oversize-packets threshold value 40
switch(config-fault-monitor-profile) # excessive-crc-errors threshold value 35
switch(config-fault-monitor-profile) # excessive-fragments threshold value 50
switch(config-fault-monitor-profile) # excessive-tx-drops threshold value 20
switch(config-fault-monitor-profile) # excessive-collisions threshold value 40
switch(config-fault-monitor-profile) # excessive-late-collisions threshold value 30
switch(config-fault-monitor-profile) # excessive-alignment-errors threshold value 50
```

Configuring the excessive link flaps fault with threshold count:

```
switch(config-fault-monitor-profile) # excessive-link-flaps threshold count 10
```

Configuring the excessive broadcasts and multicasts faults with threshold:

```
switch(config-fault-monitor-profile) # excessive-broadcasts threshold percent 40
switch(config-fault-monitor-profile) # excessive-broadcasts threshold pps 400000
switch(config-fault-monitor-profile) # excessive-multicasts threshold percent 10
switch(config-fault-monitor-profile) # excessive-multicasts threshold pps 10000
```

Removing the configured threshold value for the faults:

```
switch(config-fault-monitor-profile) # no excessive-jabbers threshold
switch(config-fault-monitor-profile) # no excessive-oversize-packets threshold
switch(config-fault-monitor-profile) # no excessive-crc-errors threshold
switch(config-fault-monitor-profile) # no excessive-fragments threshold
switch(config-fault-monitor-profile) # no excessive-tx-drops threshold
switch(config-fault-monitor-profile) # no excessive-link-flaps threshold
switch(config-fault-monitor-profile) # no excessive-broadcasts threshold
switch(config-fault-monitor-profile) # no excessive-multicasts threshold
switch(config-fault-monitor-profile) # no excessive-late-collisions threshold
switch(config-fault-monitor-profile) # no excessive-alignment-errors threshold
```

Command for applying a fault monitoring profile to a port

Syntax

```
apply fault-monitor profile <profile-name>
```

```
no apply fault-monitor profile <profile-name>
```

Description

Configures the fault monitoring profile to an interface. The profile is allowed to be modified after it is applied on a port.

The `no` form of this command removes the fault monitoring profile from an interface.

Command context

```
config-if
```

Parameters

<profile-name>

Specifies the fault monitor profile name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the fault monitoring profile to an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# apply fault-monitor profile noisy-ports
```

Configuring the fault monitoring profile to a specified range of interfaces:

```
switch(config)# interface 1/1/2-1/1/24
switch(config-if)# apply fault-monitor profile quiet-ports
```

Command for configuring VSX synchronization for a fault monitoring profile

Syntax

```
vsx-sync
```

```
no vsx-sync
```

Description

Configures VSX synchronization for a fault monitoring profile.

The `no` form of this command removes the VSX synchronization for a fault monitoring profile.

Command context

```
config-fault-monitor-profile
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring VSX synchronization for a fault monitoring profile:

```
switch(config-fault-monitor-profile)# vsx-sync
```

show fault-monitor profile

Syntax

```
show fault-monitor profile
```

```
show fault-monitor profile <profile-name>
```

Description

Shows the fault monitoring profile information. The command without the profile name displays all the available fault monitoring profiles.

Parameters

<profile-name>

Specifies the fault monitor profile name.

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying the fault monitoring information for all the profiles available on the switch:

```
switch# show fault-monitor profile
```

```
-----  
Fault monitor profile: 'noisy-ports'  
-----
```

Fault	Enabled	Threshold	Action	Auto Enable
excessive-broadcasts	yes	400000 pps	notify	--
excessive-multicasts	yes	10 percent	notify	--
excessive-link-flaps	yes	10	notify	--
excessive-oversize-packets	yes	40	notify-and-disable	80
excessive-jabbers	yes	30	notify-and-disable	100
excessive-fragments	yes	50	notify	--
excessive-crc-errors	yes	35	notify	--
excessive-tx-drops	yes	20	notify	--
excessive-collisions	yes	40	notify	70
excessive-late-collisions	yes	30	notify-and-disable	--
excessive-alignment-errors	yes	50	notify-and-disable	100

```
-----  
Fault monitor profile: 'quiet-ports'  
-----
```

Fault	Enabled	Threshold	Action	Auto Enable
excessive-broadcasts	yes	300000 pps	notify	--
excessive-multicasts	yes	100000 pps	notify	--
excessive-link-flaps	yes	20	notify-and-disable	45
excessive-oversize-packets	yes	25	notify-and-disable	20
excessive-jabbers	yes	30	notify	--
excessive-fragments	yes	60	notify	--
excessive-crc-errors	yes	25	notify-and-disable	30
excessive-tx-drops	yes	30	notify	--

Displaying the fault monitoring details for a particular profile:


```
switch# show fault-monitor profile noisy-ports
```

```
-----  
Fault monitor profile: 'noisy-ports'  
-----
```

Fault	Enabled	Threshold	Action	Auto Enable
excessive-broadcasts	yes	400000 pps	notify	--
excessive-multicasts	yes	10 percent	notify	--
excessive-link-flaps	yes	10	notify	--
excessive-oversize-packets	yes	40	notify-and-disable	80
excessive-jabbers	yes	30	notify-and-disable	100
excessive-fragments	yes	50	notify	--
excessive-crc-errors	yes	35	notify	--
excessive-tx-drops	yes	20	notify	--
excessive-collisions	yes	40	notify	70
excessive-late-collisions	yes	30	notify-and-disable	--
excessive-alignment-errors	yes	50	notify-and-disable	100

show interface fault-monitor profile

Syntax

```
show interface [<IF-NAME>|<IF-RANGE>] fault-monitor profile
```

Description

Shows the fault monitoring profile configured for the particular interface, range of interfaces, or all the interfaces.

Parameters

<IF-NAME>

Specifies the interface name.

<IF-RANGE>

Specifies the interface range.

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying all the interfaces configured with the fault monitoring profiles:

```
switch# show interface fault-monitor profile
```

```
-----  
Port      Fault Monitor Profile  
-----  
1/1/1     noisy-ports  
1/1/2     quiet-ports  
1/1/4     quiet-ports  
1/1/5     noisy-ports  
1/1/6     noisy-ports  
1/1/7     quiet-ports
```

Displaying the list or range of interfaces configured with the fault monitoring profiles:

```
switch# show interface 1/1/1-1/1/2,1/1/6 fault-monitor profile
```

```
-----  
Port      Fault Monitor Profile
```

```
-----
1/1/1    noisy-ports
1/1/2    quiet-ports
1/1/6    noisy-ports
```

show interface fault-monitor status

Syntax

```
show interface [<IF-NAME>|<IF-RANGE>] fault-monitor status
```

Description

Shows the active fault monitoring profile configured for the particular interface, range of interfaces, or all the interfaces.

Parameters

<IF-NAME>

Specifies the interface name.

<IF-RANGE>

Specifies the interface range.

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying active fault information for all the interfaces configured with the fault monitoring profiles:

```
switch# show interface fault-monitor status
```

Port	Fault	Fault Elapsed Time	Port State	Time Left
1/1/1	excessive-broadcasts	Tue Apr 14 14:29:09 UTC 2020	down	60
	excessive-jabbers	Tue Apr 15 14:29:09 UTC 2020	--	--
1/1/2	excessive-oversize-packets	Tue Apr 16 14:29:09 UTC 2020	down	--

Displaying active fault information for the list or range of interfaces configured with the fault monitoring profiles:

```
switch# show interface 1/3/1,1/3/3 fault-monitor status
```

Port	Fault	Occurring Since	Port State	Time Left
1/1/4	excessive-broadcasts	Tue Apr 14 14:29:09 UTC 2020	down	60
	excessive-jabbers	Tue Apr 15 14:29:09 UTC 2020	--	100

show running-config

Syntax

```
show running-config
```

Description

Shows the running configuration for the fault monitoring profile.

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying running configuration for the fault monitoring profiles:

```
switch# show running-config
fault-monitor profile noisy-ports
  excessive-broadcasts
  excessive-broadcasts threshold pps 10000
  excessive-multicasts
  excessive-multicasts threshold pps 10000
  excessive-link-flaps
  excessive-link-flaps threshold count 7
  excessive-oversize-packets
  excessive-oversize-packets threshold value 40
  excessive-jabbers
  excessive-jabbers threshold value 30
  excessive-fragments
  excessive-fragments threshold value 50
  excessive-crc-errors
  excessive-crc-errors threshold value 35
  excessive-tx-drops
  excessive-tx-drops threshold value 20
  excessive-collisions
  excessive-collisions threshold value 40
  excessive-late-collisions
  excessive-late-collisions threshold value 30
  excessive-alignment-errors
  excessive-alignment-errors threshold value 50
interface 1/1/1
  apply fault-monitor profile noisy-ports
```

show running-config all

Syntax

```
show running-config all
```

Description

Shows the running configuration with all options for the fault monitoring profile.

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying running configuration with options for the fault monitoring profile:

```
switch# show running-config all
fault-monitor profile noisy-ports
  excessive-broadcasts
  excessive-broadcasts threshold pps 10000
  excessive-broadcasts action notify-and-disable auto-enable 2000
  excessive-multicasts
  excessive-multicasts threshold pps 10000
  excessive-multicasts action notify
  excessive-link-flaps
  excessive-link-flaps threshold count 7
```

```
excessive-link-flaps action notify-and-disable auto-enable 2000
no excessive-oversize-packets
excessive-oversize-packets threshold value 40
excessive-oversize-packets action notify
no excessive-jabbers
excessive-jabbers threshold value 30
excessive-jabbers action notify
no excessive-fragments
excessive-fragments threshold value 50
excessive-fragments action notify
no excessive-crc-errors
excessive-crc-errors threshold value 35
excessive-crc-errors action notify
no excessive-late-collisions
excessive-late-collisions threshold value 30
excessive-late-collisions action notify
no excessive-collisions
excessive-collisions threshold value 40
excessive-collisions action notify
no excessive-tx-drops
excessive-tx-drops threshold value 20
excessive-tx-drops action notify
no excessive-alignment-errors
excessive-alignment-errors threshold value 50
excessive-alignment-errors action notify
```

The `auditors` group enables administrators to create users that can perform auditing tasks without allowing those users the authority to view or change the switch configuration.

As is the case for other users, auditors can access the switch using the Web UI, REST API, or the CLI.

Auditing tasks (CLI)

When you log on to the switch CLI as a user with auditor rights, you have access to the auditor command context only.

```
auditor>
```

The tasks that can be performed by auditors are as follows. The commands listed are the only commands auditors can execute other than session commands like `print`, `list`, and `exit`. However, auditors can use all command options except as noted. See the command description for each command for complete information about the command.

Task	Command or procedure	Example
Show event log contents	<code>show events</code>	<code>show events -a -r</code>
Show local accounting log contents	<code>show accounting log</code>	<code>show accounting log last 10</code>
Copy command output to a remote server or to a local USB drive.	<code>copy command-output</code>	<code>copy command-output "show events -a -r" tftp://10.100.0.12/file</code>



NOTE: When using the `copy command-output` command, users with auditor rights can specify the following commands only:

```
show accounting log
show events
```

Auditing tasks (Web UI)

Auditors have access to the Log page only. When you log on to the switch Web UI as a user with auditor rights, the Log page is displayed.

From the Log page, you can view and export event log entries.

The Web UI does not provide access to the accounting logs.

REST requests and accounting logs

All REST requests—including GET requests—are logged to the accounting (audit) log.

The URI of the REST API resource for accounting logs is the following:

`/rest/v10.04/logs/audit`

In an accounting log entry for a REST request:

- `service=https-server` indicates that the log entry is a result of a REST API request or a Web UI action.
- The string value of `data` identifies the REST API request that was executed.

For more information about accounting log entries, see the description of the `show accounting log` CLI command.

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation feedback

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